Designing for the Absent Grid: An MINLP Superstructure Formulation for an Off-Grid Pineapple Whole-Plant Biorefinery with a Shared Sugar Hub, Bajo Atrato (ZNI), Colombia

[NOTA: título. El anterior («Design and Optimization… Techno-Economic and Environmental Assessment») promete una optimización resuelta y una evaluación con cifras, y la tesis entrega la formulación y los protocolos (6.9, 11.2). El nuevo nombra la tensión (el supuesto oculto de red, 1.2) y el concepto que define la tesis (shared sugar hub), como sugería tu comentario del 24-sep. Si resuelves el MINLP antes de entregar, el título anterior vuelve a ser exacto. Decide tú.]

A Doctoral Dissertation (Draft)

[NOTA — PASADA DE RIGOR (5-nov-2026): copia de la «framework-highlighted». Ni el original REVISADA ni la copia «framework-highlighted» se modificaron (SHA-256 comprobado). ROSADO = lo escrito o cambiado en esta pasada; verde y turquesa siguen siendo de las pasadas anteriores; amarillo = notas. Qué hace: alinea las afirmaciones de los Cap. 6 y 11 con lo que ya está resuelto y verificado (la instancia reducida del MINLP con datos del TEM v10, certificada por enumeración, validada contra Excel en 0,08 % y con 38 comprobaciones de consistencia) y con el contraste de sus parámetros frente a Campos et al. (2020) y Seguí Gil y Fito Maupoey (2018). Cambios: 6.9 (dos párrafos nuevos), 6.11, 11.1 (una frase y un párrafo nuevo), 11.2 (primera y segunda limitación) y 11.3 (segunda dirección). Fuente de cada cifra: bajo_atrato_minlp/README.md, evidence_report.md y uncertainty_report.md, y el capítulo LaTeX tesis_tex/cap_minlp_bajo_atrato.tex. Reformula con tu voz lo rosado antes de entregar.]

Complete Draft: Chapters 1–11

[NOTA: copia «framework-highlighted» de la versión REVISADA (24-sep). El original no se modificó (SHA-256 comprobado). Turquesa = cambios con control de cambios de la pasada del 24-sep, aceptados solo en esta copia para poder editar encima (34 inserciones, 20 eliminaciones; los 13 comentarios se conservan). Verde = palabras o párrafos escritos o cambiados en esta pasada. Amarillo = notas para que verifiques o decidas. Problema principal que resuelve esta pasada: la tesis declara en 6.9 y 11.2 que el MINLP no se ha resuelto, pero el título, el esquema 1.8 y algunos cierres (Cap. 8, 10, 11) hablan como si hubiera resultados; el lector se entera en la página 111 de 215. Cambios: (1) título calibrado con la tensión de la red ausente y el hub de azúcar; (2) un párrafo en 1.6 que declara desde el inicio el estado diseño-y-metodología; (3) verbos calibrados en 1.2, 1.4, 1.8, 8, 10 y 11; (4) aperturas con problema en los Cap. 3, 5 y 9; (5) coherencia de conteos: el 24-sep añadió el objetivo (7) pero el Cap. 11 seguía diciendo «six»; (6) H1 en 11.1 alineada con la pregunta revisada; (7) las obras citadas en los Cap. 1 a 11 (30, 53, 12, 22, 8, 11, 20, 9 y 10) se cotejaron con sus resúmenes en Consensus (3-nov): corregí en verde lo que el resumen contradice y dejé en amarillo lo que solo el texto completo puede confirmar. No cambié la tesis ni la estructura de capítulos. (8) Cinco notas «LIBROS (1/5-5/5)» señalan dónde introducir Grossmann (2021), Maravelias (2021) y Burk (2026).]

Process Systems Engineering / Techno-Economic and Environmental Assessment

Bajo Atrato, Chocó, Colombia

Note on Sources

The literature cited in this draft was retrieved via academic search (Semantic Scholar, PubMed, Scopus, and arXiv indices) and is reported here with full bibliographic detail to the extent returned by that search. Prior to submission, every citation should be independently verified by the author against the original source, and reference formatting should be cross-checked against the current APA 7th edition Publication Manual and the specific formatting requirements of the receiving institution.

Chapter 1. Introduction

1.1 Background: The Pineapple Value Chain and Whole-Plant Valorization

Pineapple's global value chain concentrates value downstream while leaving biomass — and risk — upstream: processing keeps the pulp's sugar and discards 50-60% of the fruit's weight. Pineapple (Ananas comosus) ranks among the most economically important tropical fruits, with world production exceeding 25 million tonnes annually and continuing to expand across more than 80 producing countries (Hossain et al., 2016). Global markets and value chains for the crop have grown increasingly concentrated around varietal change, retailer-driven quality standards, and logistics innovation, even as the economic benefits of that expansion remain unevenly distributed across producing regions and along the chain from smallholder to exporter (Bairwa et al., 2025). This asymmetry is not incidental to the present study: it is the same structural feature — value concentrating downstream while biomass and risk concentrate upstream — that this thesis argues can be partially corrected at the processing-plant scale through whole-plant valorization.

Industrial processing of pineapple, however, remains overwhelmingly single-product. Pulp is pressed for juice or minimally processed for fresh-cut and canned fruit, while processing generates 50-60% of the original fruit's weight as peel, core, stem, and crown residue that is routinely discarded (Aili Hamzah et al., 2021; Sarangi et al., 2022). In Colombia specifically, pineapple production generates on the order of 185,000 tonnes of peel, pulp, stem, and leaf waste annually (Celis Cabeza et al., 2024) — a volume that mirrors the broader inefficiency documented across fruit-processing value chains internationally, where the fraction carrying the most immediately monetizable product (fermentable sugar in the pulp) is extracted while fractions carrying a proteolytic enzyme (bromelain), phenolic and vitamin micronutrients, dietary fiber, and lignocellulosic material are treated as waste (Sarangi et al., 2023).

[NOTA — citas del Cap. 1.1 (Consensus, 3-nov). Celis Cabeza et al. (2024): el resumen atribuye las ~185 000 t a la «producción» de piña en Colombia, no al procesamiento (corregido aquí y en 3.2); además Consensus registra ese DOI (10.18273/revion.v37n3-2024006) con un solo autor, John Wilmer Parra Llanos: revisa la autoría en la revista antes de citarlo como «Celis Cabeza et al.». Nath et al. (2023) no habla de cascada en su resumen; Louw et al. (2023), que ya citas en 2.6, sí («cascading of potential products from highest to lowest economic value»): considera citarlo aquí. El rango 50-60 %: Sarangi et al. (2022) y Campos et al. (2020) dicen ~60 %; el 50 % no aparece en los resúmenes.]

This pattern is neither a technological accident nor a simple market failure; it is, this thesis contends, a design-methodology failure. Chemical process design has historically proceeded sequentially: a target product is chosen, a flowsheet is built to produce it at minimum cost, and only the residual streams that this flowsheet cannot avoid generating are considered for secondary use, typically as an afterthought bolted onto an already-fixed configuration. Whole-plant valorization requires inverting this sequence — treating the full compositional diversity of the feedstock as the starting point of design, not as a byproduct of it. Rather than optimizing a single processing line, the whole plant is partitioned into fractions with structurally different compositions (developed quantitatively in Chapter 3), and each fraction is matched to the products for which its composition gives it the highest realizable value: a cascading approach in which higher-value, lower-volume products (bromelain, phenolic extracts) are recovered before residual streams are directed toward higher-volume, lower-value platforms (sugars, fiber, bioenergy) (Nath et al., 2023). This is the value-density logic formalized in Section 3.4 and embedded structurally in the superstructure of Chapter 6; its academic lineage in the forest-biorefinery product-portfolio-selection literature (Louw et al., 2023; Mansoornejad et al., 2010) is discussed in full in Section 2.6.

Integrated biorefinery demonstrations combining bromelain recovery with downstream sugar-platform fermentation have already shown this cascading logic works at bench scale: a single processing waste stream has been shown capable of yielding bromelain, xylooligosaccharides, and

fermentable glucose in one integrated sequence (Banerjee et al., 2021), and combined bioethanol-bromelain recovery trains have been demonstrated directly on pineapple residue (Gil et al., 2018). What these studies do not, and by their bench-scale nature cannot, resolve is the question this thesis is built around: when four such fractions, each with its own candidate technologies, are pursued simultaneously and share a common intermediate resource, what is the economically and environmentally optimal way to allocate that shared resource, and how does an off-grid energy constraint change the answer? A recent bibliometric review of the pineapple literature spanning nearly 8,000 documents found that research has concentrated overwhelmingly on chemistry, food technology, and bromelain's clinical properties, with few studies on rural extension or technology transfer to growers (Villalobos et al., 2023). Process integration and systems-level design are absent from the topics that review finds concentrated — an inference from its keyword map rather than a finding it states — and that absence is the gap this thesis addresses from the process-systems-engineering side of the literature rather than the food-science side where most prior pineapple-waste research originates.

[NOTA: Villalobos et al. (2023) — el resumen confirma 7 885 documentos y los temas dominantes (química, tecnología de alimentos, uso medicinal, bromelina clínica), pero el vacío que reporta es extensión rural y transferencia a productores, no integración de procesos. Esa parte ahora está marcada como inferencia tuya.]

1.2 The Bajo Atrato / Zonas No Interconectadas (ZNI) Context: An Off-Grid Siting Constraint as a Design Driver

Colombia's electricity system distinguishes the Sistema Interconectado Nacional (SIN) from the Zonas No Interconectadas (ZNI) — regions covering roughly half of the country's land area but a small fraction of its population, historically dependent on diesel-fired local generation at high cost and with unreliable supply (Bustos et al., 2014). Around 1.9 million Colombians, all

of them in rural areas, still lacked electricity access in 2020, and government policy has struggled for more than a decade to move ZNI communities off diesel despite sustained subsidy programs and an increasingly complex regulatory apparatus (Garces et al., 2021). System-dynamics modeling of ZNI electrification demonstrates that a transition from diesel to renewables is both technically and economically viable, and that, to make the best use of limited public funding, it should begin as soon as possible and be accompanied by a zero-diesel policy by 2040 (Garces et al., 2023). Techno-economic assessments of hybrid photovoltaic/wind/diesel systems in Colombian off-grid villages independently find that renewable-dominated configurations are cost-competitive once diesel's true operating cost — not its subsidized retail price — is accounted for in the comparison (Haghighat Mamaghani et al., 2016), and a fuzzy-AHP decision framework developed for non-interconnected rural areas of the Colombian Caribbean ranks solar photovoltaic first among six renewable alternatives, with economic and environmental criteria weighted highest, although all six score within five percentage points of one another (Moreno et al., 2022).

[NOTA — citas verificadas en Consensus (resúmenes, 3-nov). Garces et al. (2021): el resumen dice «around 1.9 million» en 2020, «all residing in rural areas»; el texto decía «more than» y «overwhelmingly». Garces et al. (2023): el resumen sí dice «demonstrate», así que dejé tu verbo, pero no dice «only under…» ni que demorar la transición agrave la carga fiscal sin mejorar la confiabilidad; reemplacé esa parte por lo que el resumen sí afirma (empezar cuanto antes, política cero diésel a 2040). Haghighat Mamaghani et al. (2016): el resumen describe el estudio (HOMER, tres veredas) pero no reporta resultados ni compara el costo real del diésel con su precio subsidiado; confírmalo en Resultados o suaviza la frase. Moreno et al. (2022) existe (IJEEP, doi:10.32479/ijeep.12913): la cito directo en lugar de «as cited in…», con su alcance real (región Caribe, 10 expertos, diferencias <5 %); falta en la lista de referencias, agrégala. Bustos et al. (2014) es un working paper de SSRN que Consensus no indexa: no verificado.]

Bajo Atrato, in the department of Chocó, sits within this ZNI context: agro-industrially promising — pineapple cultivation is already present in the region — but structurally constrained by the absence of a firm, low-cost electricity supply that most techno-economic biorefinery studies

simply assume via grid connection. This assumption is not a minor simplification in the biorefinery TEA literature; it is, in effect, a hidden boundary condition that silently excludes an entire class of sites, including most of rural Chocó, from the population of locations a conventional techno-economic feasibility study would even consider viable. Reframing the boundary condition — from 'assume grid access' to 'design for its absence' — is therefore not merely a regional adaptation of an existing methodology; it changes which superstructure configurations are feasible at all, since any configuration whose peak or continuous power demand exceeds what the site's own residue stream can generate is, by construction, inadmissible regardless of its economic merit under a grid-connected counterfactual.

This thesis treats the off-grid constraint as a design driver, not as a downstream engineering problem to be solved after the process is fixed. Concretely, this means three things. First, the biorefinery must generate its own process heat and electricity from its own lignocellulosic residue (Chapter 5); no external energy import is admitted into the reference design, consistent with the documented reality that grid extension to remote ZNI communities is frequently either technically infeasible or economically irrational relative to distributed generation (Garces et al., 2023). Second, the resulting energy balance is coupled directly into the superstructure's feasibility set (Section 5.4, Section 6.1) rather than sized independently after the product portfolio is chosen — a methodological departure from the sequential design-then-size pattern that dominates the off-grid bioenergy literature reviewed in Section 2.7. Third, the design's robustness to feedstock seasonality and composition variability must be evaluated with respect to whether energy autonomy — not only economic performance — is preserved (Section 5.4's flexibility analysis, Section 6.7's uncertainty formulation). A design that is profitable in expectation but which cannot sustain its own energy island under plausible feedstock variability is not a viable ZNI design,

regardless of its expected net present value; this is the precise distinction, elaborated methodologically in Section 2.9 and Chapter 6, between an economic risk criterion (expected value under uncertainty) and a feasibility guarantee (flexibility under uncertainty), and this thesis insists the two not be conflated.

1.3 Problem Statement: From Single-Product Processing to an Integrated, Energy-Autonomous Fractionation Biorefinery

The problem this thesis addresses can be stated formally as follows: given whole-plant pineapple biomass with structurally distinct fraction compositions, and given an off-grid siting constraint that removes the option of external electricity or heat supply, what configuration of processing lines — including which lines are built, at what scale, and how shared intermediate streams are allocated among them — maximizes economic value while remaining energy-autonomous, and while making the environmental performance of the resulting configuration an explicit, evaluated criterion rather than an afterthought computed incidentally to the economic model?

Existing literature addresses component parts of this problem, but always in isolation from the others, and the isolation is not merely a matter of scope — it reflects genuinely different methodological traditions that this thesis must bridge. Bromelain recovery from stem and core has a substantial process literature on precipitation-based and membrane-based routes, including a head-to-head comparison of ultrafiltration with aqueous biphasic extraction (Simões et al., 2021) and an optimized polysaccharide-precipitation route (Campos et al., 2019), but this literature is written from a bioprocess-engineering perspective that treats the recovered enzyme as the main product of interest and, with exceptions such as the fermentation of the ultrafiltration permeate to ethanol by Gil et al. (2018), is largely silent on what happens to the permeate or mother liquor

thereafter — precisely the stream this thesis routes to the shared sugar hub. Fruit-residue-to-bioenergy gasification systems are documented across multiple off-grid contexts, from olive-mill electrification in Morocco to rural mini-grids in India (Sánchez-Lozano et al., 2023; Chambon et al., 2020), but this literature is written from an energy-systems perspective in which the biomass feedstock is exogenous and the downstream chemical-product question simply does not arise. Superstructure-based MINLP synthesis with simultaneous economic-environmental optimization is a mature methodology in process systems engineering, demonstrated with joint NPV and GWP objectives for gasification-route hydrocarbon biorefineries (Wang et al., 2013) and, with yield, cost, and waste objectives optimized in turn, for multi-product lignocellulosic biorefineries (Zondervan et al., 2011), but this literature is written from an optimization-methodology perspective that, with rare exceptions (Kenkel et al., 2022), treats the choice of feedstock and product slate as a case-study detail rather than as the object requiring justification in its own right.

[NOTA — citas de 1.3 (Consensus, 3-nov). Simões et al. (2021) compara ultrafiltración con sistemas acuosos bifásicos, no con precipitación; Campos et al. (2019) optimiza la precipitación con carragenina sin compararla con UF. Gil et al. (2018), que ya citas en 1.1, fermentó a etanol el permeado de la UF: la frase «largely silent on the permeate» necesitaba esa excepción. Lo distintivo de la tesis sigue en pie, pero es el reparto del hub entre varias plataformas, no el reuso del permeado en sí. Zondervan et al. (2011) existe (Comput. Chem. Eng., doi:10.1016/j.compchemeng.2011.01.042, «Optimal design of a multi-product biorefinery system») pero falta en la lista de referencias; su resumen optimiza rendimiento, costo y residuos por separado, no economía y ambiente a la vez. Kenkel et al. (2022) usa MILP, lo que respalda tu contraste con el hub no convexo (1.7).]

What is missing — and what Section 2.10's differentiation table develops systematically — is the combination of all four: a whole-plant pineapple superstructure, under an off-grid energy-autonomy constraint treated as a hard feasibility condition rather than a sizing exercise performed after the fact, solved via simultaneous (not sequential) MINLP synthesis, with uncertainty and flexibility treated with the methodological precision the underlying decisions require rather than

folded together under a single undifferentiated label of 'risk.' The absence has two concrete costs. Because biorefinery techno-economic assessment silently assumes grid access, an entire class of sites, including most of rural Chocó, never enters the feasible set (Section 1.2). And because sequential design fixes each line's allocation before the next is sized, it structurally cannot see the shared-hub trade-off, returning a solution no better — and in a nonconvex system possibly worse — than the simultaneous one (Section 1.4). Establishing that no single prior study occupies this intersection is, admittedly, a claim of absence rather than presence, and Section 2.10 is correspondingly explicit about the boundary of that claim: it does not assert that no individual technical component of this thesis is novel in isolation, but that their integration under a genuinely binding off-grid constraint, solved as one simultaneous optimization problem rather than a sequence of decoupled sub-problems, is.

1.4 Why an MINLP Superstructure? Simultaneous Synthesis of Product Lines under a Shared Sugar Hub

A superstructure embeds every candidate processing configuration inside a single network representation, with binary variables activating or deactivating candidate units and technologies, and continuous variables governing flows, splits, and operating conditions. Optimizing over the superstructure simultaneously selects the configuration and its operating conditions, rather than fixing a configuration by engineering judgment first and only then optimizing its parameters. The systematic superstructure representation via state-task and state-equipment networks, formalized with Generalized Disjunctive Programming and reformulated into MILP/MINLP, is well established in the process-synthesis literature and traces directly to Yeomans and Grossmann's (1999) foundational framework, which this thesis's Chapter 6 formulation adapts and extends to a bio-based, off-grid setting the original framework was not built to address.

This distinction between simultaneous and sequential design is not academic in this problem; it is the specific mechanism by which whole-plant valorization can either succeed or silently fail to capture its own potential. The shared sugar hub means the profitability of any one line — say, the citric acid platform drawing from the hub — depends on how much sugar-bearing stream is diverted there rather than to vinegar, ethanol, or xanthan gum production, a coupling that a sequential, fraction-by-fraction design process structurally cannot see, because by the time the citric-acid line is being sized, the vinegar line's allocation has already been fixed by an earlier, independent decision. A sequential designer optimizing each line in isolation is, in effect, solving four smaller, easier problems whose combined solution is provably no better, and in a nonconvex system can be strictly worse, than the solution to the one larger, harder problem this thesis formulates.

Formally, the superstructure of this thesis is a Mixed-Integer Nonlinear Program (MINLP): binary variables y activate candidate processing lines, with associated fixed-charge, minimum-scale capital-expenditure effects (Section 6.4); continuous variables α govern splits at the fraction trains and at the hub itself; and the hub's own allocation is — as Section 6.3 makes explicit — a nonlinear (bilinear) pooling problem (Quesada & Grossmann, 1995; Misener & Floudas, 2009), since the property (sugar concentration, purity) of the pooled stream is itself a nonlinear function of which upstream streams are mixed and in what proportion. The choice of MINLP over a linear (MILP) formulation is therefore not a modeling convenience adopted for generality's sake; it is forced by the physics of the hub itself, and any formulation that linearizes this coupling away — as, this thesis will argue in Section 2.6, most prior superstructure biorefinery studies implicitly do — is not modeling a simplified version of this problem but a structurally different problem that happens to share its topology.

1.5 Research Question, Hypotheses, and Objectives

Research question: Can a simultaneous MINLP superstructure formulation of a whole-plant pineapple biorefinery, under a hard off-grid energy-autonomy constraint, identify configurations that are more profitable than a sequential, single-fraction design baseline while revealing non-degenerate economic-environmental trade-offs among them — and how sensitive is that configuration to feedstock and market uncertainty?

Hypotheses:

H1. A whole-plant superstructure that jointly optimizes fraction-level processing lines and their shared sugar hub yields a higher optimal net present value (NPV) than the sum of independently optimized single-fraction lines, because the simultaneous formulation captures hub-allocation trade-offs invisible to sequential design — consistent with reported gains from simultaneous superstructure/utility co-design in other biorefinery contexts (Kong et al., 2016).

H2. The energy-autonomy constraint (Chapter 5) is binding — that is, it materially restricts the feasible product portfolio relative to an unconstrained, hypothetically grid-connected design — and the magnitude of that restriction can be quantified via a Flexibility Index on the lignocellulosic-to-energy balance.

H3. The Pareto-optimal set generated via the ε-constraint method on NPV and global warming potential (GWP) (Section 6.6) is non-dominated and non-degenerate — that is, meaningful economic-environmental trade-offs exist among candidate configurations rather than a single configuration dominating on both criteria — following the precedent set for hydrocarbon and algal biorefinery superstructures solved the same way (Wang et al., 2013; Gong & You, 2014).

General objective: To design and optimize, via an MINLP superstructure, a whole-plant pineapple biorefinery for the off-grid Bajo Atrato context, and to assess its techno-economic and environmental performance under uncertainty.

Specific objectives: (1) Characterize the composition, availability, and seasonality of pineapple whole-plant fractions in Bajo Atrato as the quantitative basis for the superstructure (Chapter 3). (2) Develop validated process models for each fraction's candidate processing trains and for the shared sugar hub (Chapter 4). (3) Formulate and solve the off-grid energy island as a gasification-internal combustion engine (ICE) system coupled to process heat/power demand (Chapter 5). (4) Formulate the whole-plant superstructure as an MINLP, explicitly representing the hub's bilinear pooling structure and the discrete/continuous coupling of line activation (Chapter 6). (5) Generate the NPV-GWP Pareto front via ε-constraint and characterize its uncertainty via a stochastic/robust hybrid formulation and a flexibility analysis of the energy island (Chapter 6). (6) Conduct a life-cycle assessment of candidate configurations and reconcile it with the techno-economic assessment via formal multi-criteria decision-making (Chapters 7-8). (7) Synthesize the techno-economic and environmental results to test hypotheses H1-H3, and frame the recommended configuration within Colombia's ZNI policy and implementation environment (Chapters 9-10).

[NOTA: el objetivo (7) lo añadió la pasada del 24-sep (turquesa). En esta copia llevé el conteo al Cap. 11 («six» → «seven» y una frase de conclusión para el objetivo 7 en 11.1). Si rechazas el (7), devuelve esos cambios. El cambio de la pregunta de investigación (comentario del 24-sep) también sigue siendo decisión tuya.]

1.6 Scope and System Boundaries

This thesis addresses a gate-to-gate system: from the point of whole-plant pineapple biomass arrival at the processing facility to the point of shipped products and process

residues/emissions. The life-cycle assessment of Chapter 7 labels its boundary cradle-to-gate under the residue convention: the pineapple residue enters the system burden-free, since its cultivation burden would be incurred whether or not the biorefinery is built (Section 7.2). Upstream agricultural production is excluded from the process model and from the primary life-cycle assessment system boundary but is discussed qualitatively in Section 3.3 for feedstock availability and seasonality purposes; a sensitivity case allocating a share of cultivation burden to the residue is noted as a direction for Chapter 7, consistent with recommendations from biorefinery life-cycle-assessment methodology reviews that boundary choice be made explicit and justified rather than defaulted into implicitly (Ahlgren et al., 2015). This choice is not without cost: residue-based system boundaries systematically under-report the full environmental footprint of the value chain relative to a cultivation-inclusive treatment that allocates a share of cultivation impacts to the residue stream, and Chapter 7 returns to this limitation explicitly rather than presenting the residue-burden-free result as if it were the whole picture.

Geographically, the design basis (Section 3.5) reflects Bajo Atrato feedstock availability and logistics; the design is not claimed to generalize without re-parameterization to other pineapple-producing regions or to grid-connected sites, where the off-grid constraint that drives much of this thesis's methodological content would not bind and would therefore not justify the additional modeling complexity the energy-autonomy coupling introduces. Temporally, the techno-economic evaluation uses a representative annual reference scenario (Section 3.5) with seasonality treated as a source of parametric uncertainty (Section 6.7) rather than as an explicit multi-period scheduling problem; full multi-period/campaign scheduling of the biorefinery, while a natural extension given the process-scheduling frameworks reviewed for related discrete-manufacturing contexts, is noted as outside this thesis's scope and as a direction for future work.

One further boundary governs how every later chapter should be read. This dissertation is a design-and-methodology study: it formulates the superstructure and its process, energy, economic, and environmental sub-models, and it specifies the protocols by which they are to be solved, verified, and tested, but it does not yet report an end-to-end numerical solution of the MINLP (Section 6.9, Section 11.2). Chapters 6-10 therefore state formulations, tests, and their interpretation rather than numerical results, and the computation itself is the first item of future work (Section 11.3).

[NOTA: párrafo nuevo. Hoy el lector descubre en 6.9 (y en 11.2) que no hay solución numérica; hasta ahí, el título, la pregunta y el esquema 1.8 le hacen esperar resultados. Las tres frases solo repiten lo que ya dicen 6.9, 11.2 y 11.3. Cuando escribas el resumen (Abstract), que el borrador aún no tiene, conviene que diga lo mismo en una frase.]

1.7 Contributions and Novelty

This thesis's contributions, developed against the positioning established in Section 2.10, are five. First, a whole-plant, four-fraction pineapple biorefinery superstructure, as distinct from the single-fraction (pulp-only) scope of prior pineapple-processing techno-economic studies. Second, an explicit, hard off-grid energy-autonomy constraint coupled directly into the superstructure's feasibility set, rather than treated as an independent utility-sizing exercise performed after product-portfolio selection. Third, an explicit bilinear pooling formulation of the shared sugar hub, with a stated and justified relaxation/solution strategy, rather than an implicit or linearized treatment of hub allocation. Fourth, a methodologically precise separation of stochastic (expected-value, known-distribution) and robust (guaranteed-feasibility, uncertainty-set) treatments of uncertainty, applied respectively to market/price risk and to energy-autonomy feasibility, including a Flexibility Index quantification of the latter. Fifth, an integrated techno-economic and life-cycle assessment in which the final configuration selection is made via a

formally justified multi-criteria decision method applied to the NPV-GWP Pareto front, reconciled with an economic value-density interpretation — extending the integrated multi-objective-optimization/multi-criteria-decision-making methodology piloted for power-to-X and biomass-to-X processes (Kenkel et al., 2022) to a biorefinery context whose core is a genuinely nonconvex (pooling) network, which the power-to-X precedent did not need to confront.

1.8 Thesis Outline

Chapter 2 reviews the state of the art across pineapple biomass valorization, biorefinery fractionation strategies, superstructure-based process synthesis, off-grid bioenergy, techno-economic and life-cycle assessment methods, and multi-objective/multi-criteria decision methods, closing with the differentiation table that positions this thesis's contributions. Chapter 3 characterizes the feedstock and establishes the design basis. Chapter 4 develops the process models for each fraction's processing train and the shared sugar hub. Chapter 5 formulates the off-grid energy island. Chapter 6 formulates the MINLP superstructure, including its uncertainty and flexibility analysis, and states how it will be solved and verified. Chapter 7 develops the life-cycle assessment methodology. Chapter 8 develops the techno-economic assessment and the formal multi-criteria procedure by which the Pareto-optimal configurations will be narrowed to a recommended configuration. Chapter 9 reads Chapters 6-8 together and specifies the tests of hypotheses H1-H3. Chapter 10 places the recommended configuration within Colombia's ZNI policy environment and sets out an implementation roadmap for Bajo Atrato. Chapter 11 reports conclusions against each specific objective, the thesis's limitations, and directions for future research.

Chapter 2. State of the Art

The literature addresses each element of the problem stated in Section 1.3 — bromelain recovery, sugar-platform fermentations, off-grid gasification, superstructure-based MINLP synthesis, and techno-economic, life-cycle and multi-criteria assessment — but from separate methodological traditions that rarely meet. This chapter reviews each tradition in turn (Sections 2.1-2.9) and closes with a differentiation table (Section 2.10) showing where they fail to intersect: the gap this thesis occupies.

2.1 Pineapple Biomass: Composition and Uses of Pulp, Stem+Core, Peel, and Leaves/Crown

Whole-plant pineapple biomass partitions into four fractions with distinct compositions and, correspondingly, distinct valorization pathways: the pulp, rich in fermentable sugars and the basis of juice, wine, and vinegar platforms; the stem and core, which carry the highest concentration of bromelain along with phenolic compounds; the peel, which combines a substantial sugar-bearing juice fraction with insoluble dietary fiber and comparatively low bromelain content; and the leaves/crown, lignocellulosic in composition and therefore unsuited to fermentation platforms but well suited to thermochemical energy conversion (Chapter 5). Direct compositional characterization confirms this partition quantitatively: peel juice and stem juice differ markedly in phenolic content, antioxidant capacity, and vitamin C, while the corresponding press-cake flours are high in insoluble dietary fiber, with peel juice and stem juice representing approximately 17.3% and 4.8% of total pineapple weight, respectively, and their flours a further 11.4% and 3.1% (Campos et al., 2020). Fraction-level enzymatic activity likewise varies systematically: bromelain activity and purification behavior differ across core, peel, crown, and

stem fractions enough that fraction choice materially affects downstream process design, with stem tissue showing the highest activity in three direct comparisons (Huang et al., 2021; Mardawati et al., 2023; Chiarelli et al., 2024) but the lowest in a nanofiltration study that ranks flesh first (Misran et al., 2022), while the order of the remaining fractions also varies: peel exceeds core in MD2 by-products (Chiarelli et al., 2024), whereas core is the most active fraction among processing wastes that exclude stem (Banerjee et al., 2021) — a discrepancy this thesis treats not as a contradiction to be resolved by literature review alone but as a design-relevant source of parametric uncertainty to be carried explicitly into Chapter 4's process models and Chapter 6's robustness formulation, since committing to a single point estimate for bromelain content would silently understate the sensitivity of the stem+core train's economics to cultivar and post-harvest handling.

This composition heterogeneity is the structural reason a single-product design cannot capture the biomass's full value: no single process technology is simultaneously appropriate for high-value enzyme recovery, sugar fermentation, fiber recovery, and lignocellulosic energy conversion. Comparative phenolic profiling across pineapple tissues reinforces this point at a finer grain than the four-fraction partition alone would suggest: total phenolic content, flavonoid content, and the specific phenolic acid profile (catechin, gallic acid, ferulic acid, among others) differ not only between peel, core, and pulp but between the inner and outer layers of the pulp itself (Uslu et al., 2024; Nordin et al., 2023), and crown tissue is characterized by depsides of p-coumaric and ferulic acid, which peel also carries at elevated concentrations, whereas pulp is dominated by glutathione- and cysteine-derived phenolic conjugates (Steingass et al., 2015). For this thesis's purposes, this finer structure is acknowledged but not modeled explicitly — Chapter 4's process models operate at the four-fraction resolution the superstructure requires — but it is

flagged here as a limitation to be revisited should future work seek to further sub-fractionate the peel or pulp streams for premium phenolic-extract products.

[NOTA — citas de 2.1 (Consensus, 3-nov). Chiarelli et al. (2024) no reporta el corazón como la fracción más activa: el tallo basal es la más activa (717 CDU/mg), luego cáscara (217) y corazón (124); la corona no tuvo actividad. Con Huang et al. (2021) y Mardawati et al. (2023), el tallo sale primero en tres comparaciones, pero Misran et al. (2022) lo ubica último (pulpa > cáscara > corazón > corona > tallo, en retenidos de nanofiltración); el orden de corazón y cáscara también cambia (Banerjee et al., 2021 da el corazón). La discrepancia que la tesis señala es real, pero no con las fuentes que citaba. Ramli et al. (2017) es una revisión cuyo resumen no ordena fracciones: la quité de esta frase. Steingass et al. (2015): el resumen dice que la cáscara también tiene esos dépsidos en concentración elevada, lo contrario de «not dominant in peel». Faltan en la lista de referencias: Uslu et al. (2024, J. Food Safety and Food Quality, doi:10.31083/0003-925x-75-86) y Steingass et al. (2015, Anal. Bioanal. Chem., doi:10.1007/s00216-015-8811-2).]

2.2 Cascading and Whole-Plant Fractionation Strategies for Fruit Biomass

The cascading principle — recovering the highest-value, typically lowest-volume products first, and directing residual streams toward progressively higher-volume, lower-value platforms — is well established in agro-industrial biorefinery literature, and its academic formalization is most developed in the forest-biorefinery product-portfolio-selection tradition reviewed in Section 2.6. Applied to tropical agro-industrial residue more broadly, integrative biorefinery reviews covering pineapple alongside sugarcane and banana by-products in Costa Rica suggest that whole-crop fractionation strategies are transferable across fruit biomass types sharing similar lignocellulosic and sugar-bearing fraction structures (Eixenberger et al., 2022). Pineapple-specific whole-plant fractionation combining bromelain, xylooligosaccharide, and glucose co-production from a single integrated waste stream has been demonstrated at bench scale, with reported yields of approximately 1 kg bromelain, 24 kg xylooligosaccharides, and 88 kg glucose per tonne of fresh processing waste (Banerjee et al., 2021).

Such demonstrations remain, however, single-configuration proofs of concept rather than optimization-based syntheses over a full candidate technology space — the gap this thesis addresses. This distinction matters methodologically, not merely as a scope statement: a bench-scale demonstration establishes that a particular cascade is technically feasible and reports its yields under the conditions tested, but it cannot, by construction, establish that the demonstrated cascade is the economically or environmentally optimal one among the combinatorially large space of candidate cascades a whole-plant biorefinery could pursue. Answering that latter, optimization-shaped question is precisely what a superstructure formulation is for, and its absence from the bench-scale pineapple-biorefinery literature reviewed here is the specific methodological vacancy Chapter 6 fills.

2.3 High-Value Bioactives: Bromelain Recovery (Precipitation vs. Ultrafiltration) and Phenolic/Vitamin Nutraceuticals

Bromelain recovery is the highest-value-density opportunity in the stem+core fraction and has an established comparative literature contrasting precipitation-based recovery against membrane-based ultrafiltration. Direct process comparisons report ultrafiltration achieving substantially higher bromelain-specific recovery than aqueous biphasic (precipitation-type) systems under matched conditions — 93.2% versus 18.5% in one head-to-head study, although the same study found the precipitation route achieved comparable or superior total protein recovery, indicating that the two technologies are optimizing for different objectives (enzyme-specific recovery versus bulk protein recovery) rather than one being unconditionally superior to the other (Simões et al., 2021). Green polysaccharide (carrageenan) precipitation — deliberately avoiding organic solvents and inorganic salts — has been optimized to achieve 80-90% recovery yields with high purity from both stem and peel juices (Campos et al., 2019), narrowing but not

eliminating the yield gap relative to membrane-based routes while offering a materially different downstream waste profile. Two-stage ceramic ultrafiltration schemes report 94-97% enzyme recovery with 2.5- to 4.4-fold purification depending on the pretreatment applied upstream of the membrane train (Nor et al., 2016, 2018).

The trade-off most relevant to this thesis's process modeling (Section 4.3) is not, in the end, purely one of recovered yield or purity, but of downstream compatibility with the rest of the superstructure. Precipitation processes generate a solvent- or salt-laden waste stream requiring its own treatment/disposal pathway, adding a cost and environmental burden this thesis's Chapter 7 must account for even though it is external to the bromelain train narrowly construed. Ultrafiltration, by contrast, generates a permeate stream that — critically for this thesis's superstructure — is itself sugar-bearing and can be routed to the shared sugar hub (Section 4.6) rather than discarded, as demonstrated directly in an integrated bioethanol-bromelain process where microfiltration/ultrafiltration permeate was fermented to ethanol after bromelain recovery (Gil et al., 2018). This is one of the concrete mechanisms by which the whole-plant, hub-coupled superstructure (Section 2.6) captures value that an isolated bromelain-recovery process evaluation, of the kind that dominates the literature reviewed in this section, structurally cannot see: choosing ultrafiltration over precipitation is not decidable from the bromelain train's own economics alone, because the choice changes what downstream sugar-hub feedstock becomes available, which in turn changes the profitability of every line drawing from that hub. This is precisely the kind of cross-train coupling Section 1.4 argues a sequential design methodology cannot represent.

2.4 Sugar-Platform Products: Vinegar, Citric Acid, Xanthan Gum, Ethanol, and Premium Fermentations (Wine, Premium Vinegar, Brandy)

The sugar-bearing streams converging at the shared hub are candidate feedstocks for a portfolio of fermentation and chemical platforms with well-documented individual process literatures. Citric acid via Aspergillus niger solid-state fermentation of agro-industrial fruit waste is extensively documented, including specifically on pineapple waste itself: pineapple peel with sucrose and ammonium nitrate supplementation has achieved 60.6 g citric acid per kg substrate at over 90% fermentable-sugar conversion (Kareem et al., 2010), and a central-composite-optimized fermentation of pineapple waste reached 15.5 g/L of citric acid with glucose supplementation (Ayeni et al., 2019); comparative trials on other fruit residues report 66 g/kg of dry apple pomace by solid-state fermentation, rising to 128 g/kg with an ethanol inducer (Dhillon et al., 2011).

[NOTA — citas de 2.4 (Consensus, 3-nov). Ayeni et al. (2019) reporta 15,51 g/L en un medio con glucosa, no un rendimiento «comparable» a los 60,6 g/kg de Kareem et al. (2010): son unidades distintas. Dhillon et al. (2011) no es una revisión ni dice que la fermentación en estado sólido sea la ruta industrial dominante; es un estudio con orujo de manzana (la ruta industrial habitual del ácido cítrico es la sumergida: si quieres decirlo, busca una fuente). Ácido acético: Roda et al. (2017a) da 5 % y Tanamool et al. (2020) 7,2 % (6,5 % a 6 L); el 7,8 % no sale de ninguno de los dos. En la lista de referencias, las dos entradas de Roda et al. (2017) no llevan «a» y «b»: la (a) es la de Food Chemistry (vinagre), la (b) la de J. Food Process. Preserv. (vino).]

Xanthan gum via Xanthomonas campestris fermentation of agro-food waste is similarly well documented, and — notably for this thesis — has been demonstrated specifically on pineapple waste hydrolysate, with machine-learning-optimized fermentation conditions predicting a maximum yield of 10.34 g/L, an 11.9% increase over an unoptimized control (Amenaghawon et al., 2024); broader reviews of agro-food-waste xanthan production confirm that yields from diverse residues are comparable to conventional glucose/sucrose substrates once pretreatment is adequately optimized (Khosravi et al., 2026).

Vinegar production from pineapple waste via sequential alcoholic and acetous fermentation is directly documented for peel, core, and pulp fractions, with acetic acid

concentrations of 5% (Roda et al., 2017a) to 7.2% w/v, or 6.5% at 6-L scale (Tanamool et al., 2020), depending on substrate and bacterial strain, and pineapple wine from the same peel/core substrate has been characterized across yeast strains and fermentation temperatures, reaching 7-8% v/v ethanol with strain- and temperature-dependent aroma profiles (Roda et al., 2017b). Broader reviews of fruit-vinegar biotechnology confirm that acetification-system choice — surface versus submerged fermentation — is one of the most influential factors in the final vinegar's physicochemical properties (Luzón-Quintana et al., 2021). Each of these platforms has, individually, an established process literature with reasonably well-characterized yields and operating conditions; what is comparatively undeveloped, and what Section 2.6 argues is the specific contribution of a superstructure approach, is their joint techno-economic treatment as competing claimants on a single shared, finite sugar resource. Treated individually, each platform's published techno-economic potential implicitly assumes it has first claim on the available sugar; treated jointly, as this thesis's Chapter 6 requires, that assumption cannot hold for more than one platform simultaneously, and the allocation problem this creates is precisely what the pooling-based hub formulation is built to resolve.

2.5 Membrane and Green Separation Processes in Biorefineries (Micro-/Ultrafiltration, Precipitation, Spray Drying)

Membrane and green separation technologies — ceramic microfiltration, ultrafiltration, precipitation, and controlled-temperature spray drying — are increasingly favored in biorefinery design over solvent-intensive classical separations, both for downstream product quality, particularly retention of thermolabile bioactives, and for reduced auxiliary-chemical demand (Castro-Muñoz et al., 2019). Reviews of membrane technology in agri-food waste recycling emphasize that pressure-driven processes such as microfiltration, ultrafiltration, and nanofiltration

operate at low temperature, with low energy requirements and no added chemicals (Papaioannou et al., 2022); a dedicated review of membrane processes across the biorefinery value chain frames pressure-driven separation as clean in a chemical-input sense but non-trivial in energy terms once scaled beyond bench level, particularly for the ultrafiltration and nanofiltration steps common to lignin and protein recovery trains that are structurally analogous to this thesis's bromelain-recovery train (Abels et al., 2013). Ultrafiltration-specific reviews in food processing identify membrane fouling as the factor that erodes ultrafiltration's performance and raises its cost (Mohammad et al., 2012).

[NOTA — citas de 2.5 (Consensus, 3-nov). Papaioannou et al. (2022) dice lo contrario de lo que se le atribuía: procesos a 20-60 °C, con bajo requerimiento energético y sin químicos. El argumento de 2.5 («membranas = energía no trivial») queda apoyado solo en Abels et al. (2013), cuyo resumen no habla de energía, y en Mohammad et al. (2012), que habla de ensuciamiento y costo. Busca una fuente que cuantifique la energía de UF/NF a escala o suaviza el párrafo siguiente («This trade-off between energy intensity and chemical intensity…»). Papaioannou et al. (2022, Sustainability, doi:10.3390/su14031483) falta en la lista de referencias.]

This trade-off between energy intensity and chemical intensity is a first-order reason the energy island (Chapter 5) and the environmental assessment (Chapter 7) cannot be treated as independent of the separation-technology choices made in Chapter 4: a design that substitutes membrane separation for solvent precipitation throughout, on the reasonable-sounding grounds that it is 'greener' in a chemical-input sense, may in fact increase the demand placed on the off-grid energy island to a degree that the island cannot sustain, or may shift — rather than reduce — the design's environmental burden from a chemical-toxicity impact category to a global-warming-potential impact category associated with the additional electricity generation required. Chapter 4 and Chapter 7 return to this tension explicitly rather than assuming membrane separation is unconditionally preferable, which the food-science literature reviewed in this section, focused as it is on product quality rather than on-site energy constraints, has no occasion to interrogate.

2.6 Superstructure-Based Process Synthesis (MILP/MINLP): Shared-Resource Hubs, Semicontinuous Activation, Value-Density Selection

Superstructure-based synthesis is a mature methodology in process systems engineering, systematically formalized via state-task and state-equipment network representations reformulated through Generalized Disjunctive Programming into MILP/MINLP (Yeomans & Grossmann, 1999). Four bodies of literature converge on this thesis's Chapter 6 formulation, and each is treated in turn.

Shared-resource hubs as pooling problems. When a shared intermediate stream is formed by mixing multiple upstream sources whose properties — not only flow rates — matter downstream, the resulting property balance is a bilinear function of concentration and flow: the classical pooling problem, whose global-optimization treatment via reformulation-linearization and spatial branch-and-bound was established for multicomponent bilinear process networks by Quesada and Grossmann (1995) and comprehensively reviewed across five problem sub-classes and solution techniques by Misener and Floudas (2009). Combinatorially complex generalized pooling has been solved to within a verified 1.2% optimality gap via piecewise-linear reformulation-linearization-technique reformulation (Meyer & Floudas, 2006), and multiperiod extensions directly relevant to this thesis's seasonal sugar-hub allocation have been formulated as nonconvex MINLPs and solved either via a discretization-based MILP reformulation (Kolodziej et al., 2013) or via a reformulation that commercial global solvers close to optimality (Castro, 2015a). This is a genuine source of nonconvexity, not a modeling inconvenience to be assumed away for tractability, and the literature offers a spectrum of treatments — exact bilinear formulations requiring global or spatial branch-and-bound, versus discretization-based linear approximations that trade exactness for tractability, either as an inner restriction of the true feasible

region or, via a discretization-relaxation scheme, as a valid outer bound on it. Section 6.3 and Section 6.8 of this thesis adopt and justify a specific point on this spectrum for the sugar hub, and the justification given there — rather than the mere existence of a spectrum of options — is the methodological contribution this literature makes possible.

Discrete line activation via semicontinuous coupling. Candidate processing lines are coupled to a minimum operating scale and an associated fixed-charge capital-cost structure via the classical fixed-charge modeling pattern. The disjunctive choice of how to enforce this coupling — a Big-M reformulation, compact but degrading in relaxation quality whenever the M value is not derived tightly from the variable's actual bounds, versus a convex-hull reformulation, provably at least as tight but larger in variable and constraint count — is a first-order determinant of the resulting MINLP's solvability and is made explicit, rather than left as an implementation detail, in Section 6.2 and Section 6.4 of this thesis.

Superstructure synthesis with simultaneous economic-environmental optimization. This thesis's Chapter 6 approach is directly precedented by superstructure syntheses that optimize net present value and global warming potential simultaneously via multi-objective MINLP solved with the ε-constraint method. A gasification-route hydrocarbon biorefinery superstructure spanning gasification technology, cooling, hydrogen source, and Fischer-Tropsch catalyst choices produced a full NPV-GWP Pareto curve with unit production costs ranging from $3.17 to $4.43 per gasoline-equivalent gallon depending on the trade-off point selected along the front (Wang et al., 2013); the same multi-objective MINLP/ε-constraint pattern has been applied to algal biorefinery carbon-sequestration networks spanning thousands of candidate processing routes (Gong & You, 2014), to lignocellulosic Agave bagasse biorefineries with an added social-impact criterion (Álvarez del Castillo-Romo et al., 2018), to distributed multiproduct biorefinery siting under combined

economic, environmental, and social objectives formulated there as a multi-period MILP but solved via the same ε-constraint methodology this thesis adopts (Santibáñez-Aguilar et al., 2014), and — closest in spirit to this thesis's Chapter 6/8 linkage — to an integrated multi-objective superstructure optimization and multi-criteria assessment framework for power-to-X and biomass-to-X methanol production (Kenkel et al., 2022). None of these prior superstructures, however, embeds a genuinely bilinear pooling hub of the kind this thesis's sugar hub requires: most either avoid multi-stream property mixing entirely, by design choices that keep each processing route's feedstock stream physically separate through to the final product, or linearize the mixing step away as a simplifying assumption whose consequences are not examined. This is precisely the methodological gap Section 6.3 closes, and it is worth stating plainly that this gap is not a matter of degree — a slightly more careful treatment of an existing model — but of kind: none of the reviewed studies' underlying optimization problems is, mathematically, the same class of problem as the one this thesis's hub forces.

[NOTA — citas de 2.6 (Consensus, 3-nov). Meyer y Floudas (2006) verifican la solución dentro de una brecha de 1,2 %, no «global optimality». Castro (2015a) no usa discretización: propone una formulación MINLP que BARON y GloMIQO resuelven al óptimo global. Santibáñez-Aguilar et al. (2014) es un MILP multiperiodo, no un MINLP. Confirmados sin cambios: Sammons et al. (2008), Dansereau et al. (2014), Álvarez del Castillo-Romo et al. (2018), Mavrotas (2009), Marler y Arora (2010), Kim y de Weck (2005).]

Value-density selection. Distinct from the mathematical-programming literature above, a complementary heuristic used in biorefinery portfolio screening ranks candidate product lines by value generated per unit of feedstock mass or processing capacity committed. This heuristic has a substantial academic lineage in the forest-biorefinery literature, where product-portfolio selection under supply-chain constraints has been formalized as an explicit design decision rather than an afterthought: Sammons et al. (2008) developed a mathematical-optimization framework for

biorefinery product allocation combining process and economic modeling; Mansoornejad et al. (2010) integrated product-portfolio design with supply-chain design under a margins-based operating policy; Dansereau et al. (2014) developed a supply-chain-fundamentals methodology for screening biorefinery product portfolios prior to detailed process design; and Louw et al. (2023), in the most directly relevant recent synthesis, systematically reviewed the comparability of techno-economic-assessment-based product prioritization across first- and second-generation biorefineries, concluding that chemical, bioplastic, and food products such as citric acid and lactic acid are consistently favored under current technology and market conditions once methodological inconsistencies across studies are controlled for — a finding directly relevant to this thesis's own candidate-technology screening in Section 3.4. This thesis uses value-density both as an a priori screening tool, in Section 3.4, narrowing the superstructure's candidate technologies before optimization begins, and, following the reconciliation established in Section 2.9, as an economic interpretation layer applied after the formal MINLP/multi-criteria-decision-making selection (Section 8.5) — not as a substitute for either the formal optimization or the formal decision method, a distinction the forest-biorefinery literature cited here, which largely predates the mature MINLP-superstructure methodology reviewed above, does not itself need to draw but which this thesis's synthesis of both traditions requires it to make explicit.

2.7 Distributed Bioenergy for Off-Grid Sites: Gasification-ICE Systems

Downdraft gasification of lignocellulosic residue, coupled to syngas cleanup and an internal combustion engine generator set, is a well-documented distributed bioenergy configuration for off-grid and rural electrification, valued for its technological maturity and feedstock flexibility relative to more capital-intensive alternatives (Martínez et al., 2012). Techno-economic assessments report this technology achieving positive net present value and payback

periods of roughly 7-13 years across contexts as varied as Moroccan olive-mill electrification, where an 80-kWe combined heat and power gasification plant achieved a 7-8-year payback (Sánchez-Lozano et al., 2023), and distributed United States biomass/municipal-solid-waste power generation, where a 60-kW downdraft system achieved a 7.7-year payback with a positive internal rate of return under baseline feed-in-tariff assumptions (Indrawan et al., 2020). Rural mini-grid studies in India confirm that hybrid photovoltaic-biomass configurations typically outperform single-source systems on both cost and reliability, though gasifier maintenance and biomass-supply sensitivity remain the dominant risk factors identified across the survey data underlying these studies (Chambon et al., 2020); a procedure for simulating biomass gasifiers in standard microgrid design software (HOMER) has been applied to isolated rural communities in Honduras and Zambia, where islanded biomass-photovoltaic hybrid microgrids delivered a lower levelized cost of energy than grid extension (Ribó-Pérez et al., 2021).

[NOTA — citas de 2.7 (Consensus, 3-nov). Indrawan et al. (2020) existe y confirma 60 kW, 7,7 años y TIR 10,9 % (Energies, doi:10.3390/en13143703): la cito directo y falta en la lista de referencias. Ribó-Pérez et al. (2021) aplica un procedimiento a dos casos; «validated» exageraba. Tostado-Véliz et al. (2024) confirmado (Ghana, −90 % de costo). La lista tiene dos «Martínez et al. (2012)» (gasificación y antioxidantes): márcalas 2012a y 2012b.]

Most relevant methodologically to this thesis's Chapter 5-6 coupling, optimal sizing of hybrid photovoltaic-diesel-biomass off-grid plants has been solved via Benders decomposition explicitly accounting for local biomass production constraints, reducing project cost by up to 90% relative to diesel-only baselines in a benchmark off-grid Ghanaian community (Tostado-Véliz et al., 2024). This decomposition-based methodology is instructive for this thesis's own tractability strategy (Section 6.8, Section 6.9): where a monolithic simultaneous MINLP proves computationally intractable at the scale this thesis's superstructure ultimately reaches, a Benders-style decomposition separating the energy-island sizing decision from the process-superstructure

allocation decision is a documented, precedented fallback, though this thesis's default position, argued in Section 1.4, is that the two decisions should be solved simultaneously wherever computationally feasible precisely because their coupling — not their independent optimization — is the source of the value this thesis claims to capture.

Applied to this thesis's context, the relevant literature gap is not the gasification-ICE technology itself, which is mature and well characterized across the studies reviewed above, but its integration as a hard feasibility constraint within a simultaneous process superstructure (Section 1.2, Section 5.4). The studies reviewed in this section size the energy system independently of, or at best via decomposition alongside, a fixed process configuration, rather than co-optimizing energy supply and product-portfolio selection within one MINLP — precisely the coupling this thesis's Chapter 6 formulation performs. For the Colombian ZNI context specifically, techno-economic and system-dynamics studies confirm both the magnitude of the diesel-dependency problem (Bustos et al., 2014) and the viability of local renewable alternatives (Haghighat Mamaghani et al., 2016; García-Vera et al., 2020), but none couples that energy system to a co-designed agro-industrial process superstructure of the kind proposed here.

2.8 Techno-Economic and Life-Cycle Assessment Methods for Biorefineries

Techno-economic assessment methods for biorefineries — equipment sizing and costing, operating cost and revenue modeling, and profitability metrics such as net present value, internal rate of return, payback period, and minimum selling price — are well established in the process-engineering literature (Scown et al., 2021). A recent review of lignocellulosic-biorefinery techno-economic-assessment studies confirms minimum selling price as the dominant comparability metric across biofuel and high-value-product routes, with capital expenditure and feedstock cost consistently identified as the dominant sensitivity drivers across the reviewed literature (Louw et

al., 2023). Early-stage capital-cost estimation for novel biorefinery processes specifically — relevant to Section 8.1's minimum-scale capital-expenditure treatment — has been benchmarked across six heuristic costing methods, with the review cautioning explicitly that extrapolating conventional-refinery cost correlations to biorefinery equipment can violate the correlations' underlying scale and technology-maturity assumptions, sometimes producing capital-cost estimates that diverge by more than a factor of two depending on the heuristic chosen (Tsagkari et al., 2016).

Life-cycle assessment methods, standardized under ISO 14040/14044, require an explicit functional unit, system boundary, life-cycle inventory, and characterization method. Methodological reviews of biorefinery life-cycle assessment converge on the same set of unresolved issues repeatedly across more than a decade of published guidance: system-boundary inconsistency between cradle-to-gate and gate-to-gate treatments, co-product allocation method (mass, energy, economic, or system-expansion allocation), and functional-unit choice all materially change reported environmental performance and frequently render published biorefinery life-cycle assessments mutually incomparable even when they nominally assess the same technology (Ahlgren et al., 2015). A 2024 review of 59 biorefinery life-cycle-assessment studies found only 28% used primary rather than secondary or literature-derived data, and 39% conducted no uncertainty or sensitivity analysis at all — a finding this thesis treats as a direct methodological warning rather than a merely descriptive statistic, since it implies that a large fraction of the comparative benchmarks Chapter 7 might otherwise cite uncritically are themselves of uncertain reliability (Gaffey et al., 2024). System-boundary setting specifically has been shown to depend heavily on whether feedstock is treated as dedicated biomass or as residue: residue-based systems, this thesis's own case, more often omit upstream agricultural impacts from the

assessment boundary, a choice that must be stated and justified rather than defaulted into silently (Saraiva, 2017).

[NOTA — citas de 2.8 (Consensus, 3-nov). Gaffey et al. (2024): el 39 % es la proporción de estudios que NO hicieron análisis de incertidumbre o sensibilidad; el texto lo había invertido. Scown et al. (2021) describe herramientas en evolución, y Louw et al. (2023) habla de «substantial variabilities» entre TEA: «well standardized» no se sostenía. Tsagkari et al. (2016) y Saraiva (2017) confirmados; el «factor of two» de Tsagkari no está en el resumen.]

This thesis's Chapter 7 develops the life-cycle-assessment methodology as its own dedicated treatment precisely to avoid the documented pitfalls surveyed in this section — reflecting this section's positioning that techno-economic and life-cycle assessment, while conceptually parallel in that both require a system boundary and an inventory derived from the same underlying mass and energy balances (Section 4.8), warrant independent methodological chapters rather than treating environmental performance as a single Pareto-objective term computed incidentally to the economic model, which is the pattern this thesis observes across most of the superstructure literature reviewed in Section 2.6.

2.9 Multi-Objective Optimization (MOO) and Multi-Criteria Decision-Making (MCDM): How Pareto Generation and Post-Optimization Selection Work Together

Multi-objective optimization and multi-criteria decision-making address two distinct, sequential questions, and this thesis's methodology depends on keeping that distinction explicit rather than treating the two as interchangeable, as a portion of the applied biorefinery literature reviewed in Section 2.6 appears implicitly to do. Multi-objective optimization (Section 6.6) generates the set of non-dominated, Pareto-optimal configurations trading off net present value and global warming potential. The generation method matters, and it matters specifically for this thesis's problem structure: the classical weighted-sum scalarization is simple to implement but is

proven — both analytically and empirically — to miss whole regions of a nonconvex Pareto frontier and to produce poorly distributed solutions even where it does succeed in finding feasible points on the front (Marler & Arora, 2010; Kim & de Weck, 2005). Section 2.6 has already established that this thesis's shared sugar hub is inherently nonconvex owing to its bilinear pooling structure, so this limitation of weighted-sum scalarization is not a hypothetical concern raised for methodological completeness but a documented failure mode directly applicable to this problem's own feasible region. The augmented ε-constraint method (AUGMECON), which optimizes one objective while bounding the other as a parametric constraint and has been specifically designed to avoid producing weakly Pareto-optimal solutions while accelerating the overall search through redundant-iteration avoidance, is the standard, widely implemented remedy (Mavrotas, 2009), and it is therefore the methodologically correct choice for this thesis's Pareto generation (Section 6.6), independent of any preference weighting the eventual decision-maker might hold.

Multi-criteria decision-making addresses the separate, downstream question: once the Pareto-optimal set exists, which single configuration should be recommended for implementation? A wide multi-criteria decision-making literature has developed around this exact hand-off from optimization to selection, with systematic reviews categorizing methods into pairwise-comparison, distance-based, and outranking families (Azhar et al., 2021). Comparative benchmarking studies consistently find that TOPSIS and VIKOR, while both 'closeness-to-ideal' compromise methods in a loose sense, differ in normalization procedure — vector normalization for TOPSIS, linear normalization for VIKOR — and in how they treat trade-offs across criteria; VIKOR can fail to produce a full sorted ranking under certain data conditions (Zamani-Sabzi et al., 2016), while TOPSIS is comparatively more sensitive to the normalization choice itself (Opricovic & Tzeng, 2004). A formal comparison of VIKOR against the outranking methods PROMETHEE and

ELECTRE found VIKOR's compromise-ranking logic converges with outranking results in most, but not all, cases, with divergence concentrated in problems where criteria are only partially compensatory (Opricovic & Tzeng, 2007). A large benchmarking study across TOPSIS, VIKOR, COPRAS, and PROMETHEE II confirms that both the weighting method (entropy versus standard deviation, among others) and the normalization technique materially shift final rankings, sometimes by more than the choice of MCDM method itself accounts for (Sałabun et al., 2020). Outranking methods such as PROMETHEE and ELECTRE are generally preferred over pure ranking methods when criteria are not fully compensatory — that is, when a very poor score on one criterion, such as global warming potential, should not be fully offset by an excellent score on another, such as net present value, a property directly relevant to this thesis's own two-objective decision problem in Section 8.5.

Both method families require a weighting scheme to express the relative importance of the criteria under consideration. Entropy weighting derives weights objectively from the dispersion of the data itself, but is documented to be highly sensitive to how state probabilities are estimated from the underlying decision matrix, with proposed corrections addressing this instability by modifying the probability-estimation step rather than the entropy calculation itself (Mukhametzyanov, 2021); combined entropy-AHP weighting schemes that blend this objective signal with subjective expert judgment have been shown to reduce the bias either method carries alone, at the cost of requiring both an expert elicitation process and a sufficiently well-dispersed decision matrix (Chen, 2020). The Analytic Hierarchy Process derives weights subjectively via pairwise expert comparison and is well established but time-intensive and prone to consistency violations as the number of criteria grows. SWARA, the Step-wise Weight Assessment Ratio Analysis, derives weights via a rank-ordered stepwise comparison that requires less pairwise

judgment than the Analytic Hierarchy Process while retaining a comparable degree of expert control over the outcome, and in its rough-number form produced ranks identical to those of the rough Analytic Hierarchy Process and Best-Worst Method in a logistics application (Zavadskas et al., 2018).

Critically for this thesis's internal methodological consistency, the multi-criteria decision-making method selected here is not decorative literature review inserted to demonstrate familiarity with the field. It is the method actually applied in Section 8.5 to select the recommended configuration from the Pareto front generated in Section 6.6, and its output is then reconciled with, not silently replaced by, the value-density economic interpretation of Section 2.6 and Section 3.4. Where the two selection logics agree, this constitutes a robustness check on the recommended configuration; where they disagree, Section 8.5 reports and interprets the disagreement explicitly rather than quietly preferring whichever method happens to favor a pre-conceived outcome.

2.9.1 Ranking Methods (TOPSIS, VIKOR) and Outranking Methods (PROMETHEE, ELECTRE)

TOPSIS ranks alternatives by their geometric distance to a positive-ideal and negative-ideal solution simultaneously, selecting the alternative with the shortest distance to the ideal and the greatest distance from the anti-ideal (Opricovic & Tzeng, 2004). VIKOR instead constructs a compromise ranking that explicitly balances group utility, the majority-preferred outcome, against individual regret, the worst-performing criterion for the selected alternative, making it a natural fit when a decision-maker wants to avoid an alternative that is excellent on average but catastrophic on one dimension — a property directly relevant to Section 8.5's NPV-GWP trade-off, where a configuration with high NPV but very poor GWP performance is exactly the kind of outcome

VIKOR's regret term is designed to penalize and TOPSIS's pure Euclidean-distance aggregation is comparatively less equipped to flag (Opricovic & Tzeng, 2007).

PROMETHEE builds pairwise outranking, or preference, relations between every pair of alternatives across every criterion using a preference function, then aggregates these into net outranking flows, avoiding the compensatory aggregation that both TOPSIS and VIKOR perform, at the cost of requiring more parameter choices — specifically, the preference-function shape and threshold per criterion. ELECTRE similarly builds outranking relations, but through concordance and discordance indices rather than a distance metric, and is comparatively better suited to screening out clearly dominated alternatives from a large candidate set than to producing a single fully sorted ranking; comparative studies report ELECTRE's ranking completeness as its main practical weakness relative to TOPSIS and VIKOR (Zamani-Sabzi et al., 2016). An applied comparison of five methods on Latvia's renewable-energy options found TOPSIS, VIKOR, and PROMETHEE-GAIA selecting the same top alternative while MULTIMOORA and COPRAS favoured another (Zlaugotne et al., 2020) — a pattern Section 8.5 checks for explicitly when applying more than one method to this thesis's own Pareto set as a robustness cross-check, on the reasoning that agreement at the top of the ranking, where the actual implementation decision is made, matters more for this thesis's purposes than agreement across the full ranked list.

[NOTA — citas de 2.9 (Consensus, 3-nov). Zlaugotne et al. (2020) es un solo caso (energía renovable en Letonia), no «comparisons across engineering domains»; el desacuerdo «en el medio del ranking» no está en el resumen. Zavadskas et al. (2018) es una sola aplicación logística. Que VIKOR no siempre da un ranking completo lo reporta Zamani-Sabzi et al. (2016), no Opricovic y Tzeng (2004); que TOPSIS sea más sensible a la normalización no está en ninguno de los dos resúmenes. Opricovic y Tzeng (2007) comparan VIKOR con TOPSIS, PROMETHEE y ELECTRE, pero la «convergencia en la mayoría de los casos» no está en el resumen. «Chen (2020)» falta en la lista y es ambiguo: lo más probable es C.-H. Chen (2020), Entropy, doi:10.3390/e22020259 (pesos entropía-AHP con TOPSIS). Confirmados: Azhar et al. (2021), Sałabun et al. (2020), Nabavi et al. (2023), Roszkowska (2026), Mukhametzyanov (2021).]

2.9.2 Weighting Schemes (Entropy, AHP, SWARA) and Normalization

Beyond method choice, normalization — the step that rescales heterogeneous criteria, currency units for net present value, kilograms of CO2-equivalent for global warming potential, onto a comparable dimensionless scale before aggregation — is itself a documented source of ranking instability independent of the multi-criteria decision-making method or weighting scheme selected. A dedicated entropy-based analysis of normalization procedures within TOPSIS found that vector, max, linear-sum, and max-min normalization schemes can produce materially different rankings from identical input data, with the effect strongest for low-entropy, low-dispersion criteria where small numerical differences between alternatives are amplified or dampened differently depending on the normalization formula applied (Roszkowska, 2026). A systematic engineering-focused sensitivity study across eight multi-criteria decision-making methods, two weighting methods, and sixteen real decision datasets found that entropy-derived weights were consistently more sensitive to linear transformation and reciprocal reformulation of objectives than CRITIC-derived weights, and identified reciprocal objective reformulation — converting a minimize-this-criterion objective into a maximize-its-reciprocal formulation, a transformation that should be mathematically inconsequential to a well-behaved method but was not, in practice, inconsequential to several of the methods tested — as the single largest driver of ranking instability across the methods examined (Nabavi et al., 2023).

These findings motivate an explicit sensitivity check in Section 8.6: this thesis reports multi-criteria decision-making rankings under at least two normalization schemes and, where entropy weighting is used, cross-validates against an independent weighting method — the Analytic Hierarchy Process or SWARA — rather than reporting a single-normalization, single-weighting result as if it were a definitive, method-independent conclusion. This is, admittedly,

more methodological overhead than a large fraction of the applied multi-criteria decision-making literature reviewed above imposes on itself; the justification for imposing it here is that the documented instability is large enough, relative to the plausible magnitude of the NPV-GWP trade-off this thesis's own Pareto front is expected to exhibit, that a single-method result could not be defended as robust without it.

2.10 Research Gaps and Positioning of This Work (Differentiation Table vs. Prior Studies)

Synthesizing Sections 2.1 through 2.9, the literature addresses this thesis's constituent elements individually but not, this thesis argues, in the specific combination the Bajo Atrato problem requires. Table 2.1 summarizes this positioning across the six dimensions most load-bearing for the contributions claimed in Section 1.7; each row corresponds directly to a specific objective in Section 1.5 and a specific methodological chapter in this thesis's outline (Section 1.8), and the table should be read as a set of six falsifiable positioning claims — each independently checkable against the cited literature — rather than as a single undifferentiated assertion of novelty.

Dimension	**Typical prior pineapple-processing studies**	**Typical superstructure biorefinery studies (other feedstocks)**	**This thesis**
Feedstock scope	Single fraction (pulp/juice); byproducts (bromelain, fiber) treated separately (Sarangi et al., 2022)	Often whole-feedstock, but not pineapple-specific (Zondervan et al., 2011)	Whole-plant, four fractions, jointly optimized
Energy supply	Assumed grid-connected or unspecified	Occasionally off-grid, usually sized after process design (Chambon et al., 2020)	Off-grid, hard feasibility constraint co-optimized with the superstructure
Shared intermediate streams	Not applicable (single product)	Sometimes modeled, often linearized or absent (Misener & Floudas, 2009)	Explicit bilinear pooling formulation with a justified relaxation strategy
Uncertainty treatment	Rare, or single-point sensitivity analysis	Stochastic or robust, rarely both, rarely distinguished by decision timing	Explicit stochastic (market) / robust (energy feasibility) separation, with Flexibility Index quantification

Dimension	Typical prior pineapple-processing studies	Typical superstructure biorefinery studies (other feedstocks)	This thesis
Environmental assessment	Rare or absent	Often a single Pareto-objective term (GWP) without full LCA methodology (Wang et al., 2013)	Dedicated LCA chapter, methodologically independent of the TEA (Ahlgren et al., 2015)
Configuration selection	Not applicable / engineering judgment	Weighted-sum or ad hoc selection from Pareto set	Formal MCDM (ranking/outranking) reconciled with value-density interpretation (Kenkel et al., 2022)

This positioning — not any single element considered in isolation — is this thesis's claimed novelty (Section 1.7). It is worth stating explicitly what this positioning does not claim: it does not claim that bromelain ultrafiltration, citric acid fermentation, downdraft gasification, the pooling problem, the ε-constraint method, or any individual multi-criteria decision-making technique employed in this thesis is itself a novel contribution to its own sub-field — each is drawn from, and credited to, the mature literatures reviewed in Sections 2.3 through 2.9. The claimed contribution is the specific integration: that these components, most of which have previously been studied only in isolation from one another or in combinations that stop short of this problem's full scope, have not, to the best of this review's knowledge, been assembled into one simultaneous optimization problem under a binding off-grid energy constraint for a whole-plant pineapple biorefinery. Chapters 3 through 8 develop that integration in full.

Chapter 3. Feedstock Characterization and Fractionation Basis

The superstructure of Chapter 6 can be no more reliable than the feedstock numbers it is given, and for whole-plant pineapple those numbers are neither uniform nor settled: the aggregate residue figure hides fractions of very different composition (Section 3.2), and the ranking of fractions by bromelain activity changes from one study to the next (Section 3.2.2). This chapter establishes the quantitative feedstock basis on which the process models of Chapter 4, the energy island of Chapter 5, and the superstructure of Chapter 6 are all built. Its purpose is narrower than a general review of pineapple composition: every figure reported here is reported because it is subsequently used as a parameter, a bound, or a distribution in a later chapter, and the chapter is organized accordingly — from the mechanical act of fractionation (Section 3.1), through fraction-specific composition (Section 3.2), to the logistics reality of Bajo Atrato (Section 3.3), the candidate-technology screening this composition data supports (Section 3.4), and the resulting design basis (Section 3.5).

3.1 Whole-Plant Partitioning by Mechanical Fractionation: Pulp, Stem+Core, Peel, Leaves/Crown

Whole-plant fractionation begins with a mechanical separation of the harvested pineapple into its four macroscopic tissue classes — pulp, stem and core, peel, and leaves/crown — prior to any chemical or biochemical processing. This separation is not a trivial preprocessing step to be assumed costless; it is itself a unit operation with a defined equipment train, energy demand, and yield distribution that this thesis's Chapter 4 mass balances (Section 4.8) must account for explicitly. The bench-scale precedent for this separation is well documented: pineapple stems and peels can be mechanically ground and juiced separately using standard juice-extraction equipment,

yielding for each by-product a liquid juice fraction and a solid wet press-cake fraction that is subsequently dried and milled into a flour (Campos et al., 2020). This two-stream output — juice and press-cake — per input fraction is the structural template this thesis adopts for the pulp, stem+core, and peel trains in Chapter 4: each fraction train begins with a juice/solid mechanical separation, after which the juice stream is routed toward fermentation or enzyme-recovery processing and the solid stream toward fiber or lignocellulosic processing.

The leaves/crown fraction departs from this template. Unlike pulp, stem+core, and peel, which each yield an economically meaningful juice fraction, the leaves and crown are predominantly structural, lignocellulosic tissue with comparatively little extractable juice of separate economic interest at the scale this thesis considers; accordingly, this fraction is processed whole, after size reduction, directly into the delignification-pretreatment train of Section 4.5, with its own sugar-bearing hydrolysate contributing to the shared hub (Section 4.6) and its lignin-rich residue contributing to the energy island (Chapter 5). This asymmetry between the three juice-and-solid fractions and the one whole-fraction lignocellulosic stream is itself a consequence of the underlying composition data developed in Section 3.2, and it is the first of several points in this thesis where a composition fact directly determines a process-topology choice rather than merely parameterizing a topology chosen on other grounds.

3.2 Fraction-Specific Composition

Aggregate reviews of pineapple processing waste report that peel, core, crown, and pomace together constitute 50-60% of total fresh fruit weight (Aili Hamzah et al., 2021; Sarangi et al., 2022), with Colombian pineapple production specifically generating on the order of 185,000 tonnes of this residue annually (Celis Cabeza et al., 2024). This aggregate figure, however, obscures compositional differences among the fractions large enough that no single valorization

pathway could be applied uniformly across them — the central empirical fact this section documents in support of the whole-plant, multi-technology design philosophy argued in Chapter 1.

3.2.1 Pulp: Fermentable Sugars

The pulp fraction is characterized by a high content of reducing sugars and soluble fiber relative to the other three fractions, and correspondingly low crude-fiber content (Huang et al., 2021). This composition profile is precisely what makes pulp the historically default feedstock for juice, wine, and vinegar production (Section 2.4): its sugar is already in a directly fermentable form requiring minimal pretreatment relative to the lignocellulosic hydrolysis needed for the leaves/crown fraction. Because pulp is also the fraction historically retained for direct fresh, canned, and juice consumption, this thesis's design basis (Section 3.5) treats pulp allocation to the biorefinery superstructure as constrained by the fraction of harvested fruit not already committed to primary fresh/juice markets, rather than assuming the entirety of harvested pulp is available as biorefinery feedstock — a conservative assumption relative to several of the whole-plant valorization studies reviewed in Chapter 2, which typically treat all fractions as equally available residue.

A directly Chocó-grown characterization of this fermentable-sugar content is available, though it must be scoped correctly rather than substituted uncritically for the pulp-specific data above. Murcia Palacios (2013) reports HPLC sugar profiles for whole-fruit juice — pulp, core, and peel extracted together via mechanical juicer, not the isolated pulp fraction Section 3.1's mechanical separation produces — for both commercially accepted-grade and rejected-grade Chocó fruit: accepted-grade juice carried 574.64 g/L total fermentable sugar (363.29 g/L sucrose, 120.90 g/L glucose, 90.45 g/L fructose), while rejected-grade juice carried only 158.83 g/L total

(2.94 g/L sucrose, 75.62 g/L glucose, 80.27 g/L fructose) — a greater than 72% loss of total fermentable sugar, concentrated almost entirely in sucrose (over 99% degraded) rather than glucose (37%) or fructose (11%), consistent with post-harvest invertase activity hydrolyzing sucrose into its constituent monosaccharides faster than those monosaccharides are themselves consumed by spoilage organisms. This finding is used here as regionally grounded corroboration that reject-grade pineapple sugar content cannot be assumed equal to fresh, commercially graded fruit composition — a caution this thesis's own design basis (Section 3.5) already applies by carrying documented ranges rather than point estimates — rather than as a direct replacement for Huang et al.'s (2021) pulp-fraction-specific composition data, since Murcia Palacios's (2013) juice is a whole-fruit extraction, not an isolated pulp stream.

3.2.2 Stem + Core: Bromelain and Phenolics

The stem and core fraction carries the highest bromelain activity among the whole-plant fractions in several direct comparisons (Huang et al., 2021; Mardawati et al., 2023, as reviewed in Section 2.3's bromelain literature), although this thesis notes, as flagged in Section 2.1, that this ranking is not fully consistent across cultivars and extraction protocols in the literature: core ranks above peel among stem-free processing wastes (Banerjee et al., 2021) but below peel in MD2 by-products, where basal stem is the most active fraction (Chiarelli et al., 2024), others report a different ordering entirely once crown and pulp fractions are included in the comparison (Misran et al., 2022, as reviewed in the membrane-extraction literature of Section 2.3). This thesis's design basis (Section 3.5) therefore treats bromelain specific activity in the stem+core feedstock as a parameter with an explicitly stated range, not a point estimate, propagated into the stochastic yield-uncertainty treatment of Section 6.7 rather than resolved by literature adjudication alone.

[NOTA: corregido con los resúmenes de Consensus (3-nov): Chiarelli et al. (2024) no da el corazón como la fracción más activa. Misran et al. (2022) sí confirma un orden distinto (pulpa > cáscara > corazón > corona > tallo).]

Beyond the enzyme itself, the stem+core fraction — together with the peel — carries a substantial phenolic-compound burden relevant both to the nutraceutical co-product line considered in Section 4.3 and to the antioxidant capacity that differentiates pineapple by-product flours from ordinary fiber ingredients (Campos et al., 2020). Total phenolic content and antioxidant capacity vary systematically not only between stem, core, peel, and crown but between the inner and outer regions of a single tissue (Nordin et al., 2023; Uslu et al., 2024), while the dominant phenolics reported differ by study — chlorogenic, caffeic, and ferulic acids in stem and peel juices (Campos et al., 2020), catechin, gallic, and ferulic acid across core, pulp, and peel (Uslu et al., 2024), and p-coumaric and ferulic depsides with cysteine-derived conjugates in a tissue-resolved profile (Steingass et al., 2015). This thesis's process model (Section 4.3) targets a spray-dried nutraceutical powder co-product from the stem+core train as a secondary revenue stream alongside bromelain, consistent with the integrated valorization precedent of Campos et al. (2020).

[NOTA: el texto decía que los ácidos dominantes (clorogénico, cafeico, ferúlico) son «broadly consistent across studies». Esos tres son los de Campos et al. (2020); Uslu et al. (2024) reporta catequina, ácido gálico y ferúlico, y Steingass et al. (2015) dépsidos y conjugados de cisteína. Solo el ferúlico se repite.]

3.2.3 Peel: Insoluble Dietary Fiber (IDF), Low Bromelain, Approximately 67% Sugar Juice

The peel fraction combines two economically distinct sub-streams once mechanically separated into juice and press-cake: a sugar-rich juice — reported in this thesis's design basis at approximately 67% sugar content on a juice basis, while peel juice is reported as containing lower phenolic, antioxidant, and vitamin C content than stem juice in direct comparison (Campos et al., 2020) — and a press-cake fraction rich in insoluble dietary fiber, predominantly cellulose and hemicellulose, with reported insoluble fiber-rich fractions in the range of 41.8-48.0 g per 100 g of

dried peel material and lignin contents in the same fiber-rich fraction reported as high as 60.7-65.8 g per 100 g (Huang et al., 2020). Peel bromelain activity, while present, is lower than stem activity in three of the four comparisons reviewed in Section 3.2.2 and not consistently lower than core activity, which is the compositional basis for this thesis routing peel primarily toward the fiber/flour and sugar-juice trains of Section 4.4 rather than toward the enzyme-recovery train, in contrast to the stem+core fraction.

[NOTA — citas de 3.2.3 (Consensus, 3-nov). Campos et al. (2020) dice que el jugo de cáscara tiene menos fenoles, antioxidantes y vitamina C que el de tallo, pero no compara azúcares: quité esa parte. El «~67 % sugar content on a juice basis» (aquí, en 3.2.3 y en su título) no tiene fuente y no es físicamente plausible para un jugo (≈10-15 °Brix); ¿es base seca, o la fracción de jugo en la cáscara? Revísalo antes de que alimente el hub (4.6, 6.8). Huang et al. (2020, J. Food Drug Anal., doi:10.38212/2224-6614.2179) confirma 41,8-48,0 g/100 g y 60,7-65,8 g/100 g, pero falta en la lista de referencias. La cáscara supera al corazón en Chiarelli (2024) y en Misran (2022): si el ruteo de la cáscara a fibra y azúcar se mantiene, conviene justificarlo por volumen y fibra, no por bromelina.]

The peel juice's sugar content is the second, alongside the stem+core ultrafiltration permeate and the leaves/crown hydrolysate, of the three streams converging at the shared sugar hub modeled in Section 4.6, and its approximately 67% sugar composition — high relative to the other two hub-feeding streams — makes peel juice, in this thesis's preliminary screening (Section 3.4), the single largest-volume contributor to the hub's total sugar-bearing flow, a fact with direct consequences for the hub's bilinear property-balance sensitivity analyzed in Section 6.8.

3.2.4 Leaves/Crown: Lignocellulosic Composition (Lignin 24-26%)

The leaves and crown fraction is lignocellulosic in composition, consistent with its structural rather than storage function in the plant. This thesis's design basis adopts a lignin content in the range of 24-26% for the combined leaves/crown feedstock, consistent with reported pineapple-leaf lignocellulosic characterizations in this range (Saini et al., 2022), while noting that

reported values across the wider pineapple-biomass lignocellulosic literature vary meaningfully by plant part and analytical method — cellulose, hemicellulose, and lignin contents reported for MD2 pineapple biomass differ measurably between leaf, stem, and root tissue even within a single study, with leaf lignin at 22 wt% against 17 wt% in stem and 19 wt% in root (Mansor et al., 2019), and Colombian Perolera-variety peel characterization reports a comparable but not identical total lignin content of 26.40% on a dry basis (Celis Cabeza et al., 2024) — evidence that the 24-26% figure, while a defensible central estimate for this thesis's design basis, should be treated as a design parameter subject to the same cultivar- and region-specific verification this thesis recommends for bromelain activity in Section 3.2.2.

[NOTA — citas de 3.2.4 (Consensus, 3-nov). El resumen de Saini et al. (2022) no da contenido de lignina (reporta 38,1 g/L de azúcares reductores y fermentación >91 %): confirma el 24-26 % en el texto completo o cambia la fuente. Mansor et al. (2019) da 22 % de lignina en hoja de MD2, por debajo del rango de diseño; lo añadí porque refuerza tu propia advertencia. Celis Cabeza et al. (2024): 26,40 % confirmado (cáscara, no hoja).]

This lignin content is directly consequential for two of this thesis's later chapters. First, it governs the delignification-pretreatment severity required in Section 4.5's enzymatic hydrolysis train, since higher lignin content is documented to impede enzymatic accessibility to the underlying cellulose and hemicellulose, generally requiring more aggressive (and more energy-intensive) pretreatment to achieve comparable hydrolysis yields (Celis Cabeza et al., 2024). Second, and in a direct tension with the first point, the same lignin fraction is precisely the combustible fraction the energy island of Chapter 5 relies upon for gasification feedstock: lignin that is chemically removed to improve hydrolysis yield in the sugar-platform train is lignin that is no longer available to the gasifier, and this thesis's superstructure (Section 6.1) must therefore represent the leaves/crown fraction's allocation between the hydrolysis train and the direct-to-

gasification pathway as a genuine design decision with a real trade-off, not as two independent uses of two independent feedstocks.

3.3 Availability, Seasonality, and Logistics of Pineapple Biomass in Bajo Atrato

Biomass feedstock supply chains are, across the wider bioenergy and biorefinery literature, subject to a well-documented combination of spatial dispersion and temporal seasonality that materially affects delivered feedstock cost and, in the more sophisticated supply-chain-design literature, the optimal siting and scale of the processing facility itself (Atashbar et al., 2018; Eksioglu et al., 2009). Spatio-temporal variability in feedstock yield and quality has been shown, in contexts as different as United States cellulosic-biofuel supply sheds and Flemish reductive-catalytic-fractionation biorefining, to be significant enough that ignoring it can materially underestimate delivered feedstock cost relative to a model that accounts for multi-year yield and quality variation explicitly (Roni et al., 2023), and spatio-temporally explicit supply-chain models have been built to site such biorefineries (De Meyer et al., 2024). Bio-hub and distributed-depot supply-chain architectures, in which preprocessing is decentralized toward the biomass source rather than centralized at the biorefinery gate, have been shown in a United States case study to enlarge a biorefinery's feedstock draw area by 57% and its supply volume by 177% without raising delivered cost (Roni et al., 2019).

[NOTA — citas de 3.3 (Consensus, 3-nov). Roni et al. (2023) sí reporta la subestimación del costo; De Meyer et al. (2024) construye el modelo espacio-temporal pero no reporta subestimación: separé las dos. Roni et al. (2019) es un solo estudio (57,3 % de área, 177,4 % de volumen). Liu et al. (2017) dice exactamente lo que se le atribuye (la estacionalidad cambia asignación e inventario, no localización): la cito directo. Wang et al. (2020) optimiza la cadena de cosecha y logística por condado más que la localización de la planta. Confirmados: Atashbar et al. (2018), Eksioglu et al. (2009), Sahoo et al. (2016). Murcia Palacios (2013) y las cifras de Chocó (169 ha, 3 883 t) no están en Consensus: no verificadas.]

This thesis's own feedstock context differs from the large-scale, dispersed dedicated-energy-crop supply chains that dominate the literature just cited in one material respect: pineapple processing residue is a co-product of an existing, already-sited agro-industrial activity in Bajo Atrato, not a dedicated energy crop whose cultivation location is itself a design variable. This narrows the supply-chain design problem relative to the general biomass-logistics literature — this thesis does not solve a facility-location problem of the kind reviewed by Sahoo et al. (2016) or Wang et al. (2020) — but does not eliminate the seasonality dimension of that literature, since pineapple harvest, like the agricultural biomass reviewed throughout Section 3.3's cited literature, is not uniformly distributed across the calendar year. Seasonal variation in feedstock arrival rate is, accordingly, carried into this thesis's design basis not as a multi-period scheduling problem (explicitly excluded from scope in Section 1.6) but as a source of parametric uncertainty in the superstructure's uncertainty formulation (Section 6.7), consistent with the general finding in the biomass-supply-chain literature that feedstock seasonality primarily affects the allocation and inventory-cost dimension of supply-chain design rather than the facility-location dimension once the facility is already co-located with an existing processing operation (Liu et al., 2017).

Logistics within Bajo Atrato itself — road infrastructure, transport distance from cultivation areas to the processing site, and storage/spoilage constraints given pineapple's documented short shelf life of four to five days post-harvest under normal storage conditions (Hossain et al., 2016) — are treated in this thesis's design basis as constraints on the reference scenario's assumed feedstock delivery cost and reliability (Section 3.5), reflecting documented regional infrastructure limitations characteristic of Colombian Zonas No Interconectadas more broadly (Bustos et al., 2014; Garces et al., 2021). This thesis does not develop an independent geospatial logistics-optimization model of the kind reviewed above for dedicated-energy-crop

supply chains; given the co-located, single-site nature of this thesis's feedstock context, such a model's marginal contribution over a parametrically uncertain delivered-cost assumption is judged, in this design basis, to be limited relative to its modeling cost, though this judgment is revisited as a direction for future work in Section 11.3 should the biorefinery's scale grow to the point where multi-site sourcing becomes economically relevant.

This logistics constraint is not a generic concern imported from the wider biomass-supply-chain literature cited above; it has been directly documented for Chocó itself. Murcia Palacios (2013) reports that pineapple cultivated in Chocó is harvested manually and transported by river to Quibdó's commercial centers without refrigeration, so that fruit exposed to more than two to three days of unrefrigerated river transport, combined with mechanical bruising during loading and unloading and pest damage (chiefly fruit fly) during subsequent commercialization, systematically fails Colombian technical standard NTC 729-1 and is discarded before ever entering formal processing — a mechanism that directly instantiates, for this thesis's own study area, the generic post-harvest deterioration risk Hossain et al. (2016) document only at the level of the crop generally. The same source quantifies the resulting reject stream for a single municipality: approximately 12,000 kg per month of rejected whole pineapples were, at the time of that study, disposed of at Quibdó's open-air municipal dump (Murcia Palacios, 2013). This figure characterizes a stream distinct from, and additional to, the four processing-residue fractions of Section 3.1: it is whole fruit rejected prior to any commercial processing, not a peel, stem+core, or leaves/crown by-product of fruit that was processed. This thesis's design basis (Section 3.5) accordingly does not fold this reject-fruit volume into the fraction-level feedstock totals of Section 3.2, since doing so without first resolving that scope difference would misrepresent the biorefinery's feedstock system boundary; it is instead noted here as documented evidence that Bajo

Atrato's total addressable pineapple-derived feedstock plausibly exceeds the processing-residue fractions this thesis's superstructure draws on, and as a specific, quantified direction for the future-work scale-up discussion of Chapter 11.

The same source situates this reject volume within Chocó's broader pineapple production: departmental cultivation reached 169 hectares and 3,883 tonnes in 2008, on the order of 1% of Colombia's 436,044-tonne national production that year (Ministerio de Agricultura y Desarrollo Rural's Plan Frutícola Nacional, as reported by Murcia Palacios, 2013), with two locally recognized cultivars — "tabugana" (rounder, spineless, higher water content) and "la chocoana" (spined, elongated) — grown by both Afro-descendant and Indigenous communities in the region. This departmental-scale figure is small relative to the aggregate Colombian processing-residue estimates cited in Section 3.2 (Celis Cabeza et al., 2024), a scale difference this thesis's design basis (Section 3.5) treats as a further argument for sizing the reference scenario conservatively rather than assuming national-average residue availability transfers directly to the Bajo Atrato context.

3.4 Candidate Product Portfolio per Fraction and the Value-Density Decision Rule

With the four fractions' compositions established (Section 3.2), this section screens the candidate processing technologies reviewed in Chapter 2 against those compositions to establish the reduced candidate set the superstructure of Chapter 6 actually optimizes over — a necessary step, since the full combinatorial space of every technology reviewed in Sections 2.3-2.4 applied to every fraction is far larger than the resulting MINLP could tractably represent, and a portion of that space is compositionally inadmissible in any case (for instance, applying a high-bromelain-activity-dependent precipitation route to the comparatively low-bromelain peel fraction, per Section 3.2.3, would not be a sensible candidate regardless of its cost).

The screening criterion this thesis applies is value density: the value generated per unit of feedstock mass (or, equivalently, per unit of processing capacity) committed to a given candidate line, a criterion with a well-established lineage in the forest-biorefinery product-portfolio-selection literature reviewed in Section 2.6 (Sammons et al., 2008; Mansoornejad et al., 2010; Dansereau et al., 2014) and most recently systematized across first- and second-generation biorefineries by Louw et al. (2023), whose review found that chemical, bioplastic, and food products including citric acid, ethanol, and comparable platform products are consistently favored once methodological inconsistencies across techno-economic-assessment studies are controlled for. This thesis applies the value-density rule at two distinct points, and it is important to this thesis's methodological coherence (developed in Section 2.9 and returned to in Section 8.5) that these two applications not be conflated. The first application, described in this section, is a priori and heuristic: it narrows Chapter 6's superstructure to a computationally tractable candidate set before any optimization is performed, and its role is exclusionary — ruling out candidate technologies whose value density is so low, relative to competing uses of the same feedstock fraction, that their inclusion could not plausibly change the optimal solution. The second application, reserved for Section 7.6 and Section 8.5, is a posteriori and interpretive: once the superstructure has been solved and a Pareto-optimal configuration selected via formal multi-criteria decision-making, value density is used to explain why that configuration was selected in economically intuitive terms, not to select it in the first instance.

Applying this a priori screening to the composition data of Section 3.2 yields the following reduced candidate set carried forward into Chapter 4 and Chapter 6: for the pulp fraction, wine, premium vinegar, and microfiltered-juice lines (Section 4.2), selected over lower-value-density alternatives such as unprocessed juice export on the grounds that the latter's value density is

dominated by the fermented and premium-processed alternatives once bromelain-recovery co-benefits are excluded (pulp having comparatively low bromelain activity relative to stem+core, per Section 3.2.2); for the stem+core fraction, bromelain recovery (both precipitation and ultrafiltration routes, carried forward as a genuine technology choice rather than pre-resolved, given the yield/purity trade-off documented in Section 2.3) together with a spray-dried nutraceutical powder co-product (Section 4.3); for the peel fraction, insoluble-dietary-fiber flour and sugar-rich juice recovery (Section 4.4), the latter feeding the shared hub; and for the leaves/crown fraction, a split allocation between the delignification/hydrolysis train feeding the hub (Section 4.5) and direct gasification feedstock for the energy island (Chapter 5), the specific split being a genuine superstructure decision variable rather than a pre-screened choice, for the reason given in Section 3.2.4. At the shared hub itself, the pooled sugar stream is carried forward as feedstock to citric acid, xanthan gum, vinegar, and ethanol platforms (Section 2.4), citric acid and ethanol retained among the products Louw et al. (2023) rank as preferred, and xanthan gum and vinegar retained on the pineapple-specific evidence of Section 2.4, and each competing for hub allocation in the superstructure of Chapter 6 rather than being pre-allocated a fixed share.

[NOTA: la lista de productos preferidos de Louw et al. (2023) incluye ácido cítrico y etanol, pero no goma xantana ni vinagre; la frase anterior le atribuía los cuatro.]

3.5 Design Basis, Plant Capacity, and Reference Scenario Definition

The reference design basis adopted for this thesis's process modeling (Chapter 4), energy-island sizing (Chapter 5), and superstructure optimization (Chapter 6) is a single representative annual operating scenario, sized to the feedstock availability established in Section 3.3 and apportioned across the four fractions and their sub-streams according to the compositional yields established in Section 3.2. Consistent with the scope limitation stated in Section 1.6, this reference

scenario represents an annualized average rather than an explicit multi-period harvest calendar; seasonal deviation from this average is carried forward as a parametric uncertainty input to Section 6.7 rather than as a scheduling decision, and plant capacity is sized, following standard techno-economic-assessment practice for biorefineries operating against a variable agricultural feedstock supply, to a defensible average annual throughput rather than to the seasonal peak, consistent with the general finding in the biomass-supply-chain literature that peak-sizing a facility against a highly seasonal feedstock is rarely the cost-minimizing choice once storage and part-load operating costs are considered jointly (Wang et al., 2020, as reviewed in Section 3.3).

This reference scenario, together with the fraction-level composition parameters of Section 3.2, the candidate-technology set of Section 3.4, and the logistics and seasonality assumptions of Section 3.3, constitutes the complete quantitative design basis against which Chapter 4's process models are developed, Chapter 5's energy balance is closed, and Chapter 6's superstructure is parameterized. Where a parameter in this design basis carries a documented range rather than a single literature-consistent point estimate — bromelain specific activity (Section 3.2.2) and leaves/crown lignin content (Section 3.2.4) being the two most consequential examples identified in this chapter — that range, rather than its midpoint alone, is propagated explicitly into the uncertainty and flexibility analysis of Section 6.7, consistent with this thesis's broader methodological commitment, argued in Chapter 1 and Chapter 2, to treating uncertainty with the precision the underlying decisions require rather than resolving it by silent adoption of a single convenient number.

Chapter 4. Process Modeling and Simulation of the Processing Trains

A superstructure can choose only among technologies it can describe quantitatively, and many of this thesis's candidate technologies are documented only at bench scale, under feedstocks, operating conditions, and product specifications that differ from those of a Bajo Atrato plant. Before Chapter 6 can optimize anything, each candidate must be translated into a unit-operation model whose yields, flows, compositions, and energy demands are consistent with one another and with the feedstock basis of Chapter 3 — because an error in any of them propagates directly into the shared hub's bilinear balance and the energy island's sizing. This chapter builds those models, train by train: the pulp train (Section 4.2), the stem+core train (Section 4.3), the peel train (Section 4.4), the lignocellulosic train (Section 4.5), the shared sugar hub that couples all four (Section 4.6), the membrane and green-processing unit models common to several trains (Section 4.7), and the mass- and energy-balance closure and stream-reconciliation procedure that ties the whole flowsheet together (Section 4.8). Section 4.1 first establishes the modeling framework, software environment, and validation strategy within which all of the above are developed.

4.1 Modeling Framework, Software Environment, and Validation Strategy

Process simulation is the standard tool for generating the mass and energy balance data a techno-economic assessment requires, and the biorefinery literature has converged on a small set of established platforms for this purpose. Aspen Plus and SuperPro Designer are among the commercial simulators most used for biorefinery techno-economic assessment, each embedding built-in unit-operation models, physical-property databases, and costing correlations that materially reduce the modeling burden relative to a from-scratch implementation (Kasani et al., 2022). SuperPro Designer in particular has an established track record in Colombian bioprocess

techno-economic assessment directly relevant to this thesis's own geographic context: it has been validated for anaerobic digestion of sugarcane vinasse (Canizales et al., 2020), for whey syrup and protein-concentrate valorization attached to a Colombian dairy operation (Gómez et al., 2019), and for a baker's-yeast-industry vinasse biorefinery producing animal feed, fertilizer, and biofuel co-products (Gómez et al., 2024) — three precedents this thesis's own process models draw upon methodologically, particularly for the hierarchical-decomposition approach to multi-train flowsheet construction each of these studies employs. A recent review of SuperPro Designer's application to large-scale bioprocess simulation under a circular-economy framing confirms the software's continued suitability for exactly the kind of waste-to-value-added-product flowsheet this thesis's superstructure represents (Aranda-Barradas et al., 2025).

This thesis adopts SuperPro Designer as its primary simulation environment on the strength of this Colombian precedent and its explicit built-in support for batch and semi-continuous bioprocess unit operations — fermentation, enzymatic hydrolysis, membrane filtration, and spray drying among them — that dominate this thesis's four processing trains, in preference to Aspen Plus, whose built-in unit-operation library is comparatively stronger for continuous, equilibrium-stage-dominated chemical processes than for the biologically mediated batch operations this thesis's superstructure requires. This is a documented trade-off in the literature rather than an idiosyncratic choice: comparative studies of the open-source alternative DWSIM report material- and energy-balance agreement within 2-10% of Aspen Plus for a sorghum biorefinery (Angannan et al., 2025) and good agreement with commercial software for the NREL bioethanol process, while noting that general chemical-process simulators show compatibility issues with bioprocesses (Sreemahadevan et al., 2024) — the latter point, rather than any documented shortfall of DWSIM,

supports this thesis's choice of SuperPro Designer as the better-suited incumbent for this specific process portfolio rather than a general-purpose alternative.

[NOTA — citas de 4.1 (Consensus, 3-nov). Sreemahadevan et al. (2024) y Angannan et al. (2025) son favorables a DWSIM; ninguno reporta «persistent compatibility gaps» de DWSIM. Lo que Sreemahadevan sí dice es que los simuladores químicos comerciales tienen problemas con bioprocesos: reformulé el argumento sobre eso. Kasani et al. (2022) es una revisión de software para biorrefinerías de microalgas (Algal Research, doi:10.1016/j.algal.2021.102597): no ordena los simuladores por uso, y falta en la lista de referencias. Confirmados: Canizales et al. (2020), Gómez et al. (2019, 2024), Aranda-Barradas et al. (2025).]

Validation strategy. Each unit-operation model within the four processing trains (Sections 4.2-4.5) is parameterized directly from the experimental yield, purity, and operating-condition data reported in the bench-scale studies reviewed in Chapter 2 and Chapter 3 — bromelain recovery yields from Simões et al. (2021), Campos et al. (2019), and Nor et al. (2016, 2018); citric acid and xanthan gum fermentation yields from Kareem et al. (2010) and Amenaghawon et al. (2024); ceramic microfiltration flux and rejection data specific to pineapple juice from Arikrishnan et al. (2022) — rather than from generic literature defaults, consistent with this thesis's broader methodological commitment (Chapter 3) to carrying documented parameter ranges, not single point estimates, into the superstructure wherever the underlying bench-scale literature itself reports a range. Where a unit operation's governing equations are themselves well established in the process-systems-engineering literature independent of the pineapple-specific bench data — the pooling-problem property balance of Section 4.6 being the clearest example — those equations are adopted directly from their original formulation (Quesada & Grossmann, 1995; Misener & Floudas, 2009) rather than re-derived, and the pineapple-specific bench data instead populates the equations' parameters (stream compositions, flow rates) rather than their structure.

4.2 Pulp Train: Juice Extraction → Enzymatic Treatment → Ceramic Microfiltration (0.2 µm) → Wine / Microfiltered-Juice / Premium Lines

4.2.1 Juice Extraction, Enzymatic Treatment, and Ceramic Microfiltration

The pulp train begins with the mechanical juice extraction described in Section 3.1, followed by an enzymatic treatment step — pectinase and/or cellulase addition, following the enzyme-treatment precedent established for juice clarification more broadly (Changmai et al., 2019, as reviewed in Section 2.5's ceramic-membrane clarification literature) — that reduces juice viscosity and pectin content ahead of membrane processing, a step documented to substantially improve subsequent permeate flux relative to unclarified juice across the citrus, mosambi, and passion-fruit clarification literature reviewed in Section 2.5 (Nandi et al., 2009, 2010, as reviewed therein). The enzyme-treated juice is then split three ways within the candidate-technology set established in Section 3.4: a wine line (direct alcoholic fermentation of the raw or lightly clarified juice), a microfiltered-juice line (clarification without fermentation, for direct sale as a clarified beverage), and a premium line (a combination of extended fermentation and clarification targeting a differentiated, higher-unit-value product, following the premium-vinegar and premium-fermentation precedent of Section 2.4).

A more specific, pineapple-direct parameterization of this enzymatic treatment step is now available, sharpening the general Changmai et al. (2019) precedent cited above for the qualitative existence of enzyme-assisted clarification into a quantified, commercial-vendor-anchored design point. Quesada Salas (2013), an undergraduate thesis conducted for an actual pineapple-processing company at the Universidad de Costa Rica on precisely this unit operation — enzymatic maceration followed by tangential microfiltration, for clarified pineapple juice — reports a real commercial vendor envelope for its "P1" pectinase/polygalacturonase/cellulase preparation (55-

60°C, 200-300 mg enzyme per kg pulp, 30-120 minutes residence time) and a real measured maximum juice-extraction yield of 87% (%w/w) achieved by pressing pineapple puree after this enzymatic treatment. This thesis's accompanying computational proof-of-concept (Chapter 11) adopts this 87% figure directly as the pulp fraction's own juice-extraction yield, a materially higher and pineapple-specific value in place of an earlier general-fruit-pressing placeholder. The source's own selected operating point within that real envelope is not itself recoverable, however: even its raw experimental-results table reports treatments only by symbolic concentration, time, and temperature codes rather than numeric values, and its commercial enzyme supplier is anonymized throughout — confirmed, at multiple independent locations within the source document itself rather than inferred from one ambiguous passage, to be intentional confidentiality redaction protecting a sponsoring company's process know-how, not a transcription or document-conversion artifact. This thesis's process model accordingly specifies a defensible operating point within the real vendor envelope — the vendor's own minimum residence time together with a mid-envelope concentration and temperature — rather than treating the redacted company optimum as recoverable by assumption, a choice further supported by two of the source's own real, non-redacted findings: a secondary citation to Vaillant et al. (1999) establishing 60 minutes as the real maximum acceptable industry residence time for this class of treatment, against which the source's own (redacted) chosen time is described only as "prudencial," and a real qualitative finding, citing Dak et al. (2008), that favoring enzyme concentration over residence time reduces the risk of pineapple spoilage and off-flavor development during maceration.

The microfiltered-juice and premium lines both route through a ceramic microfiltration step specified at a 0.2 μm nominal pore size, consistent with the operating specification reported for semi-industrial cross-flow microfiltration of fruit beverages more broadly (Zuluaga et al.,

2024) and directly precedented for pineapple juice itself: continuous dead-end microfiltration of pineapple juice using low-cost ceramic membranes with average pore sizes in the 0.56-0.87 μm range has been directly characterized, reporting permeate flux, total-phenolic-content retention, and total-soluble-solids retention as functions of applied pressure (Arikrishnan et al., 2022) — data this thesis's process model adopts directly to parameterize the pulp train's microfiltration unit rather than relying on the citrus- or mosambi-juice literature (Nandi et al., 2009; Satyannarayana et al., 2023) as a proxy, since pineapple juice's own rheology and fouling behavior, while broadly similar in mechanism (cake filtration dominant, per the fouling-model analyses common across this literature), is not assumed identical in magnitude to other fruit juices without direct evidence. This is a specific instance of this thesis's broader modeling discipline, stated in Section 4.1: where pineapple-specific data exists, it is used in preference to analogous data from other fruit matrices, even where the two matrices behave qualitatively similarly.

A second, cross-flow pineapple-juice microfiltration characterization is directly relevant to this specification and is adopted as the primary sizing basis for the reasons given below: Laorko et al. (2010) clarify pineapple juice using membranes with nominal pore sizes of both 0.1 and 0.2 μm, the latter matching this train's own design specification exactly, in contrast to Arikrishnan et al.'s (2022) 0.56-0.87 μm dead-end ceramic membranes, which bracket rather than match the 0.2 μm design point. At the optimum operating condition Laorko et al. (2010) identify — 3.4 m/s cross-flow velocity and 0.7 bar transmembrane pressure — clarification proceeds at an average permeate flux of approximately 37 L/(m²·h), with no significant change in the pH, reducing-sugar content, or titratable acidity of the clarified juice relative to the unclarified feed. This thesis's process model accordingly adopts Laorko et al.'s (2010) 37 L/(m²·h) average flux as the microfiltration unit's sizing design point at the train's specified 0.2 μm pore size and cross-flow

configuration, reserving Arikrishnan et al.'s (2022) dead-end characterization as a secondary cross-check on the underlying cake-filtration fouling mechanism (Section 4.7) rather than as the primary sizing basis, since a cross-flow module operated at the design's own nominal pore size is the closer match to the equipment configuration this train specifies.

Real commercial equipment now anchors this train's mechanical processing stages more concretely than Section 3.1's general characterization alone. Del Fresno et al. (2022), reviewing white-grape juice-extraction methods, tabulate actual commercial machine models and their rated capacities for two operations directly relevant to this train, drawn via a carefully scoped rather than wholesale analogy: a grape "crusher"'s function — disrupting fruit tissue structure to prepare it for downstream juice release, without itself separating juice — is the real functional counterpart of this train's own milling stage, not of the pineapple-specific washing/peeling/coring operations upstream of it, since grapes are neither peeled nor cored and no equivalent real equipment for those two operations exists in this source; pressing, by contrast, is a direct and unforced analogy, since both operations press a macerated plant-tissue mash to extract juice. At this train's own processing scale, this thesis's accompanying computational proof-of-concept (Chapter 11) finds that even the smallest cataloged real commercial unit in either category runs at 5-11% of its own rated capacity — a boutique-scale mismatch relative to typical commercial wine-processing equipment discretization that is itself informative for Section 8.2's eventual equipment-selection and cost protocol, since it indicates the relevant vendor decision at this scale is which smallest available real unit to select, not a continuous capacity-optimization problem. Pressing technology is further selected on real, documented quality grounds rather than availability alone: Del Fresno et al. (2022) report that continuous screw presses and decanter centrifuges, favored in high-volume wineries for speed, yield juice with higher turbidity, higher pH, more extracted phenolics, and

lower resulting ester formation than pneumatic presses, which extract juice gently at low pressure at the cost of a slower, discontinuous cycle — a trade-off this thesis resolves in favor of the pneumatic press, consistent with the premium-line product-differentiation hypothesis this train's own value-density screening (Section 3.4) already presumes commercially realizable. Neither Del Fresno et al. (2022) nor this thesis reports a capital cost for either selected equipment model, so equipment cost for these two stages, like the washing/peeling/coring operations that remain entirely unresolved at the equipment-model level, is carried forward to Section 8.2 as a concrete, disclosed action item rather than estimated by analogy.

4.2.2 Clarification Target and the SO2-Free Design Intent

The train's 0.2 µm microfiltration specification conflicts with a clarification target the white-winemaking literature treats as a fermentation requirement, and the conflict must be resolved before the fermentation parameters below are set. Vernhet (2022), reviewing must clarification for white winemaking, establishes a real target turbidity of 100-200 NTU ahead of fermentation: clarifying beyond this range depletes the phytosterols and lipids that a must's own suspended solids supply, nutrients yeast require to maintain cell-membrane integrity, particularly under the anaerobic conditions this train's own low-oxygen milling and closed, isothermal fermentation design specifically creates. A 0.2 µm microfiltration step of the kind this train specifies removes essentially all suspended solids, driving turbidity far below this real industry target — a previously undisclosed risk to fermentation kinetics and yeast viability that also means Moresi's (1989a, 1989b) fermentation kinetics, developed and calibrated on a must that was not comparably ultra-clarified, are applied below to a juice more thoroughly clarified than his own calibration basis, an additional, compounding validity limitation Section 6.7's uncertainty treatment should carry forward. Two mitigations — a coarser pre-clarification stage sized to leave the real 100-200 NTU

target rather than near-zero turbidity, following the static-settling and centrifugation physics Vernhet (2022) also reviews, or supplementation of the clarified juice with a commercial yeast-hull or sterol product, a real, established enological remedy for exactly this lipid-depleted-must scenario — are identified here as concrete design options rather than resolved, since neither has yet been incorporated into this train's process model.

This train's own 0.2 µm microfiltration specification was originally motivated, in part, by a design intent to avoid sulfur dioxide (SO2) addition entirely, in pursuit of a differentiated, niche market positioning consistent with the premium-line product-differentiation hypothesis already introduced above. Checking that intent against the white-wine-technology literature surfaces a real, material gap rather than confirming it outright: SO2 serves three distinct real functions in conventional winemaking — antimicrobial, antioxidant, and antioxidasic, this last inhibiting the polyphenol oxidase (PPO) enzyme responsible for enzymatic browning (Loira et al., 2022) — and while a 0.2 µm membrane can plausibly retain intact microorganisms at or above its own pore size, PPO is a dissolved enzyme molecule orders of magnitude smaller than 0.2 µm, passing through the membrane un-inactivated; microfiltration alone therefore does not replicate SO2's antioxidasic function, and enzymatic browning could still occur without SO2 even downstream of this train's own clarification step. Ultra-High-Pressure Homogenization (UHPH, 300-400 MPa) is a real, nonthermal must-treatment technology, approved by the International Organisation of Vine and Wine (OIV Resolution OENOMICRO 16594B), that closes this gap directly: real published results on white must report total elimination of yeast and bacteria, a 90% reduction in polyphenol-oxidase activity relative to an SO2-treated control, and 1.5 times the antioxidant capacity of an untreated control, all achieved without SO2 (Loira et al., 2022, citing Loira, Morata, Bañuelos, et al., 2018, and Bañuelos et al., 2020, both secondary citations not independently verified by this thesis). The

same source's own published methodology combines UHPH treatment with subsequent 0.22 µm filtration — essentially this train's own microfiltration design — which is why this thesis's process model now inserts a UHPH treatment step immediately ahead of microfiltration, sized, per this thesis's accompanying computational proof-of-concept (Chapter 11), against a real commercial unit (Ypsicon Advanced Technologies SL, rated to 10,000 L/h, of which this train's own throughput represents under 10%) rather than an idealized or unsized addition to the flowsheet. As with the milling and pressing equipment above, no capital cost for this equipment is available from the cited source, and UHPH cost accordingly joins the same Section 8.2 action-item list. This addition addresses SO2's antioxidasic and antimicrobial functions specifically; it does not resolve the phytosterol-depletion risk identified above, which concerns the must's nutrient content rather than its microbial or enzymatic-oxidative load, and the two findings should not be read as substitutes for one another.

4.2.3 Wine Fermentation and Acetification Kinetics

Wine fermentation parameters — yeast strain, temperature, and resulting ethanol yield — are adopted from the pineapple-specific fermentation characterization of Roda et al. (2017b), which reports ethanol yields in the 7-8% v/v range across three Saccharomyces cerevisiae strains and three fermentation temperatures, with strain and temperature jointly determining the resulting aroma profile relevant to the premium-line product differentiation this thesis's value-density screening (Section 3.4) presumes is commercially realizable.

This thesis's process model additionally draws on a second, Chocó-department fermentation-kinetics characterization, directly relevant to this train's residence-time and yield parameters rather than to its aroma profile: Murcia Palacios (2013) fermented undiluted Chocó-grown pineapple juice with commercial lyophilized S. cerevisiae under controlled temperature

(30°C) and pH (4.5), reaching 65.31 g ethanol/L in 12.5 hours — a substrate-to-product yield of 0.41 g ethanol per g fermentable sugar consumed, equivalent to 80.39% of the 0.511 g/g stoichiometric maximum, and, per the comparison Murcia Palacios (2013) himself draws against the pineapple-fermentation literature, broadly consistent with the 82-89% theoretical yields reported elsewhere for S. cerevisiae strains fermenting fresh pineapple juice (Alain et al., 1987, as cited in Murcia Palacios, 2013). The same study found that temperature and pH control reduced the time to reach an equivalent final ethanol titer from approximately 72 hours (uncontrolled, ambient conditions) to 12.5 hours — a sixfold reduction in required residence time this thesis's process model adopts as the justification for specifying controlled-temperature, controlled-pH fermentation vessels in the wine line's equipment train, rather than the uncontrolled ambient fermentation a lower-capital-cost design might otherwise default to, since the resulting reduction in required fermenter residence time (and hence fermenter count or working volume, following the sizing logic of Moresi, 1989b) plausibly offsets the additional temperature- and pH-control capital and operating cost, a trade-off Section 8.2's capital-cost protocol will need to quantify explicitly once vendor-quotation data is available rather than assumed favorable on the strength of this kinetic result alone. Because Murcia Palacios's (2013) substrate was whole-fruit juice rather than the isolated pulp-fraction juice this train processes (the same scope distinction noted in Section 3.2.1), this yield and kinetic result is adopted here as regionally grounded evidence for achievable S. cerevisiae fermentation performance on Chocó-grown pineapple sugars generally, not as a pulp-fraction-specific substitute for Roda et al.'s (2017b) strain-and-aroma characterization.

[NOTA — citas de 4.2 (Consensus, 3-nov). El artículo existe (J. Ferment. Technol., 1987, doi:10.1016/0385-6380(87)90146-4), pero Consensus registra al autor como «Kouakou Alain»: es probable que el apellido sea Kouakou y la cita correcta «Kouakou et al., 1987». El resumen habla de jugo fresco, no de jugo de enlatadora, y da 82 % y 89 % como

máximos (el 77 % no aparece). Falta en la lista de referencias. Laorko et al. (2010) confirmado exactamente (0,2 µm, 3,4 m/s, 0,7 bar, 37 L/m²·h); Arikrishnan et al. (2022) y Nandi et al. (2009, 2010) también. Zuluaga et al. (2024) existe (Sep. Sci. Technol., doi:10.1080/01496395.2024.2315609; CFM semiindustrial de 0,2 µm) pero falta en la lista. Satyannarayana et al. (2023): Consensus registra «Tangential microfiltration of lime and pineapple juices…» con primer autor R. Kumar; revisa la autoría. Los capítulos de White Wine Technology (Morata, 2022), Advances in Vinegar Production (Bekatorou, 2020), Ullmann y Moresi (1989) no tienen resumen en Consensus: no verificados en contenido.]

The premium line's further acetification of the wine output — the specific mechanism by which Section 3.4's "premium vinegar" designation for this line is realized — draws its kinetic parameterization directly from Roda et al.'s (2017a) pineapple-waste acetification data: a pineapple-waste wine (7% v/v ethanol, itself obtained via the saccharification-plus-alcoholic-fermentation route of Roda et al., 2014, 2016) inoculated with Acetobacter aceti and acetified for 30 days at 32°C reaches 5% w/v (50 g/L) acetic acid. This is a real, pineapple-specific kinetic characterization rather than an analogous one — a stronger evidentiary basis than the wine line's own Murcia Palacios (2013) endpoint enjoys relative to the broader fermentation literature, since Roda et al.'s (2017a) substrate is drawn from the same feedstock class (pineapple processing waste) this train's premium line itself processes. Chaining this kinetics onto the wine line's own directly measured ethanol output (Murcia Palacios, 2013) — using that titer as the acetification stage's initial substrate, and closing the substrate balance with the exact ethanol-to-acetic-acid reaction stoichiometry rather than an additional fitted parameter — yields a simulated titer that remains within a plausible band around the independent pineapple-vinegar titers Tanamool et al. (2020) report (72 g/L acetic acid within approximately 16 days at laboratory scale, 65 g/L within approximately 25 days at 6 L scale); this is a cross-check, not a calibration, since the two characterizations were obtained independently.

This real current-practice characterization is not, however, the only relevant benchmark for this line's eventual equipment specification. Industrial submerged vinegar fermentation — the

Frings Acetator system, in continuous industrial use since the 1940s and responsible for more than 60% of the vinegar produced worldwide (Emde, 2017) — reports a mean specific productivity of 40-60 kg acetic acid per m³ of broth per day for its repeated fed-batch ("single-stage high-strength process") mode, with conversion reported qualitatively as near the theoretical yield and final acidities of up to 23% w/v (Emde, 2017). This is roughly an order of magnitude faster than Roda et al.'s (2017a) real pineapple-waste result, though for a different substrate class (alcohol or wine vinegar, not pineapple-waste) and industrial rather than laboratory-scale equipment; a directly comparable mechanistic kinetic model for wine-vinegar acetification exists (García-García et al., 2020, after Jiménez-Hornero et al., 2009) but its published parameter set covers growth-rate and inhibition constants only, not the production coefficients a full simulation would additionally require, so it is not used here to simulate a titer, only as an independent order-of-magnitude cross-check on achievable acid-tolerance limits. Whether pineapple-waste vinegar could approach the Frings-scale industrial productivity in a properly designed submerged acetator, rather than the small-scale, static conditions Roda et al.'s (2017a) data reflects, is treated in this thesis as an open, genuinely undecided design question, not resolved by assumption. It is carried into the reduced-scope computational proof-of-concept accompanying this thesis (see Chapter 11's discussion of computational execution as future work) as two separate, competing candidate technologies for this fermentation step — current practice and an optimized acetator — rather than as a single collapsed kinetic model. That proof-of-concept finds the two technologies land on structurally different capacity constraints: current-practice acetification remains kinetics-limited (its approximately 30-day cycle time bounds annual throughput well below feedstock availability), while the optimized-acetator scenario becomes feedstock-limited at the same annual ceiling the wine line itself reaches. This is a capacity-sizing consequence, not merely a unit-price difference,

that Section 6.6's capital-budget treatment should carry forward explicitly when comparing the two acetification routes against each other and against the wine line.

4.2.4 Fermenter Sizing and Fill-Time Verification

That capacity-sizing finding rests, in the proof-of-concept's initial run, on a single reference-sized reactor per route; once the pulp train's target processing scale is raised to a value no single such reactor can plausibly serve (Chapter 11's future-work discussion of the proof-of-concept's own scale-up exercise), sizing however many parallel reactor trains, or how large a single train, each route requires becomes itself a genuine discrete structural decision, additional to the technology-choice decisions Section 6.4-6.5 already formalize with binary variables. The proof-of-concept resolves this train-count decision without introducing further binary variables, following the "NLP superstructure" principle Corsano et al. (2009) formalize for exactly this class of problem — their own worked example, a fermentor-network unit-duplication decision, is structurally close to this train's own wine and vinegar reactors. Because installed reactor volume and annual capacity are linearly related at a fixed cycle time (the same volume-independent-cycle-time assumption underlying the dynamic-similarity reasoning this thesis applies elsewhere), and because the six-tenths-rule capital-cost exponent this thesis applies throughout (the economy-of-scale cost structure Duncan and Reimer document generally) is strictly concave, the cost-minimizing train count follows directly from a closed-form argument rather than requiring an integer-programming search over candidate configurations — a formal methodological grounding for, rather than a replacement of, the same fermenter-count-and-volume trade-off logic Moresi (1989b) already supplies for the wine line's own sizing above. Section 6.10 develops this benchmark, and its limits, at the superstructure level.

The cost-minimizing train count just derived is not, however, the configuration this thesis's process model ultimately specifies for the wine line's own fermentation equipment, because a competing real concern was identified and prioritized ahead of pure capital-cost minimization: filling one large, cost-optimal fermenter at this train's own processing rate extends the duration the must sits partially exposed to air during filling, a real must-oxidation risk distinct from, and not addressed by, the microfiltration- and UHPH-based must-protection measures developed above, which govern microbial and enzymatic-oxidative load rather than filling-time exposure. This thesis's process model accordingly specifies SMALLER, MORE NUMEROUS, equally sized fermenters in place of the cost-optimal single large vessel, at a disclosed capital-cost premium over that optimum quantified directly rather than left implicit: this thesis's accompanying computational proof-of-concept (Chapter 11) finds that reducing the reference fermenter size from a cost-optimal single vessel to twelve identical, smaller vessels raises reactor capital expenditure while cutting each vessel's own fill time roughly twelvefold, a trade-off Section 8.2's capital-cost protocol should weigh against the real, but not yet independently quantified, reduction in oxidation risk this achieves, since no dissolved-oxygen or browning-index threshold tying fill duration to oxidation outcome has yet been established for this specific must.

This same fill-time concern extends one stage further downstream than the fermenter specification alone addresses, and checking whether it does so is itself a genuine verification exercise rather than a formality. Twelve parallel fermenters fed from this train's own continuous upstream throughput fill and complete their cycles in a staggered sequence, not in lockstep, since a plant processing juice at a steady rate naturally reaches each successive vessel's capacity at intervals of the single-vessel cycle time divided by the vessel count — the same steady-state throughput relationship that already determines the vessel count in the first instance. An earlier

iteration of this thesis's aging-tank-farm sizing logic instead modeled all twelve fermenters as discharging their full combined volume into one aging tank simultaneously, once per cycle — a modeling assumption that, left unexamined, would have silently undone the fill-time rationale just established one processing stage downstream, since an aging tank receiving twelve fermenters' combined output at once faces the identical slow-fill exposure the fermenter resizing was specifically adopted to avoid. Correcting this to the physically consistent staggered-discharge assumption does not change the total aging-tank volume required, since that quantity depends only on aggregate throughput and the fixed 90-day dwell time, not on how that throughput is chunked into discharge events, but it does change the discharge granularity the underlying capital-cost optimization is sensitive to: this thesis's accompanying computational proof-of-concept (Chapter 11) finds that staggered discharge reduces the aging-tank farm from nineteen smaller tanks to twelve larger ones, each now receiving the output of exactly twenty successive, staggered fermenter batches, at a real capital-cost saving of approximately 14% relative to the uncorrected lockstep assumption. This is the concrete, quantified answer to how many fermenters should feed each aging tank at this train's own scale, and it stands as a specific, worked instance of the more general hierarchical-versus-simultaneous coupling limitation Section 6.10 discloses for this thesis's dynamic-kinetics-to-capacity modeling chain: an unexamined assumption bridging two hierarchically sized stages can silently defeat a design decision made at the upstream stage, and only checking the coupling explicitly, as done here, catches it.

Smaller fermenters are not the only defense against fill-time oxidation, nor the cheapest: prefermentation antioxidant dosing complements them at a fraction of their capital premium, although it acts on a different mechanism. Kilmartin (2022) documents real prefermentation antioxidant chemistry (sulfur dioxide, ascorbic acid, glutathione) available to white winemaking,

including a real legal dosing ceiling for glutathione of 20 mg/L (OIV-OENO Resolution 445-2015). At this ceiling, this thesis's accompanying computational proof-of-concept (Chapter 11) finds that glutathione dosing would cost on the order of one order of magnitude less annually, even under a wide and deliberately unsourced illustrative unit-cost bracket, than the annualized capital-cost premium already carried for smaller fermenters. The two measures are not strict substitutes: smaller fermenters reduce oxidation exposure by shortening fill TIME, while antioxidant dosing does not alter fill time at all but instead chemically scavenges oxygen and oxidation byproducts once present. Whichever underlying mechanism is judged the more material real risk to this train's must — a question this thesis's design-stage methodology does not itself resolve — is the question that should determine which measure, or combination, Section 8.2's eventual capital-budget protocol actually specifies; glutathione dosing, being a real additive distinct from sulfur dioxide with its own independent OIV approval, does not itself conflict with this train's no-SO2 product-differentiation intent, though whether that intent extends to excluding all chemical additives, glutathione included, or specifically to sulfur dioxide alone, remains a product-positioning question outside this thesis's own design-stage scope.

4.2.5 Premium Line: Kinetic Basis, Heat Removal, and Open Design Items

The pulp train's "premium line," introduced above as a candidate technology but not previously quantified numerically, is now given a real mechanistic basis: Moresi's (1989a) directly transcribed white wine fermentation kinetics (his Eq. 9, a Monod-type growth rate modified for substrate and ethanol inhibition and ethanol-driven cell decay) combined with Moresi's (1989b) own real, reported temperature-cycling policy for an "open" (out-vessel cooling) fermentation system: ferment adiabatically from an inoculation temperature of 15°C until the liquor reaches a 22°C threshold, then circulate the must through an external heat exchanger back to 15°C over a

real reported 4-hour refrigeration phase, and repeat. Applied to this train's own real pineapple sugar concentration (158.83 g/L, Murcia Palacios, 2013) and extended to a 5-day cycle for aroma and quality differentiation — a duration chosen for that quality rationale, not derived from Moresi's own cost-minimizing ~78-81 hour stopping point (his Table 3), which this thesis's process model would otherwise adopt directly — the proof-of-concept accompanying this thesis (Chapter 11) finds two results that must be read together, not separately: a REAL, price-independent yield advantage (99.2% sugar conversion versus the fast line's partial conversion, yielding 15.4% more ethanol per tonne of sugar processed, since pineapple's own comparatively low sugar concentration lets the extended, temperature-moderated cycle run essentially to completion well within the 5-day window regardless of exactly how long that window is set), and a REAL cost penalty, smaller than a shorter-duration reading of Moresi's own kinetics might suggest but still material (reactor and 90-day aging-tank-farm capital expenditure approximately 78% higher than the existing line's, since a cycle 9.6 times longer than the existing 12.5-hour line still needs proportionally more installed reactor volume to process the same annual feedstock target, net of the staggered-discharge aging-tank-farm correction developed above, which this premium line's own reactor-train count also benefits from). Reconciling these into a single verdict requires a price: the proof-of-concept computes a break-even wine price of approximately $486.64 per tonne — a 21.7% premium over the existing line's illustrative $400 per tonne price — at which the premium line's own net-present-value contribution merely MATCHES, not exceeds, the existing fast line's. Whether pineapple wine produced under real, differentiated aroma-preservation conditions could command a price premium of that magnitude is a market question this thesis's design-stage methodology cannot itself answer, and Section 6.10's benchmarking discipline is applied here without exception: the

finding is reported as a quantified economic bar the quality hypothesis must clear, not as evidence that it does.

Whether this premium line's fermenters should be jacketed in-vessel or served by the external heat exchanger Moresi's (1989b) own cycling policy assumes above is itself a real, quantifiable design question his own published results answer directly, and answering it surfaces a finding that shifted materially once this train's own feedstock scale was itself corrected. Moresi (1989b) reports jacketed, closed-vessel isothermal operation as his own cost-optimal configuration below approximately 60,000 hL of wine per vintage, with an external plate heat exchanger serving multiple fermenters becoming cost-optimal above that volumetric threshold, a conclusion he additionally grounds structurally in an exchanger-sharing economy that requires many simultaneous fermenters (his own case studies span four to ten exchangers serving five to sixty-nine fermenters) to be worthwhile at all. At this train's own annual output of 78,300 hL/yr — itself revised upward from an earlier 58,500 hL/yr estimate once the pulp fraction's real juice-extraction yield (87%, established above) replaced an earlier, lower placeholder — this train now sits ABOVE Moresi's own volumetric crossover, where his own results point toward an external exchanger on scale alone; his own structural argument, by contrast, continues to favor jacketed vessels, since this premium line's own cost-optimal reactor-train sizing requires only two fermenters, far short of the five-to-sixty-nine-fermenter range his exchanger-sharing economy presumes. Both of Moresi's own real arguments no longer point unambiguously to the same configuration, a genuine tension this thesis discloses rather than resolves by picking whichever conclusion is more convenient; the isothermal, jacketed alternative's own economics, computed independently, are in any case nearly identical to the cycling policy's at this scale and duration, so

cycling's principal remaining justification is the aroma/quality hypothesis itself, which the mechanistic kinetic model applied here does not directly represent.

Two limitations bound what the premium-line kinetic model can predict. First, Moresi's (1989a, 1989b) own kinetic model does not represent nitrogen availability as a state variable at all — his own reported must nitrogen concentration (1.2 g/L) functions in this thesis's proof-of-concept only as a validation reference point, not as an input any growth or inhibition term depends on — which is defensible for his own grape must specifically, since real reported grape-must yeast-assimilable-nitrogen (YAN) averages across several independent regional studies (136-213 mg/L; Sablayrolles & Mouret, 2022) sit well below the real approximately 400 mg/L threshold above which nitrogen availability has no further reported effect on fermentation kinetics, meaning Moresi's own must was comfortably non-nitrogen-limiting. This train's own pineapple juice has no established real YAN measurement in this thesis, and if it falls materially below that same non-limiting threshold — plausible, since fruit musts other than grape are not consistently nitrogen-rich, though this has not been checked — applying Moresi's kinetics unmodified could overstate this train's real achievable fermentation rate and conversion, a validity limitation Section 6.7's uncertainty treatment should carry forward until a real pineapple-juice YAN value is established. Second, the 90-day aging tank farm this premium line shares in kind with the existing wine line, now sized per the staggered-discharge correction above, ages wine on its own fermentation lees at a scale — twelve-to-thirteen stainless-steel tanks in the 179,000-199,000 L range — that Comuzzo et al. (2022) identify as a real risk regime for reductive off-odor (hydrogen sulfide) development, since larger tanks present a lower lees-contact surface-to-volume ratio and make efficient lees resuspension harder to achieve than in smaller vessels. The real, documented mitigation — periodic lees resuspension via bottom nitrogen injection or in-tank mixing — has not been costed

or incorporated into this thesis's aging-tank-farm model, and is disclosed here as an open design item contingent on confirming that aging on lees, rather than aging a previously racked and clarified wine, is in fact this train's intended aging policy.

Real industrial installations further bound the plausible range for this comparison. A pilot-scale Frings acetator's annual throughput runs at roughly one hundred times its own working volume, while a production-scale, two-acetator installation processing 5 t/day of wine reaches a 10-day acetification cycle at approximately 4% acid formation per day per acetator (Bekatorou, 2020) — both consistent in order of magnitude with the optimized-acetator productivity Emde (2017) reports above. Separately, an economic analysis of pineapple-waste vinegar production specifically identified saccharification, not acetification, as the dominant cost driver, accounting for more than 90% of total production cost (Roda et al., 2017a, as reported in Kandylis, 2020) — a finding this thesis's own capital- and operating-cost breakdown (Section 8.2) should carry forward explicitly rather than assume, by default, that the acetification step dominates this line's operating expenditure.

4.3 Stem+Core Train: Bromelain Recovery (Green Polysaccharide Precipitation vs. Ultrafiltration) and Spray-Dried Nutraceutical Powder

Consistent with Section 3.4's screening, the stem+core train retains both candidate bromelain-recovery technologies identified in Chapter 2 as a genuine superstructure decision rather than pre-resolving the choice: green polysaccharide (carrageenan) precipitation, parameterized from Campos et al. (2019) at 80-90% recovery yield with low auxiliary-chemical input, and two-stage ceramic ultrafiltration, parameterized from Nor et al. (2016, 2018) at 94-97% enzyme recovery with 2.5- to 4.4-fold purification depending on the upstream enzymatic-pretreatment step applied. This thesis's process model represents both as parallel candidate

technologies activated by a binary decision variable in the superstructure (Section 6.2), each with its own capital-cost, operating-cost, and downstream-stream-composition consequences — the latter being the specific mechanism, argued in Section 2.3, by which the choice propagates into the shared sugar hub's feed composition, since only the ultrafiltration route's permeate is routed to the hub (Section 4.6), while the precipitation route's mother liquor is directed to on-site waste treatment rather than to the hub, following the process topology demonstrated for the integrated bromelain-bioethanol precedent of Gil et al. (2018). Unlike the pulp train's within-technology reactor-train-count decision (Section 4.2), this is a discrete choice BETWEEN two competing technologies, each carrying its own fixed capital charge rather than a continuously sizeable cost, and it accordingly retains the binary-variable disjunctive treatment of Section 6.5 rather than the binary-free NLP-superstructure treatment Corsano et al. (2009) supply for the former — a distinction Section 6.10 develops explicitly at the superstructure level.

The bromelain-depleted retentate or precipitation cake, together with a share of the stem+core juice not routed to bromelain recovery, feeds a spray-dried nutraceutical powder co-product line, consistent with the integrated valorization precedent established by Campos et al. (2020). Spray-drying process parameters — inlet air temperature, feed flow rate, and carrier-agent (typically maltodextrin) concentration — are adopted from the response-surface-optimized fruit-bioactive spray-drying literature reviewed in Section 2.5 and Section 4.7 below (Homayoonfal et al., 2022; Jafari et al., 2022), rather than from a pineapple-specific spray-drying study, since none of the pineapple-byproduct literature reviewed in Chapter 2 reports a directly comparable spray-drying optimization; this is flagged explicitly as a parameter carried by analogy rather than by direct pineapple-specific evidence, and Section 6.7's uncertainty treatment accordingly assigns this

sub-model's yield and quality parameters a wider uncertainty band than the bromelain-recovery parameters proper, which do have direct pineapple-specific support.

4.4 Peel Train: Fiber/Flour (IDF) and Sugar-Rich Juice Recovery

The peel train follows the same juice/press-cake mechanical split described in Section 3.1 and parameterized compositionally in Section 3.2.3: a sugar-rich juice stream, reported at approximately 67% sugar content on a juice basis and routed directly to the shared sugar hub (Section 4.6), and a press-cake stream processed into an insoluble-dietary-fiber flour, following the drying-and-milling procedure demonstrated for pineapple peel and stem press-cake by Campos et al. (2020) and consistent with the fiber-rich-fraction composition (41.8-48.0 g insoluble dietary fiber per 100 g dried material) reported for pineapple peel specifically (Huang et al., 2020, as reviewed in Section 3.2.3).

Because peel bromelain activity is lower than stem activity in most of the comparisons reviewed, though not consistently lower than core activity (Section 3.2.2, Section 3.2.3), this thesis's candidate-technology screening (Section 3.4) does not carry a bromelain-recovery option forward for the peel train, in contrast to the stem+core train — a modeling simplification directly traceable to the composition data of Chapter 3 rather than an arbitrary scope restriction, and one that keeps the peel train's process model comparatively simple relative to the stem+core and lignocellulosic trains: essentially a mechanical separation followed by, on the solid side, drying and milling, and on the liquid side, direct routing to the hub with no intermediate biochemical processing step of its own.

[NOTA: misma corrección que en 3.2.3. Con la cáscara por encima del corazón en Chiarelli (2024) y Misran (2022), excluir la recuperación de bromelina del tren de cáscara ya no se sigue solo de la composición: conviene justificarlo también por volumen, costo o dilución, o declararlo como simplificación de alcance.]

4.5 Lignocellulosic Train: Delignification Pretreatment and Enzymatic Hydrolysis

The leaves/crown fraction's lignocellulosic composition (24-26% lignin, Section 3.2.4) requires a delignification pretreatment step ahead of enzymatic hydrolysis to achieve an economically meaningful sugar yield, consistent with the broad consensus across the lignocellulosic-pretreatment literature that residual lignin physically blocks cellulose accessibility, non-productively adsorbs cellulase enzymes, and sterically hinders hydrolysis (Loow et al., 2016; Cai et al., 2023). This thesis's process model adopts a dilute-alkali (sodium hydroxide) delignification step as the reference pretreatment technology, consistent with the broad finding that alkaline pretreatment preferentially targets the lignin fraction (in contrast to dilute-acid pretreatment, which preferentially targets hemicellulose) and is among the two most-reported pretreatment technologies for achieving high sugar yields at comparatively low cost across the lignocellulosic-biomass literature generally (Loow et al., 2016).

Pretreatment severity — alkali concentration, temperature, and residence time — and the resulting delignification extent and downstream enzymatic glucose yield are parameterized from the general dilute-alkali pretreatment literature reviewed above rather than from a study specific to pineapple leaves/crown, since the pineapple-leaves bioethanol precedent reviewed in Section 2.2 (Saini et al., 2022) reports a hydrothermal, not alkaline, pretreatment route; this thesis's Section 6.7 uncertainty treatment accordingly propagates a wider parametric range for this sub-model's sugar yield than for the fraction-composition parameters of Section 3.2, which do carry direct pineapple-specific support. It bears restating, following Section 3.2.4's own observation, that this train's lignin removal is in direct tension with the energy island's feedstock demand (Chapter 5): lignin sent to delignification and subsequently to the hydrolysis train is lignin unavailable for direct gasification, and the superstructure of Chapter 6 represents the leaves/crown fraction's split

between these two uses as a genuine optimization decision rather than a pre-fixed allocation, consistent with Section 3.4's screening methodology.

The resulting lignocellulosic hydrolysate — a sugar-bearing stream dominated by glucose with a lesser xylose fraction, consistent with the general finding that alkaline delignification followed by cellulase/hemicellulase hydrolysis yields a glucose-dominant hydrolysate (Loow et al., 2016) — is the third of the three streams converging at the shared sugar hub modeled in Section 4.6, alongside the peel juice (Section 4.4) and the stem+core ultrafiltration permeate, where that recovery route is the one activated (Section 4.3).

4.6 The Shared Sugar Hub: Pooling of Peel Juice, Hydrolysate, and UF Permeate

The shared sugar hub is the structural core of this thesis's whole-plant superstructure and the specific locus of the bilinear nonconvexity introduced conceptually in Chapter 1 and positioned against the pooling-problem literature in Chapter 2. This section develops the hub's governing equations explicitly, adapting the classical pooling-problem formulation (Quesada & Grossmann, 1995; Misener & Floudas, 2009) and its multiperiod extensions (Castro, 2015a; Kolodziej et al., 2013) to this thesis's specific three-input, multi-output structure.

Let $k \in K$ = {peel juice, lignocellulosic hydrolysate, stem+core UF permeate} index the three candidate input streams to the hub, each with flow rate F_k (mass or volume per unit time) and sugar concentration π_k (mass fraction), both established in Sections 3.2, 4.4, and 4.5 as fraction-composition and process-yield parameters. Let $o \in O$ = {citric acid line, xanthan gum line, vinegar line, ethanol line} index the four candidate output platforms established in Section 2.4 and Section 3.4, each drawing a flow $F_{H,o}$ from the hub. The hub itself is represented as a single pool with an endogenous sugar concentration C_H, a continuous decision variable of the superstructure (Section 6.2) rather than a parameter, since its value depends on which upstream

input streams are actually present and in what proportion — itself a function of the upstream superstructure's own binary and continuous decisions (Section 4.3's bromelain-recovery-route choice and Section 4.5's leaves/crown allocation split chief among them).

$$\text{Hub flow balance:} \quad \Sigma_{k \in K} F_k = \Sigma_{o \in O} F_{H,o}$$

$$\text{Hub property balance:} \quad \Sigma_{k \in K} \pi_k \cdot F_k = C_H \cdot \Sigma_{o \in O} F_{H,o} \quad \text{[bilinear]}$$

$$\text{Output spec bound:} \quad \pi^L_o \cdot F_{H,o} \leq C_H \cdot F_{H,o} \leq \pi^U_o \cdot F_{H,o} \quad \forall\, o \in O$$

The hub property balance is the bilinear equation — the product of the endogenous concentration variable C_H and the endogenous total-outflow variable $\Sigma_o F_{H,o}$ — that makes this system's feasible region genuinely nonconvex, exactly the structural feature Section 1.4 and Section 2.6 argue a linearized or implicit treatment of hub allocation, as this thesis contends much of the prior superstructure biorefinery literature adopts, would silently misrepresent. The output-specification bound is written here in a big-M-gated form consistent with the flow-activation-gated specification-satisfaction pattern documented for multiperiod blending systems (Castro, 2015a; Kolodziej et al., 2013): a specification constraint only binds when the corresponding output flow $F_{H,o}$ is active, since an inactive line drawing zero flow from the hub trivially satisfies any concentration bound and should not be permitted to render the superstructure infeasible on that account.

Because this bilinear system is embedded within a larger MINLP carrying discrete line-activation and technology-selection decisions elsewhere in the superstructure (Section 6.1-6.4), Section 6.3 and Section 6.8 develop the specific solution strategy adopted for this hub — a choice between retaining the exact bilinear formulation for global-optimization solution via spatial branch-and-bound (Quesada & Grossmann, 1995), and a discretization-based linear relaxation trading exactness for tractability (Castro, 2015a) — and justify that choice against the scale the

full superstructure ultimately reaches. This chapter's role is limited to establishing that the hub is correctly and exactly formulated in the first instance, prior to any relaxation decision made downstream in Chapter 6; conflating chapter-4-level model formulation with chapter-6-level solution strategy would risk quietly resolving the tractability question by simplifying the model before the model has even been fully specified, precisely the shortcut this thesis's Section 2.6 critiques in the prior superstructure biorefinery literature.

4.7 Membrane and Green-Processing Unit Models (MF, UF, Precipitation, Controlled-Temperature Spray Drying)

Four green-processing unit-operation classes recur across the four processing trains developed in Sections 4.2-4.5: microfiltration (pulp train juice clarification, Section 4.2), ultrafiltration (stem+core bromelain recovery, Section 4.3), precipitation (the competing stem+core bromelain-recovery route, Section 4.3), and controlled-temperature spray drying (stem+core nutraceutical powder, Section 4.3). This section consolidates the generic unit-operation models this thesis applies to each, parameterized by the fraction-specific bench data cited within each train's own section above.

Membrane unit operations (microfiltration and ultrafiltration) are modeled via a permeate-flux/transmembrane-pressure relationship consistent with the cake-filtration mechanism identified as dominant across the fruit-juice ceramic-membrane clarification literature (Nandi et al., 2009, 2010; Arikrishnan et al., 2022), with flux decline over the course of a filtration run represented via the Hermia pore-blocking model framework standard in this literature (Satyannarayana et al., 2023). Membrane fouling — rather than membrane capital cost — is treated, consistent with the general finding of the membrane-processing literature reviewed in Section 2.5 (Mohammad et al., 2012), as the primary determinant of the achievable flux and hence of the membrane area, and

therefore capital cost, required to process a given throughput; this thesis's process model accordingly sizes membrane area from a fouling-adjusted average flux rather than from the clean-membrane initial flux, a conservative choice relative to sizing methodologies that do not make this distinction.

Two directly quantified data points now anchor this fouling-dominated sizing logic for the two membrane unit operations spanning this train pairing, both drawn from genuinely pineapple-specific characterizations. For the pulp train's microfiltration unit, Laorko et al.'s (2010) pineapple-juice cross-flow characterization (Section 4.2) supplies the average, fouling-adjusted flux this thesis's sizing procedure requires directly, at the train's own 0.2 μm design pore size. For the stem+core train's ultrafiltration unit, a four-paper series by Nor et al. (2015, 2016, 2017, 2018) directly characterizes exactly this separation — crude pineapple waste mixture extract processed through a two-stage tubular ceramic ultrafiltration train (75 kDa, then 10 kDa, Membralox membranes) — and identifies, via non-linear regression of the real flux-decline curves against four candidate Hermia (1982) fouling-mechanism models, cake filtration as the confirmed dominant mechanism for the first (pre-filtration) stage and pore-blocking mechanisms (intermediate at low cross-flow velocity, complete at higher velocity) for the second (concentration) stage (Nor et al., 2017).

This thesis's process model accordingly sizes the pre-filtration stage's fouling-adjusted flux directly from the confirmed cake-filtration closed-form solution. For the concentration stage, the same closed form is used only as a time-course SHAPE approximation, calibrated to Nor et al.'s (2016) own reported flux-decline endpoints (initial and steady-state flux, elapsed time): the pore-blocking mechanism's own closed-form solution, as transcribed from the reviewed scanned source, produced a diverging, unphysical flux under calibration and was rejected rather than silently used,

a limitation disclosed here and flagged for direct verification against the primary journal typesetting before this sizing basis is treated as final. Both membrane unit operations are now grounded in pineapple-specific data, superseding an earlier version of this thesis's process model that relied on Rai et al.'s (2006) resistance-in-series characterization for mosambi (Citrus sinensis) juice as an analogous stand-in for the ultrafiltration unit. Section 6.7's uncertainty treatment should accordingly narrow the ultrafiltration unit's previously wider uncertainty band for its flux-sizing parameter, though the disclosed pore-blocking-equation limitation above means the concentration stage specifically should not yet be treated as equally certain as the pre-filtration stage or the microfiltration unit.

Precipitation unit operations are modeled as a stoichiometric or near-stoichiometric addition of precipitating agent (carrageenan, for the green polysaccharide route adopted in this thesis per Section 4.3) followed by solid-liquid separation, with recovery yield and purification fold parameterized directly from Campos et al. (2019).

Spray drying is modeled via the standard atomization-drying-collection sequence, with inlet air temperature, feed flow rate, and carrier-agent (maltodextrin) concentration as the primary design variables governing both powder yield and bioactive-compound retention, consistent with the response-surface-methodology optimization approach that dominates the food-bioactive spray-drying literature (Homayoonfal et al., 2022) and the fruit-product-specific review of spray-drying encapsulation outcomes (Jafari et al., 2022), which documents that maltodextrin concentration and inlet temperature jointly, and often with an interaction effect neither factor alone predicts, determine both process yield and the retention of thermolabile bioactive compounds — a joint dependency this thesis's process model represents via the same response-surface-derived

functional form rather than treating temperature and carrier concentration as independently optimizable parameters.

4.8 Mass and Energy Balances; Stream Reconciliation

With each processing train's unit operations specified (Sections 4.2-4.7), this section closes the flowsheet-wide mass and energy balance and establishes the stream-reconciliation procedure this thesis applies before any balance data is passed forward to Chapter 5's energy-island sizing or Chapter 6's superstructure parameterization.

Mass and energy balance closure follows the standard steady-state conservation framework — accumulation equals inflow minus outflow plus generation minus consumption, specialized to zero accumulation under this thesis's steady-state, representative-annual-scenario design basis (Section 3.5) — applied systematically at every unit operation and at every stream-splitting or stream-mixing node in the flowsheet, including, critically, the hub's own bilinear property balance developed in Section 4.6, which is itself a mass balance on the sugar content of the pooled stream rather than a separate accounting device. Because this thesis's flowsheet draws its unit-operation parameters from a heterogeneous set of bench-scale literature sources (Sections 4.2-4.7), each reporting yields and compositions under its own experimental conditions, the resulting flowsheet-wide balance is not guaranteed to close exactly on first assembly — an over-determined or under-determined system in the sense familiar from the data-reconciliation literature, where the number of independently measured or literature-sourced parameters does not automatically equal the number of degrees of freedom the flowsheet's topology requires (Tamhane & Mah, 1985).

This thesis addresses this via a stream-reconciliation procedure adapted from the data-reconciliation methodology developed in the process-systems-engineering and mineral-processing literatures for exactly this class of problem: measured or literature-sourced stream values are

adjusted by weighted least-squares estimation subject to the flowsheet's mass and energy conservation constraints, with the weighting reflecting this thesis's own judgment (Section 4.1, Section 3.2) of which parameters carry direct pineapple-specific experimental support versus which are carried by analogy from related fruit or lignocellulosic matrices (Hodouin, 2010; Tamhane & Mah, 1985). Parameters with direct pineapple-specific support — bromelain recovery yields, ceramic-microfiltration flux for pineapple juice specifically, fraction-level composition data — are assigned tight reconciliation weights, permitting little adjustment from their literature-reported values; parameters carried by analogy — the lignocellulosic pretreatment severity of Section 4.5, the spray-drying parameters of Section 4.3 and Section 4.7 — are assigned looser weights, permitting the reconciliation procedure to adjust them more freely to close the overall balance, consistent with the general principle in the data-reconciliation literature that measurement (or, in this thesis's case, literature-sourced parameter) uncertainty should directly determine how much a given stream value is trusted relative to the constraint system as a whole (Rafiee & Behroozsarand, 2016).

The reconciled mass and energy balance resulting from this procedure is the specific quantitative output this chapter delivers forward: it parameterizes the energy-island sizing of Chapter 5 (via the lignocellulosic fraction's split-dependent net calorific feedstock available for gasification, Section 3.2.4 and Section 4.5) and the superstructure's own stream-flow and stream-composition parameters in Chapter 6, including, centrally, the hub's three input-stream flow rates and sugar concentrations (F_k, π_k for $k \in K$) that populate the bilinear property balance developed in Section 4.6.

Chapter 5. Off-Grid Energy Island (ZNI)

The energy island must run on the same leaves/crown fraction the sugar platform would send to hydrolysis: lignin removed to improve hydrolysis yield is lignin the gasifier no longer receives (Section 3.2.4), so the plant's own power supply is a design decision (Section 5.4), not a fixed input. This chapter formulates the off-grid energy island that Chapter 1 established as a hard feasibility constraint on the superstructure, not a downstream sizing exercise. Section 5.1 specifies the gasification–internal-combustion-engine (ICE) technology chain; Section 5.2 develops the electricity and process-heat generation model together with the diesel-backup role for aerated fermentation; Section 5.3 maps how that energy supply integrates with the process trains of Chapter 4; and Section 5.4 develops the energy-autonomy constraint's coupling to the superstructure via the Flexibility Index formalism of Swaney and Grossmann (1985).

5.1 Downdraft Gasification of Leaves/Crown → Syngas Cleanup → Internal Combustion Engine

Downdraft gasification of the leaves/crown fraction not routed to the delignification/hydrolysis train (Section 4.5, Section 3.2.4) is this thesis's reference thermochemical conversion technology, consistent with its documented maturity and feedstock flexibility for small- to medium-scale distributed power generation (Martínez et al., 2012). Figure 5.1 presents the resulting block-flow diagram: feedstock enters the downdraft gasifier, producing a raw syngas stream that requires cleanup before it can be fed to the internal combustion engine.

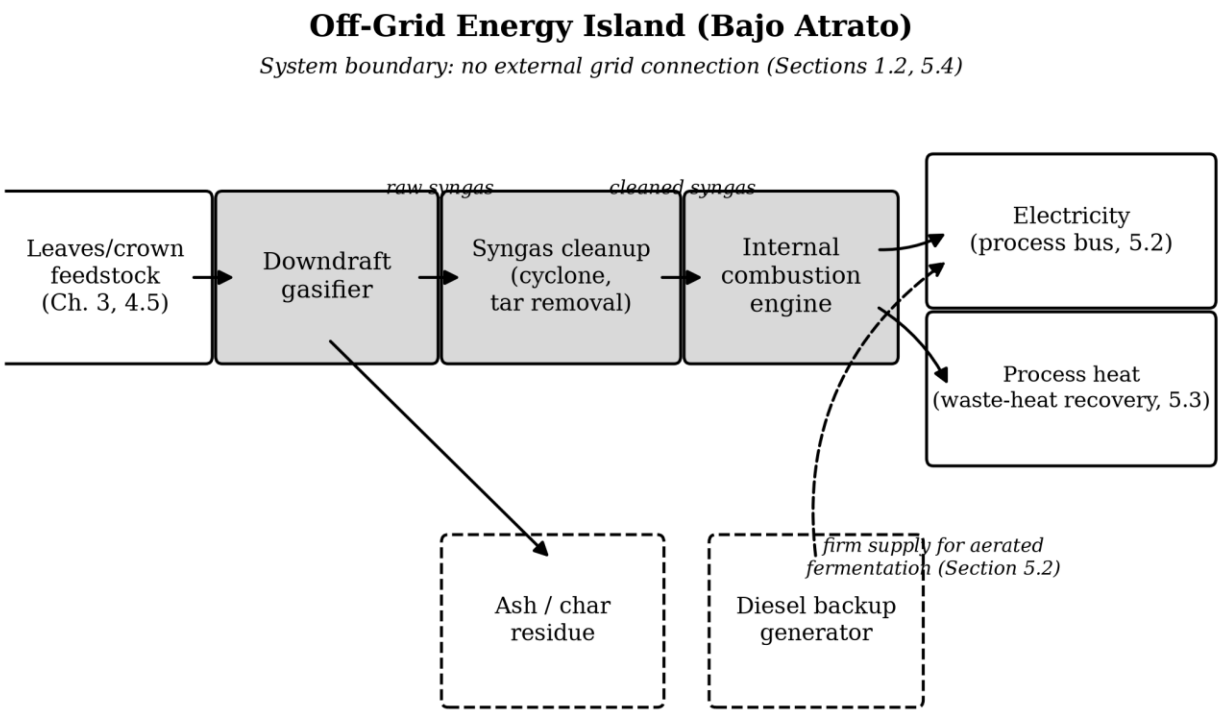


Figure 5.1. Off-grid energy island block-flow diagram: downdraft gasification of leaves/crown feedstock through syngas cleanup to internal-combustion-engine power generation, with diesel backup for firm aerated-fermentation supply.

This is an original diagram developed for this thesis, synthesizing the gasification–ICE technology chain documented by Martínez et al. (2012), the syngas cleanup literature of Section 5.1, and the diesel-backup role established in Section 5.2.

Syngas cleanup is not an optional refinement but a prerequisite for reliable engine operation: raw producer gas carries tar, particulates, and trace contaminants that cause pipeline fouling, downstream corrosion, and catalyst deactivation if not removed (Cortazar et al., 2023). Tar in particular is documented as the most persistent contaminant across the biomass gasification literature, capable of condensing in fuel lines and clogging filters well downstream of the gasifier itself (Rios et al., 2017; Abdoulmoumine et al., 2015). This thesis's reference cleanup train follows the primary/secondary strategy distinction established in the tar-removal review literature: primary measures (operating-parameter control within the gasifier itself — equivalence ratio, temperature)

reduce tar formation at the source, while secondary measures (cyclone separation for particulates, wet or dry scrubbing for residual tar) polish the raw syngas to the quality required downstream (Rios et al., 2017; Cortazar et al., 2023; Andersson et al., 2017). Given this thesis's off-grid, distributed-scale context — far below the industrial-scale gas-conditioning installations documented by Andersson et al. (2017) — a comparatively simple cyclone-plus-scrubber train is adopted as the reference cleanup technology, consistent with the small-scale gasification-ICE precedents reviewed in Section 2.7 (Chambon et al., 2020; Ribó-Pérez et al., 2021), rather than the catalytic hot-gas-cleanup systems that industrial-scale installations can justify.

[NOTA — citas del Cap. 5 (Consensus, 3-nov). Cortazar et al. (2023) menciona incrustación, corrosión y desactivación de catalizadores, no degradación del lubricante del motor; si quieres decirlo, busca una fuente sobre motores con gas pobre. Cortazar revisa solo las estrategias primarias: la distinción primaria/secundaria la formula Rios et al. (2017), que añadí a esa cita. Confirmados: Abdoulmoumine et al. (2015), Andersson et al. (2017, escala industrial), Swaney y Grossmann (1985), Grossmann et al. (1983), Ong et al. (2017). Sadhukhan et al. (2014) es un capítulo de libro, sin resumen en Consensus.]

The cleaned syngas fuels a spark-ignition or dual-fuel internal combustion engine generator set, the same technology chain documented across the off-grid techno-economic assessments reviewed in Section 2.7 (Sánchez-Lozano et al., 2023; Indrawan et al., 2020). Ash and char residue from the gasifier (Figure 5.1) is a secondary output stream this thesis does not valorize further within the energy island itself, consistent with the scope established in Section 1.6, though its disposal or soil-amendment potential is noted as a direction for the environmental assessment of Chapter 7.

5.2 Electricity and Process-Heat Generation; Diesel Backup for Aerated Fermentation

The ICE generator set produces two useful outputs this thesis's energy balance tracks separately: shaft power converted to electricity for the process bus, and waste heat recoverable from the engine jacket and exhaust for process-heat duties (Figure 5.1). This dual-output

characterization follows the combined-heat-and-power framing standard across the off-grid biomass gasification literature (Sánchez-Lozano et al., 2023; Chambon et al., 2020), and both outputs are sized from the same gasifier feedstock allocation established in Section 4.5 and Section 6.1 — there is no independent fuel source for electricity versus heat within the energy island itself.

A single exception to this all-biomass rule is deliberately built into the reference design: a diesel backup generator provides firm (uninterruptible) supply specifically for the aerated fermentation steps of the sugar-hub output platforms (citric acid, xanthan gum; Section 2.4, Section 4.6), because aeration failure during an active fermentation batch risks culture loss and, with it, the batch's entire feedstock and processing investment — a risk asymmetry that justifies a firm-power exception to this thesis's otherwise strict off-grid design philosophy (Section 1.2). This is a deliberate, narrowly scoped exception, not a retreat from that philosophy: the diesel backup's role is restricted to aerated-fermentation continuity, sized to that duty alone rather than to general plant backup, and its presence and cost are represented explicitly in the superstructure (Section 6.1) and in the environmental assessment (Chapter 7) rather than left as an unaccounted convenience. This mirrors the documented practice in ZNI electrification more broadly, where diesel is retained as a bounded, declining-role backup during a renewable transition rather than eliminated outright on day one (Garces et al., 2023).

5.3 Heat/Power Integration with Process Units (Enzymatic, Membranes, Pretreatment, Hydrolysis)

Figure 5.2 maps how the energy island's electricity and process-heat outputs are distributed across the process trains developed in Chapter 4: the pulp train's enzymatic treatment and ceramic microfiltration (Section 4.2), the stem+core train's ultrafiltration, precipitation, and spray drying

(Section 4.3), the leaves/crown train's delignification pretreatment and enzymatic hydrolysis (Section 4.5), and the sugar-hub output platforms' aerated fermentation (Section 4.6, Section 2.4).

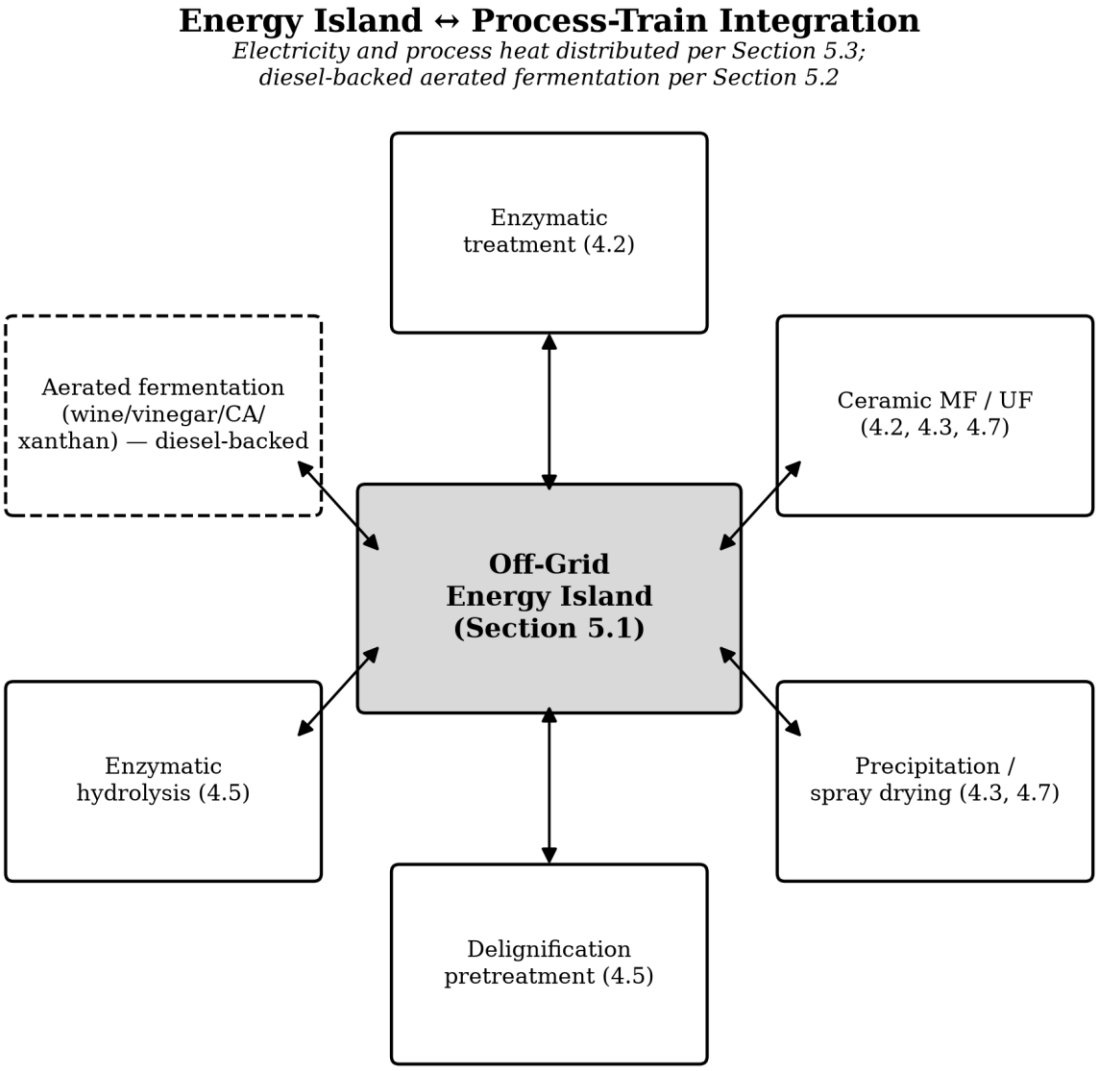


Figure 5.2. Energy island–process train integration: electricity and process-heat distribution across the pulp, stem+core, and lignocellulosic trains, with the diesel-backed aerated-fermentation exception shown at left.

This is an original diagram developed for this thesis; the six process-unit demand nodes are drawn directly from the trains formulated in Sections 4.2–4.6, and the heat-integration principle follows the Total Site / process-graph biorefinery precedent of Ong et al. (2017) and Sadhukhan et al. (2014).

This thesis treats the electricity and process-heat buses in Figure 5.2 as two separate but jointly constrained resources, following the Total Site heat, mass, and power integration methodology developed for biorefinery process graphs (Ong et al., 2017): every process unit's electricity and process-heat demand is drawn from a common pool sized by Section 5.1's gasifier-ICE output, and the resulting integration problem — which units draw how much of each utility, and at what temperature level for the heat demand specifically — is structurally the same class of problem addressed by Pinch Analysis and Total Site heat integration in the wider biorefinery process-integration literature (Sadhukhan et al., 2014). This thesis does not, however, perform a full Pinch Analysis heat-exchanger-network synthesis for the process-heat bus; consistent with the scope established in Section 1.6, process-heat demand is aggregated at the level of Figure 5.2's six demand nodes rather than resolved into an explicit heat-exchanger network, a simplification the Total Site literature itself documents as reasonable at the conceptual-design stage this thesis occupies, where rough energy targets — not a fully synthesized exchanger network — are what a first-pass superstructure optimization requires (Ong et al., 2017).

[NOTA — LIBROS (3/5), picos de la isla energética. Un balance anual promedio no demuestra autonomía hora a hora. Maravelias (2021, §4.6-4.7, pp. 111-121; §7.2.1, p. 164) modela la electricidad como recurso compartido con capacidad y costo variables en el tiempo; con esa lógica se verifica si la acetificación aireada, la hidrólisis y las membranas, al coincidir, exceden la salida del gasificador. Burk (2026, §9.3, pp. 147-148) da el orden de magnitud de la carga: calor metabólico ≈ 17 000 kJ/kg de O2 (o 13 000 kJ/kg de biomasa), aireación de 0,1-2,0 vvm (≈0,5 en equipos grandes, ≈1,0 en pequeños), que fija el compresor. Son cifras genéricas: etiquétalas como parámetros por analogía (4.1). Afecta directamente la prueba de H2 y el tamaño del respaldo diésel.]

Among the six demand nodes, membrane operations (ceramic microfiltration, ultrafiltration) and enzymatic hydrolysis are, consistent with Section 4.7's unit-operation models, predominantly electricity-consuming (pumping, agitation) with comparatively modest process-heat demand, while delignification pretreatment and spray drying are heat-intensive, consistent

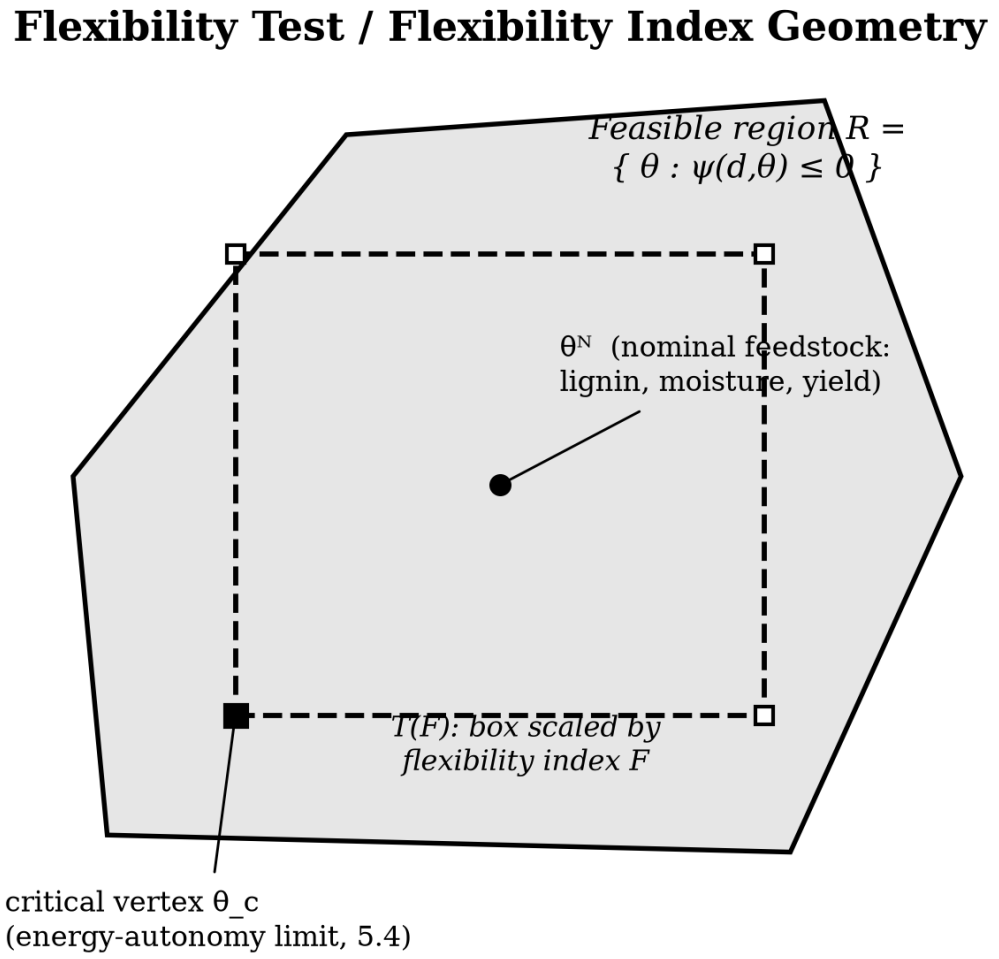
with the general finding that thermal pretreatment and thermal drying operations dominate a lignocellulosic biorefinery's process-heat budget (Loow et al., 2016; Jafari et al., 2022). This asymmetry is a specific, quantified input to Section 6.1's superstructure formulation: the leaves/crown fraction's own allocation decision (delignification/hydrolysis versus direct gasification, Section 3.2.4, Section 4.5) determines simultaneously how much process heat the pretreatment train demands and how much gasifier feedstock remains available to supply it — a second, heat-side instance of the same feedstock-allocation coupling Section 4.6 develops on the sugar-hub side.

5.4 Energy Autonomy Constraints and Their Coupling to the Superstructure

Chapter 1 (Section 1.2) established that this thesis treats energy autonomy as a hard feasibility condition, not merely a favorable economic outcome. This section formalizes that condition using the Flexibility Index framework of Swaney and Grossmann (1985), extended within a broader optimization-under-uncertainty strategy for flexible process design (Grossmann et al., 1983), and connects it explicitly to the superstructure's own decision variables.

The underlying question is not whether the energy island balances at the nominal design point — Section 5.1–5.3's models are constructed to ensure that it does — but whether it continues to balance across the plausible range of feedstock variability documented in Chapter 3: lignin content (24–26%, Section 3.2.4), moisture content, and the resulting gasifier yield all carry a parametric range rather than a single point estimate, consistent with this thesis's broader methodological commitment (Section 3.5) to propagate documented ranges rather than resolve them by silent adoption of a midpoint. Figure 5.3 illustrates the resulting geometry: a feasible region R in the space of these uncertain feedstock parameters θ, within which the energy island's design d (gasifier and ICE capacity, fixed at design time) can always be operated feasibly by proper

adjustment of control variables z — and a flexibility test asking whether a box T(F) of expected parameter deviations, scaled by a flexibility index F, lies entirely within that feasible region.



Adapted from the vertex-search formulation of Swaney & Grossmann (1985); applied here to the energy island's feedstock–autonomy feasibility set (Section 5.4).

Figure 5.3. Flexibility test and Flexibility Index geometry applied to the energy island's feedstock–autonomy feasibility set: a nominal feedstock point θ^N, an expected-deviation box T(F), and the critical vertex θ_c at which the energy-autonomy constraint first binds.

This is an original diagram developed for this thesis, adapting the vertex-search geometry of Swaney and Grossmann (1985) to the specific feedstock parameters (lignin content, moisture, yield) established in Chapter 3.

[NOTA — LIBROS (1/5), flexibilidad no convexa. Grossmann (2021, p. 151) advierte que si la región factible es no convexa los puntos críticos pueden no ser vértices; §15.8 (p. 153) da las condiciones para que sí lo sean y §15.9 (pp. 156-160) la estrategia de conjunto activo (Grossmann y Floudas, 1987), que no depende de esa suposición. Con el hub bilineal y el balance energético acoplado, la búsqueda por vértices de esta figura podría no encontrar el θc real. Propuesta: un párrafo que (a) declare la suposición de vértice y sus condiciones, y (b) adopte la formulación de conjunto activo o la deje como verificación en 6.9. La ec. 15.33 (p. 153), sensibilidad de F respecto al diseño, sirve para dimensionar el respaldo diésel de 5.2 con criterio. Protege la prueba de H2.]

Formally, following Swaney and Grossmann (1985), the flexibility test evaluates $\chi(d) = \max_{\theta \in T} \min_z \max_j f_j(d,z,\theta)$, where f_j indexes the set of constraints the energy island must satisfy — chief among them, that gasifier output plus diesel-backup supply (Section 5.2) meets process electricity and heat demand (Section 5.3) at every point in the uncertainty box. $\chi(d) \leq 0$ certifies the design is feasible across the full box $T(F)$; $\chi(d) > 0$ identifies both infeasibility and the specific critical parameter combination θ_c at which it first occurs. Where the underlying constraints are quasiconvex in the control variables z and one-dimensionally quasiconvex in the uncertain parameters θ — a property this thesis does not assume without verification, consistent with the documented failure mode in which vertex search silently misses a non-vertex critical point in genuinely nonconvex feasible regions — the critical point can be found cheaply by evaluating only the 2^p vertices of the box, where p is the number of uncertain parameters; where that property cannot be established, the exact active-set mixed-integer reformulation of the flexibility test, avoiding the vertex assumption entirely, is the fallback this thesis adopts.

[NOTA: la expresión max–min–max es la prueba de flexibilidad de Halemane y Grossmann (1983, AIChE J.); Swaney y Grossmann (1985) definen el índice F, el mayor escalamiento de la caja de incertidumbre que sigue siendo factible. Si conservas la fórmula, atribúyela a Halemane y Grossmann (1983) o di «following Halemane and Grossmann (1983) and Swaney and Grossmann (1985)». No lo verifiqué en Consensus: es conocimiento estándar del área; confírmalo en las fuentes.]

The coupling to the superstructure (Section 6.1) is direct and two-directional. In one direction, the leaves/crown allocation split between the delignification/hydrolysis train and direct gasification (Section 3.2.4, Section 4.5, Section 5.3) is itself a superstructure decision variable that determines d — specifically, how much feedstock capacity the energy island's design can draw upon — so the flexibility test of this section must be evaluated at whatever allocation the superstructure's optimization proposes, not at a single pre-fixed design. In the other direction, Section 6.7's uncertainty and robustness formulation for the superstructure's economic objective (NPV under price and yield uncertainty) is kept methodologically distinct from this section's flexibility test, following the distinction established in Section 2.9: the economic uncertainty treatment asks what configuration performs best in expectation or under a risk-adjusted criterion, while this section's Flexibility Index asks a categorically different question — whether a given fixed design can always be operated feasibly at all, independent of what that operation costs. A configuration that is optimal in expectation but which fails this section's flexibility test at any economically plausible feedstock deviation is not, on this thesis's own methodological terms established in Chapter 1, an admissible solution to the design problem, regardless of its expected profitability.

[NOTA — LIBROS (5/5), un solo problema en dos etapas. Maximizar el VPN esperado exigiendo que, para todo θ en T, exista una operación factible (autonomía energética) es exactamente la programación en dos etapas con factibilidad garantizada de Grossmann (2021, §15.2, pp. 142-143, ecs. 15.3-15.4); en p. 142 cita la relación entre flexibilidad y optimización robusta (Zhang et al., 2016). Formularlo así une 5.4 y 6.7 y da respaldo a la contribución 4 (estocástico frente a robusto; Grossmann §14.1, pp. 129-130, y §14.5, p. 139). Si el modelo por escenarios es intratable: método L-shaped o Benders (§14.3, p. 134). Complemento: Burk (2026, §13.6, p. 231) permite leer H1 como el VPN incremental de integrar el hub. Guía completa: Tesis_Bajo-Atrato_guia_insercion_Grossmann_Maravelias_Burk_2026-11-04.md.]

Chapter 6. Superstructure Formulation and MINLP Optimization

The preceding chapters established, in turn, the feedstock and market case for a pineapple whole-plant biorefinery (Chapters 1-3), the unit-operation-level process models for each of its four candidate processing trains and their shared sugar hub (Chapter 4), and the off-grid energy island that must supply all of it without external grid backup (Chapter 5). This chapter integrates those pieces into a single mathematical program: a mixed-integer nonlinear program (MINLP) that simultaneously selects which candidate technologies to build, how to split each fraction's flow among them, and how much of each output platform's capacity to install, in order to maximize (or, in the multiobjective formulation of Section 6.6, jointly optimize) the plant's techno-economic and environmental performance. Sections 6.1-6.5 build the formulation: the superstructure and its notation, the hub, permeate, and premium splits that couple the four trains, and the discrete activation and exclusivity logic. Sections 6.6-6.8 state the objectives, the robustness treatment under price and yield uncertainty, and the solution strategy for the hub's nonconvexity; Sections 6.9-6.10 set out how the model will be verified and benchmark its paradigm against Corsano et al. (2009); and Section 6.11 closes the chapter by returning to the hub problem it began with.

6.1 Superstructure Construction: Fraction Trains, the Sugar Hub, and Candidate Technologies

A superstructure, in the sense established by Yeomans and Grossmann (1999) and adopted throughout this thesis, is a network representation that embeds every candidate process configuration under consideration as a subgraph of a single larger graph, so that process synthesis becomes a matter of selecting which subset of that graph's nodes and arcs is active, rather than a matter of separately formulating and comparing a finite list of discrete alternatives one at a time.

This thesis's superstructure is built directly from the four processing trains and shared hub already developed at the unit-operation level in Chapter 4, together with the off-grid energy island of Chapter 5: it is not a new process representation invented for this chapter, but the formal MINLP encoding of a structure Chapters 4 and 5 have already established qualitatively.

Figure 6.1 depicts the resulting network. The pineapple whole-plant feedstock (Chapter 3) splits deterministically, by mechanical/physical separation, into four fractions — pulp, stem+core, peel, and leaves/crown — following the composition data of Section 3.2. Three of the four fraction trains then present a genuine superstructure decision: the pulp train's three output lines (wine, microfiltered juice, premium; Section 4.2) are all constructed as parallel candidate technologies whose relative flow allocation the optimization determines; the stem+core train's two candidate bromelain-recovery routes (precipitation versus ultrafiltration; Section 4.3) are mutually exclusive alternatives formalized as a disjunction in Section 6.5; and the leaves/crown fraction's split between the lignocellulosic train (delignification and hydrolysis, feeding the hub) and direct gasification (feeding the energy island of Chapter 5) is a continuous allocation variable α, not a discrete either/or choice, since both uses can coexist at any nonzero split. The peel train, by contrast, carries no superstructure decision of its own: Section 4.4 established that its mechanical-press split into fiber/flour and sugar-rich juice is fixed by the fraction's own composition, not a technology choice, so it enters the superstructure as a fixed-topology subgraph rather than a decision node.

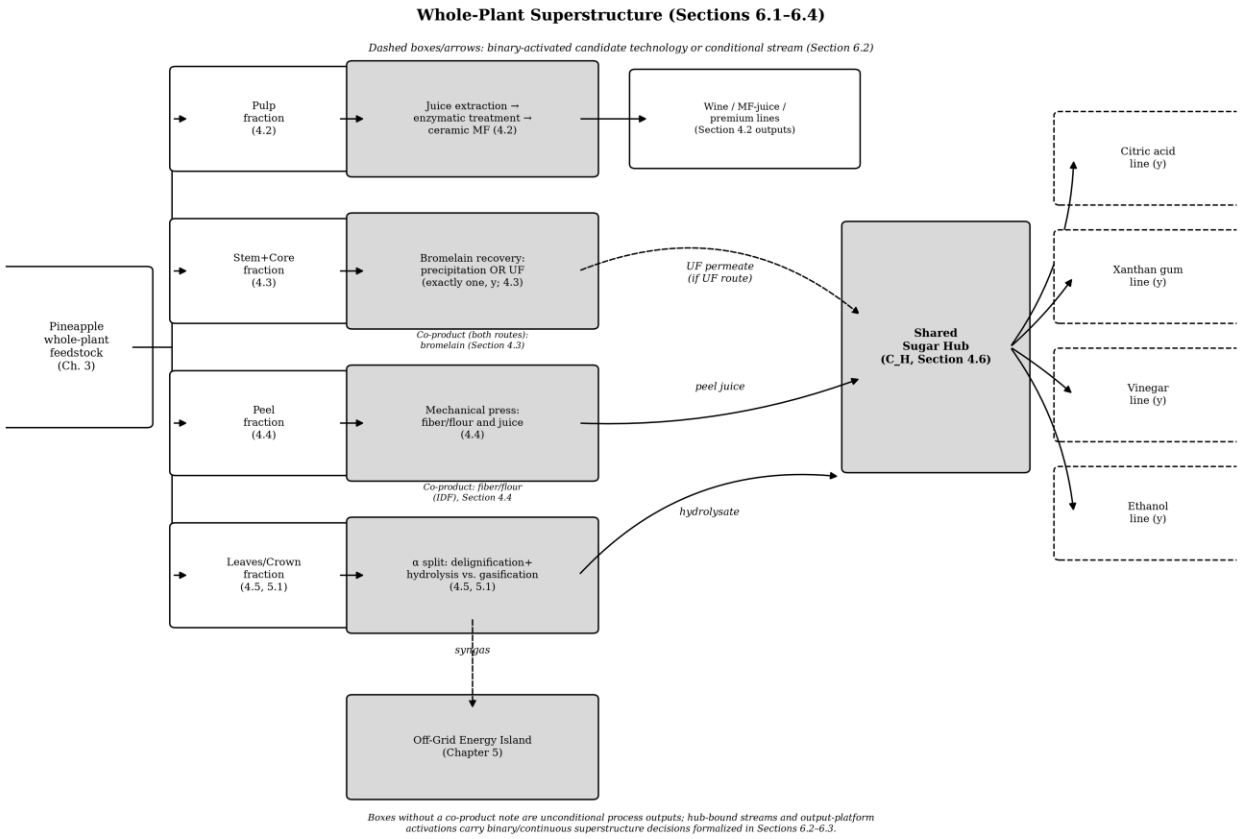


Figure 6.1. The whole-plant MINLP superstructure: four fraction trains with their candidate technologies, the shared sugar hub, and the four output platforms it feeds, together with the leaves/crown fraction's coupling to the off-grid energy island of Chapter 5.

This is an original diagram developed for this thesis, synthesizing the process-train structure of Chapter 4 and the energy-island structure of Chapter 5 (Figures 5.1-5.2) into a single superstructure representation, following the general superstructure-network conventions of Yeomans and Grossmann (1999).

Three of the four fraction trains' candidate technologies converge at the shared sugar hub already formulated in Section 4.6: the peel train's sugar-rich juice (always present, since the peel train carries no activation decision), the lignocellulosic train's hydrolysate (present in proportion to the leaves/crown allocation split α), and the stem+core train's ultrafiltration permeate (present only if the ultrafiltration route, rather than precipitation, is the active bromelain-recovery technology). The hub itself then feeds four output platforms — citric acid, xanthan gum, vinegar, and ethanol fermentation lines (Section 2.4, Section 3.4) — each a candidate technology whose construction is a binary decision. The pulp train's three lines do not route through the hub at all:

Section 4.2 established that this train's sugar content is committed to fermentation or clarification within the pulp train's own boundary, so its coupling to the rest of the superstructure is limited to shared utilities (Section 5.3) and shared capital budget (Section 6.6), not to the shared sugar pool.

This chapter's central methodological claim — stated in Chapter 1 and defended against the prior superstructure biorefinery literature in Chapter 2 (Section 2.6) — is that the hub cannot be represented as a fixed-ratio blending node or bypassed via a pre-resolved allocation without misrepresenting the underlying physics and, in consequence, biasing the optimization's technology and capacity decisions. Section 6.3 restates why: the hub's governing equation is bilinear in two decision variables (concentration and flow), not linear in one, and that nonconvexity is not an artifact of this thesis's specific modeling choice but an inherent property of pooling any two or more compositionally distinct streams under an endogenous blend composition, as the pooling-problem literature (Quesada & Grossmann, 1995; Misener & Floudas, 2009; Meyer & Floudas, 2006) has established for four decades.

6.2 Sets, Parameters, and Variables

This section fixes the notation this chapter's remaining sections build on, extending rather than re-deriving the hub notation already introduced in Section 4.6. Sets: let $f \in FR = \{\text{pulp, stem+core, peel, leaves/crown}\}$ index the four fraction trains; $j \in J_f$ index the candidate technologies available within fraction train f (e.g., $J_{\text{stem+core}} = \{\text{precipitation, ultrafiltration}\}$); $k \in K = \{\text{peel juice, hydrolysate, UF permeate}\}$ index the three candidate hub-input streams, as in Section 4.6; and $o \in O = \{\text{citric acid, xanthan gum, vinegar, ethanol}\}$ index the four hub-output platforms, also as in Section 4.6.

Parameters, largely inherited from Chapter 3's composition data and Chapter 4's process-model yields: F^0_f, the total mass or volume flow of fraction f available from the feedstock

(Section 3.2); π_k and the concentration bounds π^L_o, π^U_o of Section 4.6; unit capital-cost coefficients K_j and c_j for each candidate technology j, following the fixed-charge cost structure standard in MINLP process-synthesis formulations (Section 6.4); and the uncertain-parameter nominal values and ranges θ^N, θ^L, θ^U carried forward from Section 3.5's documented composition ranges and used identically in both this chapter's Section 6.7 robustness formulation and Chapter 5's Flexibility Index test — the same parameter data, deployed for two different and non-substitutable purposes, per the distinction Section 5.4 already insists on and Section 6.7 reiterates below.

Variables divide, as in any MINLP process-synthesis formulation (Grossmann, 2002), into continuous and binary/discrete. Continuous variables: F_j, the flow processed by candidate technology j; α, the leaves/crown allocation split between the lignocellulosic train and gasification, $\alpha \in [0,1]$; F_k and $F_{H,o}$, the hub's input and output flows as in Section 4.6; C_H, the hub's endogenous blend concentration; and V_j, the installed capacity of each candidate technology, which the fixed-charge cost model of Section 6.4 ties to the binary activation decision. Binary/discrete variables: $y_j \in \{0,1\}$, whether candidate technology j is constructed at all — this is the variable Section 6.1's superstructure figure marks with dashed boxes — and, where a discrete choice among mutually exclusive alternatives exists (the stem+core bromelain-recovery route being the clearest instance), a special-ordered-set-style constraint $\Sigma_{j \in J'} y_j = 1$ over the relevant exclusive subset $J' \subset J_f$, formalized in Section 6.5.

6.3 Shared-Resource Allocation: Hub Split, Permeate Split, and Premium Split

Section 4.6 established the hub's governing property balance — flow balance, the bilinear property balance, and the output-specification bound — as a model-formulation matter, prior to and independent of any solution-strategy decision. This section develops the allocation-decision

layer of the superstructure that sits on top of that formulation: the specific splits — of the hub's pooled sugar stream among the four output platforms, of the stem+core train's flow between the UF permeate (hub-bound) and retentate (bromelain product) once the UF route is active, and of the pulp train's premium line's flow relative to its wine and microfiltered-juice siblings — that the optimization must determine jointly with the discrete technology-selection decisions of Section 6.4, because none of these splits can be fixed in advance without pre-resolving exactly the allocation question this thesis's superstructure exists to answer.

The hub split is the output-side allocation already partially specified by Section 4.6's output-specification bound; this section adds the corresponding flow-balance and activation-gating structure. Let $\beta_o \geq 0$ denote the fraction of the hub's total pooled outflow routed to output platform o, so that $\Sigma_{o \in O} \beta_o = 1$ and $F_{H,o} = \beta_o \cdot \Sigma_{o' \in O} F_{H,o'}$. Because β_o is only meaningful for an output platform that is actually constructed, it is coupled to the binary activation variable y_o via the big-M-gated bound $0 \leq \beta_o \leq y_o$, which forces β_o to zero whenever platform o is not built — the specific mechanism, referenced generally in Section 4.6, by which an inactive output line is prevented from either drawing hub flow or (via the output-specification bound of Section 4.6) spuriously constraining the hub's feasible concentration.

The permeate split governs the stem+core train once the ultrafiltration route is the active bromelain-recovery technology (Section 4.3, Section 6.5): let $\gamma \in [0,1]$ denote the fraction of the stem+core train's post-hydrolysis flow directed to the ultrafiltration step's permeate stream (hub-bound) as opposed to its retentate (the bromelain-product stream), with the split itself parameterized from the recovery-yield data of Nor et al. (2016, 2018) rather than treated as a free continuous decision unconstrained by that bench-scale evidence: γ is bounded within the empirically documented recovery-yield range, not left to range over the full unit interval,

consistent with this thesis's broader modeling discipline (Section 4.1) of carrying documented parameter ranges rather than treating every physically plausible value as equally available to the optimizer. Because the permeate split is only physically meaningful when the ultrafiltration route itself is active, γ's own upper bound is additionally gated by the route-selection binary: $\gamma \leq y_{UF}$, so that the precipitation route's model correctly sets the permeate flow, and hence the hub's UF-permeate input, to zero.

The premium split governs the pulp train's internal allocation among its three candidate lines (Section 4.2): let $\delta_{wine}, \delta_{MF}, \delta_{premium} \geq 0$ denote the fractions of the pulp train's total juice flow routed to each line, with $\Sigma\ \delta = 1$ and each δ bounded above by its corresponding activation binary, following the identical big-M-gating pattern used for the hub split. Unlike the hub split and permeate split, the premium split does not feed the shared sugar hub at all — Section 6.1 already established that the pulp train's coupling to the rest of the superstructure runs through shared utilities and shared capital budget, not through shared sugar pooling — so its constraints are structurally independent of Section 4.6's bilinear hub formulation and can be, and are, modeled as an ordinary linear allocation without introducing any additional nonconvexity of their own.

All three splits share a common structural property worth stating explicitly, because it is the property that makes this section's constraints genuinely part of the superstructure's optimization problem rather than a bookkeeping exercise performed after the fact: each split interacts multiplicatively with at least one binary activation decision (via the big-M gating) and, in the hub split's case, with the hub's own endogenous concentration variable C_H (via Section 4.6's bilinear property balance). The three splits are therefore not separable sub-problems that could be solved independently of the discrete technology-selection decisions of Section 6.4 and then substituted into a simpler continuous model — a simplification this thesis's Section 2.6 critique of the prior

biorefinery-superstructure literature identifies as a recurring and consequential shortcut, and one this chapter's formulation is constructed specifically to avoid.

6.4 Discrete Technology and Line Activation: Semicontinuous Coupling and Minimum-Scale CAPEX

Every candidate technology j in the superstructure — each processing route within a fraction train, each hub-output platform — carries a binary activation decision y_j and a fixed-charge cost structure coupling that binary to the technology's installed capacity and capital cost, following the classical fixed-charge/semicontinuous modeling pattern documented across the MILP and MINLP process-synthesis literature (Grossmann, 2002, Section 3; Duncan & Reimer's asset-balance treatment of Chapter 35, applied here at the level of an individual candidate technology rather than a whole plant). Two constraints jointly enforce this coupling. First, the semicontinuous flow bound: $0 \le F_j \le V_j$ and $V_j \le V_j^{max} \cdot y_j$, which forces the installed capacity V_j, and hence any flow through technology j, to zero whenever $y_j = 0$, and permits V_j to range up to its maximum permitted scale V_j^{max} whenever $y_j = 1$. Second, the minimum-scale bound: $V_j \ge V_j^{min} \cdot y_j$, which prevents the optimization from activating a technology ($y_j = 1$) while installing a capacity so small that the technology's fixed-charge cost component could not plausibly be justified — a constraint with real economic content, since Section 6.6's objective function would otherwise have no structural reason to avoid installing a technology at an arbitrarily small, uneconomical scale merely to draw a small allocation of flow, a degenerate solution the minimum-scale bound is specifically constructed to rule out.

The capital-cost contribution of technology j to the objective function of Section 6.6 is then $CAPEX_j = K_j \cdot y_j + c_j \cdot V_j$, the standard fixed-charge cost form: a fixed component K_j incurred whenever the technology is built at all, plus a variable component c_j proportional to

installed capacity, consistent with the economy-of-scale cost structure Chapter 4 (Section 4.1) and the broader chemical-process-economics literature (Duncan & Reimer, Chapter 35) both document. Because K_j is multiplied by the binary y_j rather than added unconditionally, an inactive candidate technology correctly contributes zero capital cost to the objective, and the discrete technology-selection problem is, in this specific and precise sense, a capital-budgeting problem embedded within the larger flow-allocation and hub-formulation problem of Sections 6.2-6.3.

This semicontinuous structure is what gives the superstructure its combinatorial character and is the specific source, together with the hub's bilinear property balance (Section 4.6, Section 6.3), of the model's overall MINLP classification: the y_j binaries interact with the continuous flow and capacity variables multiplicatively (in the semicontinuous bound) and, for the hub-bound technologies, indirectly through the bilinear hub balance as well, so that fixing all binaries reduces the remaining problem to a continuous but still nonconvex NLP (owing to the hub bilinearity), while relaxing all binaries to [0,1] produces a continuous relaxation whose quality — and hence the tightness of the big-M values chosen for V_j^{max} and the various gating constraints of Section 6.3 — directly determines how much computational effort Section 6.8's solution strategy must expend to close the resulting optimality gap.

6.5 Optional Mutual-Exclusivity Constraints

Certain subsets of candidate technologies within a fraction train are not merely optional but mutually exclusive by construction, and this thesis's superstructure represents that exclusivity explicitly as a disjunctive constraint rather than relying on the semicontinuous bounds of Section 6.4 alone to discourage (without formally preventing) the optimizer from activating both alternatives simultaneously. The clearest instance, already flagged in Section 6.1 and Section 6.3,

is the stem+core train's bromelain-recovery choice: precipitation and ultrafiltration are alternative technologies for the same separation task, drawing from the same upstream stream, and constructing both would not merely be uneconomical but would double-count the stem+core fraction's bromelain content, since the two routes are not complementary partial recoveries but competing complete-recovery alternatives applied to the same feed.

Formally, following the Boolean-logic conventions of generalized disjunctive programming (Grossmann, 2002, Section 6; Raman & Grossmann's original GDP formulation as reviewed therein), this is expressed as the disjunction Ω_{precip} ∨ Ω_{UF}, exactly one of which must be true, with each disjunct term Ω_j carrying its own local constraints (the unit-operation model of the corresponding technology, from Section 4.3) and its own cost term. The two standard reformulations of this disjunction into an algebraic MINLP constraint are the Big-M reformulation, y_{precip} + y_{UF} = 1 together with each disjunct's constraints individually gated by its own binary via a sufficiently large constant M, and the convex-hull (variable-disaggregation) reformulation, which introduces disaggregated copies of each disjunct's continuous variables and is provably at least as tight as the Big-M relaxation for any finite M (Grossmann, 2002, Section 6.2) — the same tightness trade-off this thesis's own process-systems-engineering methodological foundation (drawing on the Grossmann GDP literature broadly) establishes as the single most consequential modeling choice for any disjunctive constraint of this kind. This thesis's Section 6.8 solution-strategy discussion returns to this choice explicitly, since the exclusivity constraint's reformulation quality interacts with, rather than sits independently of, the hub's own McCormick-relaxation and spatial branch-and-bound treatment: a loose Big-M on the exclusivity disjunction can degrade the overall relaxation's tightness even where the hub's own bilinear relaxation is itself tight.

Two further candidate-technology subsets carry a weaker, non-exclusive form of interaction that this thesis's superstructure represents through ordinary capacity and utility-sharing constraints rather than a disjunction: the pulp train's three lines (Section 4.2, Section 6.3) can coexist at any nonzero combination of activations, since they are complementary product-differentiation strategies rather than competing solutions to the same separation task, and the four hub-output platforms (Section 6.1, Section 6.3) are likewise non-exclusive, constrained jointly only by the hub's own total available pooled flow (Section 4.6's flow balance) and by Chapter 5's shared utility capacity (Section 5.3), not by any structural requirement that only one platform be built.

6.6 Objective Functions and the ε-Constraint Strategy for the NPV-GWP Pareto Front

This thesis evaluates the superstructure against two objectives established in Chapter 1 as jointly constitutive of the dissertation's research problem and formally introduced in the multiobjective-optimization framework reviewed in Section 2.9: the economic objective, expected Net Present Value (NPV) over the project horizon, aggregating the revenue, operating-cost, and capital-cost (Section 6.4) terms across all active technologies and output platforms; and the environmental objective, cradle-to-gate Global Warming Potential (GWP) per unit of combined output value, computed from the life-cycle inventory Chapter 7 develops independently of this chapter's optimization formulation. These two objectives are not merely difficult to maximize simultaneously in practice; they are formally in tension by construction, since higher-NPV configurations plausibly favor higher-throughput fossil-diesel-backed operation (Section 5.2) at the margin, while lower-GWP configurations plausibly favor configurations that minimize diesel-backup reliance even at some economic cost — the specific trade-off this thesis's Section 1.3

research question anticipates and this section's Pareto-front generation is designed to characterize rather than resolve by an arbitrary a priori weighting.

This thesis adopts the ε-constraint method (Mavrotas, 2009), and specifically its augmented variant (AUGMECON), for Pareto-front generation, in preference to the weighted-sum method reviewed and rejected for this purpose in Section 2.9. The weighted-sum method's well-documented failure mode — its inability to locate Pareto-optimal points lying in a nonconvex region of the objective space, however the weights are chosen (Kim & de Weck, 2005; Marler & Arora, 2010, as reviewed in Section 2.9) — is not a hypothetical concern for this thesis's specific problem: the hub's bilinear nonconvexity (Section 4.6, Section 6.3) propagates directly into the feasible region over which both the NPV and GWP objectives are defined, so a nonconvex Pareto frontier cannot be ruled out a priori, and a method whose guarantees depend on convexity is not a safe default choice here even setting aside the ε-constraint method's independently documented general advantages in Pareto-front resolution and computational efficiency (Mavrotas, 2009).

Formally, the ε-constraint method solves a sequence of single-objective MINLPs of the form max NPV subject to GWP $\leq \varepsilon$, for a systematically varied sequence of ε values spanning the range between the environmental objective's value at the NPV-optimal solution and its value at the GWP-optimal solution (the two anchor points of the Pareto front), with the augmented variant's second-objective slack term added to the primary objective to guard against generating weakly, rather than properly, efficient solutions — a documented failure mode of the unaugmented ε-constraint method that Mavrotas (2009) specifically addresses. Each ε-parameterized instance is itself a full instance of this chapter's MINLP superstructure model (Sections 6.1-6.5), so the Pareto front's computational cost scales with the number of ε grid points times the per-instance solution

cost characterized in Section 6.8 — a scaling relationship this thesis's Section 6.9 verification protocol explicitly plans to characterize empirically rather than assume favorable.

6.7 Robustness Formulation: Expected NPV and CVaR under Price and Yield Uncertainty

Section 5.4 established, and this section reiterates as a standing methodological commitment rather than a point made once and then set aside, that this thesis treats the energy island's operability under feedstock-parameter uncertainty (the Flexibility Index test) as categorically distinct from the superstructure's economic performance under market and yield uncertainty (this section's subject): the former asks whether a fixed design can always be operated feasibly, independent of cost; the latter asks what configuration performs best economically, in expectation or under a risk-adjusted criterion, given that prices and yields are not known with certainty at design time. Conflating the two — treating a design's expected profitability under price uncertainty as evidence of its operational robustness, or treating its Flexibility Index as a substitute for an economic risk assessment — is precisely the error this thesis's methodological framework, established in Chapter 1 and restated in Section 5.4, is constructed to avoid.

This thesis's superstructure carries price uncertainty (in the four output platforms' product prices and the diesel-backup fuel price) and yield uncertainty (in the bench-scale recovery and conversion yields carried forward from Chapter 3 and Chapter 4, several of them explicitly flagged in Sections 4.3 and 4.5 as parameters with wider uncertainty bands owing to reliance on analogous rather than pineapple-specific bench data) as a finite set of discrete scenarios $s \in S$, each with an associated probability p_s, following the two-stage stochastic programming framework of Section 2.9 and the broader stochastic-programming literature it draws on (Grossmann's L-shaped-method treatment, as reviewed in Section 2.9): here-and-now decisions (the binary technology-selection and capacity-installation decisions of Section 6.4, which must be fixed before uncertainty resolves,

since construction cannot be deferred scenario-by-scenario) and wait-and-see recourse decisions (the continuous flow-allocation splits of Section 6.3 and the diesel-backup dispatch of Section 5.2, which can be adjusted within the fixed installed capacity once a scenario's actual prices and yields are observed).

The expected-NPV formulation of the economic objective is then max $E_s[NPV(y, V, x_s; s)] = \Sigma_{s\in S}\ p_s \cdot NPV_s$, where x_s denotes the scenario-specific recourse decisions and NPV_s the resulting scenario NPV given the shared here-and-now decisions (y, V). This is the natural risk-neutral extension of Section 6.6's deterministic NPV objective and is the formulation this thesis treats as the baseline economic-uncertainty treatment. Where a risk-averse rather than risk-neutral criterion is warranted — and this thesis's Section 1.2 framing of energy autonomy as a categorical requirement, not merely a preference, suggests that a downside-weighted criterion may better reflect the actual decision context of a Bajo Atrato-scale investment than a pure expectation would — this thesis additionally formulates the objective using Conditional Value at Risk (CVaR) at a specified confidence level (Rockafellar & Uryasev, 2000; surveyed by Filippi et al., 2019, for uses well beyond finance, energy and supply-chain applications among them): $CVaR_\beta(NPV) = \zeta - (1/(1-\beta)) \cdot \Sigma_{s\in S}\ p_s \cdot \max(0, \zeta - NPV_s)$, where ζ is an auxiliary variable representing the value-at-risk threshold and β the confidence level, jointly optimized alongside the superstructure's own decision variables following Rockafellar and Uryasev's (2000) linear reformulation of the CVaR objective, which keeps this addition from introducing any nonlinearity beyond what the hub's own bilinear formulation (Section 4.6) already contributes.

[NOTA — citas del Cap. 6 (Consensus, 3-nov). El artículo de 2000 en Journal of Risk (doi:10.21314/jor.2000.038) es de Rockafellar y Uryasev; la lista de referencias lo atribuye solo a «Uryasev, S.»: corrige la entrada. Filippi et al. (2019) revisan aplicaciones en cadena de suministro, programación, redes, energía y medicina; «process-systems economic risk assessment» no aparece en el resumen. Confirmados: Duran y Grossmann (1986), Sahinidis (1996), Bongartz y Mitsos

(2017), Corsano et al. (2004, 2006b, 2007; 2006a: superestructura NLP de fermentadores; 2006c: complejo multiplanta frente a planta multipropósito, con levadura torula, levadura de panificación y brandy). Tsoukalas y Mitsos (2014) no apareció directamente, pero un artículo de 2023 confirma sus relajaciones McCormick multivariadas.]

CVaR is deployed here specifically because it penalizes the lower tail of the NPV distribution — the scenarios in which prices are unfavorable and yields disappoint simultaneously — rather than only the distribution's mean, which is the property that makes it the natural companion to expected-NPV for a risk-sensitive design decision without requiring the modeler to specify a full utility function, consistent with the broader case for CVaR's applicability outside pure portfolio-finance contexts documented by Filippi et al. (2019). This thesis reports both the expected-NPV-optimal and the CVaR-optimal configurations for the same ε-constraint grid of Section 6.6, rather than committing to a single risk posture a priori, since the two configurations' possible divergence — a design that performs best in expectation but poorly in the CVaR tail, or vice versa — is itself informative to a Bajo Atrato investment decision in a way that either metric reported alone would not be.

6.8 Nonconvexities, Linearization, and Solution Strategy

This chapter's superstructure is, by construction, a nonconvex MINLP: nonconvex owing to the hub's bilinear property balance (Section 4.6, Section 6.3), and mixed-integer owing to the technology-selection and route-exclusivity binaries of Sections 6.4-6.5. Nonconvexity is consequential in a way that simply having integer variables is not: a convex MINLP (linear or convex-nonlinear objective and constraints, with the nonconvexity confined entirely to the integer restrictions) is solvable to global optimality by Outer-Approximation (Duran & Grossmann, 1986), Generalized Benders Decomposition, or the Extended Cutting-Plane method, each exploiting the same alternating-subproblem structure reviewed in Section 2.7 and Grossmann's unified treatment

(as reviewed in Section 2.7), but the presence of a genuinely nonconvex continuous relaxation — as the hub's bilinear term introduces — means that a local NLP solver embedded within any of those methods' subproblems can converge to a local, rather than global, NLP solution, and, more consequentially, that the methods' own convergence guarantees to a global MINLP optimum no longer hold without additional machinery.

This thesis's solution strategy therefore combines two complementary techniques rather than relying on either alone. First, McCormick relaxation (as originally developed and extended to the multivariate case relevant here by Tsoukalas & Mitsos, 2014, and to reduced-space process-flowsheet optimization specifically by Bongartz & Mitsos, 2017) replaces the hub's exact bilinear term C_H · Σ_o F_{H,o} with a pair of convex/concave envelope constraints — valid outer bounds on the bilinear term's value given interval bounds on each of its two factors — yielding a convex (indeed, linear, given McCormick's envelopes are themselves linear inequalities) relaxation of the otherwise nonconvex hub balance. Because a fixed McCormick relaxation's tightness degrades as the underlying variables' bounding intervals widen, this thesis further adopts a piecewise-McCormick refinement specifically developed for bilinear pooling-type problems by Castro (2015b), partitioning each bilinear term's variable domain into sub-intervals and selecting, via additional binary variables, which partition is active — trading a larger but tighter MILP relaxation for the single, looser relaxation a non-piecewise McCormick treatment would otherwise require.

[NOTA: «as originally developed» no cita a nadie. Las envolventes de un término bilineal son las de McCormick (1976, Mathematical Programming 10:147-175), que no está en la lista. Para un producto de dos variables basta la versión original; la extensión multivariada de Tsoukalas y Mitsos (2014) sirve para funciones compuestas. Considera citar McCormick (1976) y dejar Tsoukalas y Mitsos como extensión. No verificado en Consensus: es conocimiento estándar.]

Second, where the piecewise-McCormick relaxation's remaining gap does not close to within this thesis's specified optimality tolerance through Outer-Approximation iteration alone,

spatial branch-and-bound (Quesada & Grossmann, 1995, applying the same bilinear-relaxation machinery to their original pooling-problem formulation; the global-optimization framing reviewed generally in Section 2.7) bisects the offending variable's domain and re-solves the resulting tighter sub-relaxations recursively, converging in the limit to a certified global optimum of the original nonconvex problem, at the cost of a combinatorial growth in the number of relaxation sub-problems that must be solved as the domain is refined. Figure 6.2 depicts the resulting solution-strategy loop: an Outer-Approximation master/subproblem iteration, following Duran and Grossmann's (1986) original algorithm and Quesada and Grossmann's (1992) LP/NLP-based branch-and-bound refinement of it, with the McCormick-relaxation-plus-spatial-branch-and-bound machinery specifically handling the hub's bilinear term at each iteration where the standard OA linearization alone would not suffice.

[NOTA — LIBROS (2/5), el hub como pooling/blending. (a) Grossmann (2021, §9.4, p. 89, Fig. 9.4): en problemas no convexos las linealizaciones de OA pueden cortar el óptimo global; el lazo OA + McCormick de la Fig. 6.2 no garantiza optimalidad global por sí solo, y eso es lo que justifica BARON (Apéndice A) con branch-and-bound espacial: dilo explícitamente. (b) Grossmann (§12.3, pp. 111-113): las envolventes de McCormick (cita a McCormick, 1976) son exactas solo en los límites; deriva límites ajustados de flujo y concentración desde el Cap. 4. (c) Maravelias (2021, §11.1.2, p. 263): el hub usa la formulación p (concentración); existe la q (proporcional), que conviene probar en 6.9, y §11.3 (pp. 269-274) da una aproximación MILP por discretización como verificación cruzada. (d) Maravelias (p. 264): el pooling es estático y el blending es su extensión multiperíodo; es la formulación exacta de la estacionalidad que 3.3 trata como incertidumbre. Para instancias de prueba en 6.9: Grossmann, Apéndice B (pp. 171-179), problemas de pooling de Misener y de Castro.]

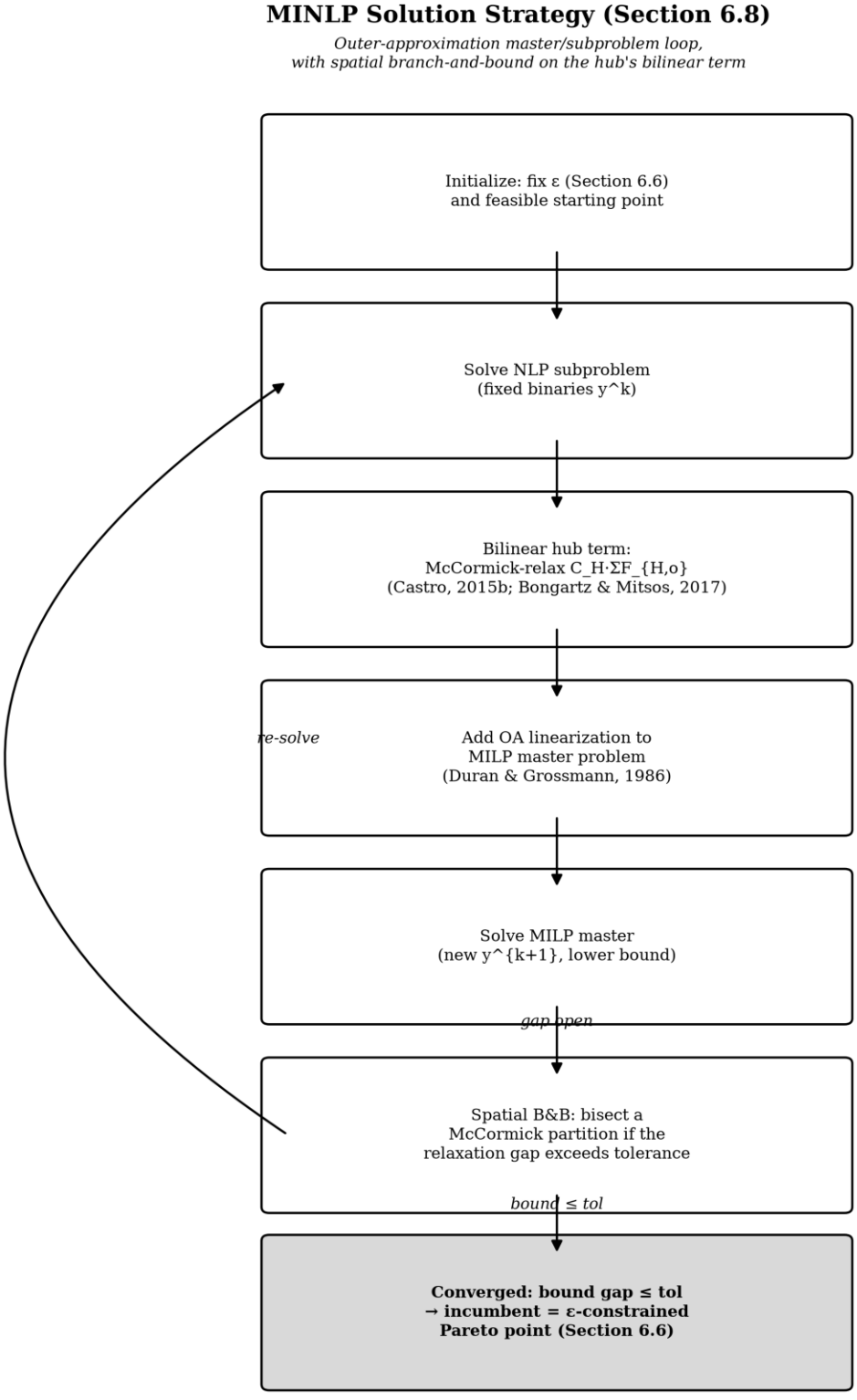

MINLP Solution Strategy (Section 6.8)
Outer-approximation master/subproblem loop,
with spatial branch-and-bound on the hub's bilinear term
Initialize: fix ε (Section 6.6)
and feasible starting point
Solve NLP subproblem
(fixed binaries y^k)
Bilinear hub term:
McCormick-relax C_H·ΣF_{H,o}
(Castro, 2015b; Bongartz & Mitsos, 2017)
re-solve
Add OA linearization to
MILP master problem
(Duran & Grossmann, 1986)
Solve MILP master
(new y^{k+1}, lower bound)
gap open
Spatial B&B: bisect a
McCormick partition if the
relaxation gap exceeds tolerance
bound ≤ tol
Converged: bound gap ≤ tol
→ incumbent = ε-constrained
Pareto point (Section 6.6)

Figure 6.2. The MINLP solution strategy this thesis adopts: an Outer-Approximation master/subproblem loop (Duran & Grossmann, 1986), with McCormick relaxation and spatial branch-and-bound applied to the hub's bilinear property-balance term whenever the relaxation gap exceeds the specified tolerance.

This is an original diagram developed for this thesis, synthesizing the Outer-Approximation algorithm of Duran and Grossmann (1986) and the piecewise-McCormick/spatial branch-and-bound treatment of bilinear pooling terms developed by Quesada and Grossmann (1995) and Castro (2015b) into a single solution-strategy flowchart specific to this thesis's hub formulation.

This thesis's computational implementation adopts BARON (Sahinidis, 1996) as the global-optimization solver of record for the resulting nonconvex MINLP, on the strength of BARON's built-in branch-and-reduce algorithm — combining the McCormick-relaxation and spatial-branch-and-bound machinery described above with automatic range-reduction techniques the software applies internally — rather than a from-scratch implementation of the Outer-Approximation/McCormick loop of Figure 6.2, which this thesis's own solution-strategy discussion above is intended to make legible and auditable even though the actual computation is delegated to established, independently validated global-optimization software rather than a bespoke solver. Where BARON's default settings prove insufficient to close the optimality gap within a practical time budget at the full superstructure's scale, this thesis's fallback, consistent with the general MINLP solution-strategy literature (Grossmann, 2002, Section 8), is to relax the global-optimality requirement to a specified, reported optimality gap rather than an unreported or silently accepted local solution — a distinction Section 6.9's verification protocol makes an explicit reporting requirement, not an implementation detail left unstated.

6.9 Model Verification and Computational Performance: Protocol

This thesis is, at the present stage, a design and methodology dissertation rather than a completed computational study: the superstructure model formulated in Sections 6.1-6.8 has not yet been solved end-to-end against a specific numerical parameterization, and this section accordingly states the verification protocol this thesis commits to applying once that

implementation is undertaken, rather than reporting computational results (solve times, optimality gaps, realized NPV or GWP values) that do not yet exist. Reporting such figures prematurely, or by extrapolation from a partial or simplified instance, would misrepresent the model's actual verified behavior and would violate this thesis's own methodological discipline, stated repeatedly across Chapters 3-5, of carrying only what has actually been established rather than what would be convenient to claim.

The verification protocol has three tiers. First, degrees-of-freedom and structural checks: confirming, for the fully specified model, that the count of independent equality constraints does not exceed the count of decision variables in a way that would render the model over-determined, following the systematic degrees-of-freedom accounting procedure Duncan and Reimer (Chapter 32) establish for mass-balance models generally and this thesis extends to the full MINLP's combined mass-balance, cost, and logical-constraint structure; and confirming that every binary variable's semicontinuous and disjunctive gating constraints (Sections 6.4-6.5) correctly force all coupled continuous variables to zero in the corresponding inactive case, checked by direct substitution rather than solver output alone, since a solver could in principle return a numerically-near-zero but nonzero value that a purely output-based check might not catch.

Second, unit and sub-block validation: solving each fraction train's technology-selection sub-problem in isolation (fixing the other three trains' decisions at a representative feasible point) and confirming that the sub-problem's optimal solution reproduces the qualitative technology choice the corresponding bench-scale literature of Chapter 3 and Chapter 4 would predict at the literature's own reported operating conditions — for instance, confirming that the stem+core sub-problem selects ultrafiltration over precipitation at parameter values matching the yield and purification advantages Nor et al. (2016, 2018) report for ultrafiltration, and precipitation at

parameter values matching the lower-auxiliary-input advantage Campos et al. (2019) report for that route — a sub-block sanity check that does not by itself certify global optimality of the full superstructure but does certify that no unit-level model formulation error is silently driving an implausible technology selection.

Third, small-instance validation against known pooling-problem benchmarks: because this thesis's central nonconvexity is a pooling problem in the specific mathematical sense established by Quesada and Grossmann (1995) and Misener and Floudas (2009), this thesis's solution-strategy implementation (Section 6.8) will be validated first against small published pooling-problem instances with known global optima from that literature, before being applied to this thesis's own full-scale hub with its problem-specific parameter values — a standard verification step in the global-optimization literature generally (the benchmark-library validation approach Bonami et al. and the broader MINLP-solver literature, as reviewed in Section 2.7, apply to solver development) and one that isolates any implementation error in this thesis's own McCormick-relaxation and spatial-branch-and-bound treatment from any error specific to this thesis's own problem instance. Only once all three verification tiers pass will this thesis report computational performance (solve time, optimality gap, and the resulting Pareto front of Section 6.6) as an established result rather than a projected one; Chapter 8's techno-economic assessment and Chapter 9's synthesis both depend on that reporting being genuine rather than anticipated, and neither chapter's own claims should be read as assuming a specific numerical outcome of this verification protocol in advance of its actually being carried out.

One reduced instance of the model has nonetheless been solved and taken through an adapted version of this protocol, and it is reported here because its results bound what the full model can claim. The instance keeps six product lines of the pulp and stem+core trains (wine,

premium wine, microfiltered juice, vinegar, press-cake flour, and bromelain), the shift and operator decisions, and an integer unit count for every equipment item, but it omits the hub's bilinear balance, the energy island, and the GWP objective; its yields and costs come from the pulp train's techno-economic workbook (TEM v10). Because Bonmin returns local optima on this nonconvex problem, each of its 112 structural combinations (240 with premium wine) was solved separately and the best retained: a certificate exhaustive over structure but local within each combination. Its evaluator reproduces twelve cases measured in the workbook to within 0.08 %, and 38 solver-free checks confirm, by direct substitution as the first tier requires, that every certified design satisfies the balances and market caps and that no design of the weekly harvest-window extension exceeds its annual bound.

What the reduced instance establishes is narrow but concrete. Under the excise tax of Law 1816 of 2016, no configuration is profitable: the best plant loses USD 3.8 million in NPV. Without the tax, wine with bromelain reaches USD 2.3 million, and premium wine at USD 15 per litre turns the taxed case positive (USD 1.5 million). Checked against two independent measurement studies of pineapple by-products (Campos et al., 2020; Seguí Gil & Fito Maupoey, 2018), five of the instance's parameters lie outside the measured ranges. Propagating those ranges leaves the wine line profitable in every one of 2,000 samples, but not the bromelain line: at USD 30 per kg it pays only if ultrafiltration concentrates the stem+core liquor at least eightfold, against a factor near two measured at laboratory scale. The verification also caught two errors in the certificate itself, combinations left unsolved by an interrupted run and a local optimum both starting points had missed, neither of which the solver reported. The reduced instance therefore supports this section's premise, that a solved number is only as strong as the verification and the evidence behind it, rather than substituting for the full computation this section commits to.

[NOTA: cifras de la instancia reducida v10 (5-nov-2026). Sin impuesto = escenario S1 (vino 600 kL + bromelina 19,5 t, 1 turno, ESAL); premium a 15 USD/L = S6 (vino 469 kL + premium 150 kL + bromelina). Los cinco parámetros fuera de rango: pulpa 0,35 (medido 0,48–0,50), PCC 0,65 con corona (0,37–0,52), líquido del PCC 0,55 (0,578–0,684), actividad del polvo 1.446 GDU/g (340–805) y eficiencia de fermentación 0,80 (0,70–0,75). El umbral exacto es una fracción de retenido de la UF de 0,125 (factor 8; polvo de 1.290 GDU/g); Seguí midió 0,53. A 60 USD/kg la bromelina se sostiene en todo el rango medido. Si la tesis debe citar las tablas, están en el capítulo LaTeX (Tablas tab:res, tab:evid y tab:unc). Falta decidir si las coronas llegan a la planta (cambia la fracción de PCC de 0,65 a ≈0,52).]

6.10 Benchmarking This Chapter's Modeling Paradigm: Corsano et al.'s NLP Superstructure and Multiplant Complex Framework

This chapter's modeling choices — binary-variable technology selection (Sections 6.4-6.5) combined with McCormick-relaxation and spatial-branch-and-bound treatment of the hub's bilinearity (Section 6.8) — are not the only superstructure paradigm documented in the process-systems-engineering literature for problems of this general class. Corsano, Montagna, Iribarren, and Aguirre (2009) develop and apply, across a monograph synthesizing their own peer-reviewed research program (Corsano et al., 2004, 2006a, 2006b, 2006c, 2007), two related but distinct superstructure frameworks for batch and semicontinuous agro-industrial process synthesis, worked through a Sugar Cane Complex case study — a shared upstream mother plant (a Sugar plant) feeding derivative product lines (Torula Yeast and Brandy/Bakery Yeast) via shared material and energy resources. That case study's structure — one shared upstream resource-producing stage feeding several downstream, technologically distinct derivative product lines — is close enough to this thesis's own hub-and-outputs architecture (Section 6.1) to serve as a genuine benchmark for this chapter's own modeling paradigm, not merely a superficially similar analogy drawn from a different problem class.

Corsano et al.'s (2009) first framework, developed for the synthesis, design, and operation of a single batch plant (their Chapter 3, addressing a fermentor-network case study), embeds every candidate structural alternative — the number of processing stages in series, and their in-phase or out-of-phase duplication — directly into one continuous nonlinear program, and relies on the unit-sizing cost function's concavity (a capital-cost exponent strictly less than one, the same economy-of-scale structure this thesis's own CAPEX treatment uses; Section 6.4) to drive every non-optimal alternative's installed size to zero at the optimum. This lets their model avoid binary variables, and hence MINLP branch-and-bound, entirely for that class of decision. Their second framework (their Chapter 5), the Multiplant Complex (MC), formalizes the mother-plant/derivative-plant architecture itself: an explicit produced/supplied/exported resource balance across all derivative plants, jointly optimized alongside each derivative plant's own synthesis, design, and operation decisions, with no decomposition or heuristic sequencing applied.

Set against this benchmark, three aspects of this chapter's own paradigm are better suited to this thesis's specific problem structure than a direct application of Corsano et al.'s (2009) techniques would be. First, this chapter's binary/MILP treatment of technology selection (Sections 6.4-6.5) correctly represents a genuine fixed-charge decision — a lump-sum capital cost K_j incurred if, and only if, technology j is built at all — that Corsano et al.'s binary-free mechanism cannot represent correctly: their concave-cost argument drives a continuous SIZE variable to zero when an alternative is suboptimal, which is mathematically sound for exactly the class of decision they apply it to (how many units, of what size, within an already-chosen technology) but does not extend to a discrete choice AMONG technologies each carrying its own fixed charge, since a fixed charge relaxed to a continuous [0,1] activation variable is linear, not concave, in that variable, and a linear relaxation has no structural reason to self-select an extreme point the way a concave power-

law cost does. This distinction is not one Corsano et al. (2009) themselves draw explicitly — their own applications never required it, since their case studies pose within-technology unit-duplication decisions, not between-technology fixed-charge choices of the kind this thesis's stem+core bromelain-recovery disjunction (Section 6.5) and pulp-train routing decision represent. Second, this chapter's hub confronts a genuine bilinear nonconvexity (a pooling problem, Section 6.3, Section 6.8) with a global-optimization treatment carrying a certifiable optimality gap (McCormick relaxation, spatial branch-and-bound, and BARON's branch-and-reduce algorithm); Corsano et al.'s (2009) own case studies are solved with CONOPT2, a local NLP solver, inside GAMS, so their reported optimal solutions carry no global-optimality guarantee — a limitation their own text does not flag, and one this thesis's Section 6.9 verification protocol is specifically constructed to avoid repeating. Third, this chapter integrates an explicit environmental objective and a Pareto-front treatment (Section 6.6) and a formal economic-uncertainty treatment under CVaR (Section 6.7); Corsano et al.'s (2009) models throughout are single-objective (annualized profit or total annual cost) and deterministic, with neither multiobjective trade-offs nor parameter uncertainty within their scope.

Set against the same benchmark, two aspects of Corsano et al.'s (2009) approach expose genuine, disclosed gaps in this thesis's own treatment rather than confirming this chapter's paradigm as unconditionally superior. First, Corsano et al. (2009) discretize their derivative plants' unit-level dynamics (fermentor mass-balance differential equations) directly into the SAME nonlinear program that decides plant structure and sizing, so that batch cycle time and reactor volume are optimized jointly, in one simultaneous solve. This thesis's own multiscale coupling for the pulp train's fermentation and acetification steps (Section 4.2; implemented in the reduced-scope computational proof-of-concept's kinetics and reactor-sizing modules, see Chapter 11's

discussion of computational execution as future work) is HIERARCHICAL rather than simultaneous: the dynamic kinetic simulation is solved first and reduced to a fixed scalar cycle time and titer, and reactor volume or train count is sized afterward against that fixed value, which cannot capture any trade-off between cycle time and reactor size the underlying kinetics might in principle admit. No such trade-off is evident in this thesis's current first-order-approach kinetic models (Section 4.2), but a richer kinetic model fitted to a fuller experimental time course could reveal one this thesis's current architecture would not detect. This narrowing is adopted here for solver tractability — this thesis's spatial-branch-and-bound treatment of the hub's own bilinearity (Section 6.8) is already a substantial computational undertaking, and embedding additional discretized differential-algebraic constraints directly into that same nonconvex MINLP would compound rather than merely add to its difficulty — not because the simultaneous approach is judged methodologically wrong.

This narrowing is not merely a limitation this chapter apologizes for; it is also, independently, a named methodology in its own right. Kallrath (2021, Ch. 14) formalizes "Polylithic Modeling and Solution Approaches" (PMSAs): solving a mixed-integer or nonconvex problem intractable as a single monolithic model instead as a SEQUENCE of linked models, in which "model m+1 can use results of the first m models" to initialize variables or tighten bounds, explicitly recommended "when the problem cannot be solved directly." This thesis's own dynamic-kinetics-to-design-point-to-capacity-to-MINLP chain (Section 4.2; the accompanying computational proof-of-concept) is a direct instance of this pattern, not an ad hoc shortcut, and Kallrath's own stated criterion for when PMSAs are the right choice matches this chapter's own stated reason above (Section 6.8's spatial-branch-and-bound treatment of the hub's bilinearity is already substantial; compounding it with embedded differential-algebraic constraints was avoided

on tractability grounds identical to Kallrath's). This second, independent precedent does not resolve the genuine trade-off against Corsano et al.'s (2009) simultaneous embedding identified above -- joint cycle-time/reactor-size optimization remains unavailable here -- but it does establish that the choice actually made has two, not zero, peer-reviewed methodological homes, rather than resting on this thesis's own tractability argument alone.

[NOTA: las citas textuales de Kallrath (2021, cap. 14) vienen de un libro que Consensus no indexa: verifica las palabras exactas y añade el número de página. Lo mismo vale para Corsano et al. (2009, Nova Science), de donde salen las cifras de 6.10 (~19 % de inversión, 15 frente a 9 trabajadores): Consensus solo tiene el artículo de 2006c sobre el mismo caso; si las cifras están allí, cítalo a él.]

That the hierarchical, rather than simultaneous, coupling identified above is not merely a hypothetical concern is itself demonstrated by a concrete instance this thesis's own verification work subsequently uncovered within that same dynamic-kinetics-to-capacity chain, one worth recording here rather than only in Chapter 4's own account of it (Section 4.2). Sizing the pulp train's wine-fermentation reactor count and then, in a separate, later step, sizing the downstream aging-tank farm from that fixed reactor-train decision — the same hierarchical pattern this section has just discussed at the level of dynamic kinetics feeding a fixed design point — initially carried an unexamined assumption bridging those two stages: that all parallel fermenter trains discharge their combined output into the aging-tank farm in lockstep, once per fermentation cycle, rather than in the staggered sequence a continuous upstream feedstock throughput actually produces. Left unchecked, this assumption would have silently undone the very design decision the upstream stage was adopted to secure: multiple smaller fermenters were specified specifically to shorten each vessel's own fill time and reduce must-oxidation exposure (Section 4.2), yet a lockstep aging-tank-farm model would have exposed the downstream aging tank to an equivalent, undiminished slow-fill exposure, one hierarchical step removed rather than eliminated. Correcting this to the

staggered-discharge assumption the underlying throughput logic actually implies both reduced the resulting aging-tank-farm capital cost by a real, quantified margin and, independently, resolved a previously open question — how many fermenter batches should feed each aging tank — into a specific, derived number (Section 4.2). This is offered here not as a defect specific to this thesis's own implementation, but as a general, transferable methodological caution for any hierarchical modeling chain of the kind Kallrath's (2021) Polylithic Modeling framework, and this thesis's own dynamic-kinetics-to-design-point-to-capacity chain, both rely on: fixing an upstream stage's output and passing it forward as an input to a separately solved downstream stage is a defensible tractability trade-off, per the case already made above, but each such hand-off carries its own, separate coupling assumption that must be checked explicitly, on its own terms, rather than inherited by default from whatever is most convenient to compute — precisely because a hierarchical chain, unlike a simultaneous formulation, provides no structural guarantee that an assumption convenient at one stage remains consistent with a design rationale established at another.

Second, and more consequential for this chapter's own central architectural claim: Corsano et al. (2009) explicitly benchmark their Multiplant Complex architecture (separate, dedicated derivative plants) against a Multiproduct Plant (MP) alternative (shared equipment producing multiple products in the same plant) for the identical Sugar Cane case study, and report that the two are economically close but not identical — total investment cost of $2,610,750 for the MC configuration against $2,188,500 for the MP configuration (the MP configuration approximately 19% cheaper to build), against similar total partial benefits ($3,654,000 MC versus $3,636,000 MP) — while also finding the two configurations imply materially different labor demand (15 workers for the MC configuration against 9 for the MP configuration, by their own stated staffing

assumptions), a factor neither of their cost models, nor this thesis's own, prices at all. This thesis's Chapter 6 superstructure, by contrast, commits from the outset (Section 6.1) to a single shared, hub-centered architecture and never poses, as an explicit alternative configuration to be benchmarked, the corresponding question for this thesis's own case: would a hypothetical set of separate, dedicated processing trains — one per output platform, each independently sized and costed, without a shared bilinear hub at all — outperform, underperform, or merely closely match the shared-hub architecture this chapter already commits to as its sole superstructure? Corsano et al.'s (2009) own result, a non-negligible but not decisive cost difference alongside a labor-demand difference their model does not price, demonstrates that this comparison is not resolvable by architectural intuition alone even in a structurally similar problem, and this thesis has not run the analogous comparison for its own hub. This is disclosed here as a genuine, currently open scope limitation of Chapter 6's superstructure formulation, not as a defect in the hub formulation itself (Section 6.3's case for the hub's bilinear representation, against a fixed-ratio-blending or pre-resolved-allocation simplification, remains independently sound on its own terms and is not what this comparison calls into question) but as an unexamined alternative TOPOLOGY this thesis's superstructure construction (Section 6.1) has not benchmarked against the one it adopts.

Third, and shared rather than distinguishing: labor and human-resource cost is not modeled anywhere in this thesis's Chapter 6 formulation, an omission Corsano et al. (2009) explicitly disclose in their own Multiplant Complex case study after finding, as noted above, that it is not a negligible factor in a comparison between architecturally different configurations of otherwise similar profitability. This thesis carries the same omission, undisclosed until this section, and for the same underlying reason Corsano et al. (2009) do not resolve it either: neither the MC/MP comparison literature nor this thesis's own reduced-scope treatment (Chapter 6's stated scope,

Section 6.2) treats staffing as a decision variable or even a fixed cost input, so a route or technology choice implying a materially different equipment count — this thesis's own pulp-train vinegar routes are one such instance, since the current-practice acetification route requires substantially more parallel reactor trains than the optimized-acetator alternative to reach the same annual capacity, per the accompanying computational proof-of-concept's reactor-train sizing (Chapter 11) — is not currently penalized in this thesis's objective function for the corresponding difference in labor demand, exactly as Corsano et al.'s (2009) own MC-vs-MP comparison was not.

Taken together, this benchmarking exercise is a genuine methodological contribution to this chapter's rigor, though a specific and bounded one: it subjects this chapter's modeling choices to an explicit comparison against an independent, peer-reviewed superstructure methodology applied to a structurally similar problem, confirms that this chapter's binary/MILP and global-optimization machinery is the methodologically correct choice for the specific fixed-charge and bilinear-nonconvexity structure this thesis's superstructure actually has (rather than an unexamined default), and converts two limitations that would otherwise have remained implicit — the hierarchical rather than simultaneous dynamic-to-plant coupling, and the untested dedicated-versus-shared architecture question — into explicit, disclosed scope statements. It does not, and should not be read to, certify this chapter's numerical results, which Section 6.9's verification protocol governs independently. Running Corsano et al.'s (2009) style of MC-versus-MP architectural comparison for this thesis's own hub-and-outputs case is accordingly recommended in Chapter 11 as a concrete, well-scoped future-work item in its own right, additional to and sequenced after the computational execution Chapter 11 already identifies as the most immediate priority. Section 9.6 carries this benchmark's second limitation forward into this thesis's own hypothesis-testing protocol, stating explicitly what confirming Hypothesis H1 would, and would

not, establish about this chapter's specific hub-and-outputs topology relative to the untested, more deeply integrated alternative this section has just identified; Section 11.2 records the resulting scope qualification as a limitation in its own right, not only as a future-work item.

6.11 From Formulation to Assessment

This chapter began from the claim that the shared sugar hub cannot be designed one line at a time: the profitability of each hub-fed platform depends on how much sugar the others take, a coupling that sequential design fixes before it can be seen (Section 1.4). The formulation developed here represents that coupling directly — as a bilinear pooling balance embedded in a mixed-integer program that selects technologies, splits flows, and installs capacity in a single decision, under the energy-autonomy constraint of Chapter 5 — and states in advance how the resulting nonconvex MINLP will be solved (Section 6.8) and verified (Section 6.9). What the full formulation cannot yet say is what any configuration is worth; the reduced instance of Section 6.9 says it only for six product lines without the hub, and shows how much that answer rests on two measured parameters. Its two objectives are, so far, placeholders: the GWP term awaits the life-cycle methodology of Chapter 7, and the NPV term awaits the cost and revenue content of Chapter 8. The next two chapters supply them, in that order.

Chapter 7. Life Cycle Assessment

Section 2.8 established that this thesis treats life-cycle assessment (LCA) as a dedicated methodological treatment, structurally independent of the techno-economic assessment of Chapter 8, precisely because the two disciplines' shared reliance on the same underlying mass- and energy-balance data (Section 4.8) has repeatedly tempted the prior superstructure biorefinery literature into conflating them — collapsing environmental performance into a single incidental Pareto term rather than developing it with the same methodological care as the economic model. This chapter develops that independent treatment in full: Sections 7.1-7.3 fix the goal, system boundary, functional unit, and allocation approach; Sections 7.4-7.6 specify the inventory, the impact-assessment method, and the biogenic-carbon treatment; Sections 7.7-7.8 set the diesel-only comparative baseline and the uncertainty protocol; and Section 7.9 shows how the chapter's GWP metric feeds the Chapter 6 optimization without becoming a mere input to it.

7.1 Goal and Scope Definition

Following the goal-and-scope definition requirement of ISO 14040 and ISO 14044 (Finkbeiner et al., 2006) — the first and, per the standard's own logic, most consequential step of any LCA study, since every subsequent methodological choice this chapter makes is answerable to the goal stated here — this thesis states its LCA's goal as follows: to quantify the cradle-to-gate global warming potential, and a supporting set of midpoint environmental impact categories, of the pineapple whole-plant biorefinery configurations the Chapter 6 superstructure generates, in order to (a) supply the GWP objective Section 6.6's ε-constraint Pareto generation requires, and (b) support a comparative environmental assessment of the off-grid energy-autonomous design against a conventional diesel-only reference scenario for the same product portfolio (Section 7.7).

This is an attributional, not consequential, LCA in the sense the allocation literature distinguishes (Schrijvers et al., 2016; Ekvall et al., 2001): it describes the environmental burdens attributable to this thesis's specific product system as designed, rather than the marginal consequences of a change in production volume on the wider market — a choice this thesis's Section 7.3 justifies further when it selects economic allocation, an attributional-LCA-consistent method, over system expansion, the consequential-LCA-consistent alternative.

The intended audience is twofold: this thesis's own examining committee and the broader academic literature on off-grid agro-industrial biorefineries, for whom the assessment must be methodologically transparent and independently reproducible in the sense Gaffey et al. (2024) find lacking in a majority of the 59 biorefinery LCA studies their review examined; and, secondarily, a prospective Bajo Atrato investment decision-maker, for whom the comparative diesel-baseline result (Section 7.7) is the single most decision-relevant output this chapter produces. Because this is a design-stage, not an operating-facility, assessment, this thesis's LCA is necessarily a prospective assessment built on the process models of Chapter 4 and Chapter 5 rather than on measured plant data — a limitation this chapter states explicitly here rather than allowing it to be discovered implicitly in Section 7.8's uncertainty treatment, since Vance et al. (2022) identify exactly this gap between novel-system LCA and measured-system LCA as a first-order methodological challenge for bioeconomy assessment generally, not a shortcoming specific to this thesis's own case.

7.2 System Boundary and Functional Unit

This thesis adopts a cradle-to-gate system boundary, illustrated in Figure 7.1: the assessment begins at the point the pineapple whole-plant feedstock (leaves/crown, stem+core, peel, pulp; Section 3.1) enters the superstructure of Chapter 6, and ends at the biorefinery's own

gate, with each of the superstructure's product and co-product streams as an assessed output. This boundary choice follows directly from the residue-feedstock framing Section 2.8 already established as this thesis's own case, following Saraiva's (2017) finding that residue-based systems more defensibly exclude upstream agricultural cultivation burden from the assessment boundary than dedicated-crop systems do, since the pineapple plant material processed here is a residue of an existing agricultural system (fresh-fruit production) whose own cultivation burden is not incurred because of this biorefinery investment and would be incurred, in full, whether or not this thesis's biorefinery is ever built. This thesis states that exclusion explicitly, per Gaffey et al.'s (2024) broader finding that biorefinery LCA studies too often leave such boundary decisions unstated rather than justified, and per this thesis's own standing methodological discipline (Section 4.1, Section 6.9) of making every such choice auditable rather than defaulted into silently.

Cradle-to-Gate LCA System Boundary (Section 7.2)

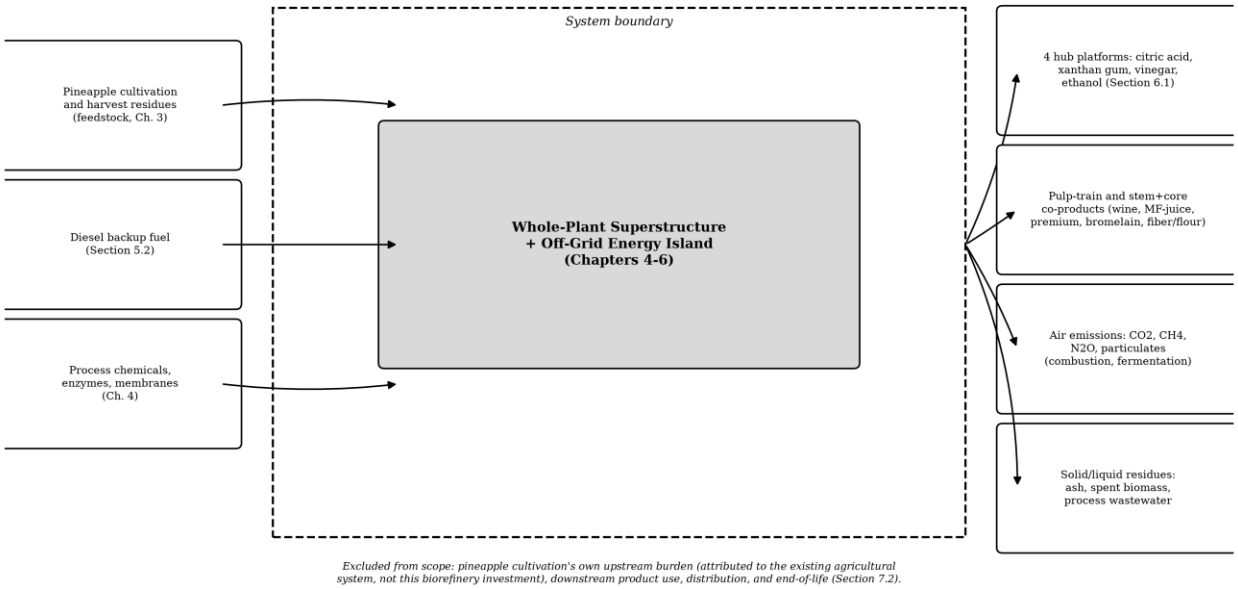


Figure 7.1. The cradle-to-gate system boundary this thesis's LCA adopts, showing the inputs crossing into and the outputs crossing out of the Chapter 4-6 superstructure and energy-island model, and the life-cycle stages explicitly excluded from scope.

This is an original diagram developed for this thesis, following the cradle-to-gate boundary convention standard in the biorefinery LCA literature (Righi, 2018; Gaffey et al., 2024) and applying it to this thesis's own superstructure structure established in Chapter 6 (Figure 6.1).

Downstream life-cycle stages — product distribution, consumer or industrial use, and end-of-life — are excluded from this thesis's boundary, consistent with the gate-to-gate-adjacent scope common across the biorefinery LCA literature Section 2.8 reviews, and justified here specifically because this thesis's four hub-output platforms and pulp-train co-products (Section 6.1) serve heterogeneous downstream markets (food ingredients, industrial biopolymers, fuel) whose use-phase and end-of-life burdens are not comparable on a common basis without a product-specific downstream study each of which lies outside this dissertation's scope; Gaffey et al. (2024) report that only 14% of the biorefinery LCA studies they reviewed extended to product end-of-life at all, confirming this thesis's cradle-to-gate choice is the field's prevailing, not an idiosyncratic, convention.

The functional unit is 1 tonne of pineapple whole-plant feedstock processed, rather than a mass, energy, or economic unit of any single output product. This choice follows directly from this thesis's own multi-output superstructure structure (Section 6.1): because the relative output mix among the four hub platforms and the pulp-train, stem+core, and peel co-products is itself an optimization output of Chapter 6, not a fixed recipe, a product-based functional unit (e.g., 1 kg of citric acid) would implicitly fix the very allocation the superstructure is designed to determine, precisely the methodological trap Sills et al. (2020) document for multi-product algal biorefineries and explicitly warn against: a fuel- or product-based functional unit can change even the sign of a comparative LCA conclusion depending on which co-product handling method is paired with it. A feedstock-input functional unit avoids this coupling, remains stable across every configuration

the Chapter 6 Pareto front generates, and is the functional-unit choice this thesis's Section 7.3 allocation treatment is specifically constructed to be consistent with.

7.3 Multifunctionality and Allocation Approach

The superstructure's multi-output structure — up to eight simultaneously marketable product and co-product streams across the four fraction trains and the shared sugar hub (Section 6.1) — makes this thesis's system a textbook multifunctional process in the LCA sense, and the allocation literature counts the choice of how to partition environmental burden among those outputs among the most debated methodological decisions in LCA, with a potentially large influence on final outcomes: Cherubini et al. (2011) show for a lignocellulosic biorefinery case study that switching among mass, energy, exergy, economic, and system-expansion allocation changes the environmental burden assigned to each product, and Amponsah et al. (2023) report a marine biorefinery's GWP ranging from 1.2-4.5 kg CO2-eq/kg under economic allocation to 14.2-41.5 kg CO2-eq/kg under system expansion without any change to the underlying process model whatsoever, and Gaffey et al.'s (2024) broader review confirms allocation-method inconsistency as one of the field's most persistent comparability failures.

[NOTA — citas de 7.3 (Consensus, 3/4-nov). Cherubini et al. (2011) dice que la asignación está «among the most debated» y tiene una influencia «potentially large», no que sea «the single most consequential decision», y su resumen no cuantifica el «several-fold»; esa magnitud la da Amponsah et al. (2023), que añadí. Grassauer/Dominguez Aldama (2023): Consensus lista como primer autor a F. Grassauer; revisa la autoría. Su 35 % es de los ACV atribucionales, no del total, y el mismo estudio señala que la asignación económica es incoherente con la jerarquía de ISO 14044 y que el enfoque más usado es «zero-burden» (40 %). Schrijvers et al. (2016): en un ACV atribucional, la «system expansion» es ampliar la unidad funcional; la sustitución de un producto desplazado es propia del ACV consecuencial. Tu sensibilidad por «system expansion» con producto desplazado (7.3) es, en sus términos, sustitución: decláralo o justifícalo.]

This thesis adopts economic (revenue-based) allocation as its primary approach, partitioning each fraction train's and the hub's shared upstream burden among its co-products in proportion to each product's share of total revenue at the biorefinery gate, using the same market-price data Chapter 8's techno-economic assessment independently develops (a shared data dependency this thesis states explicitly rather than treats as a methodological coincidence, since it means Chapter 7's allocation keys and Chapter 8's revenue model are constructed from a single consistent price dataset, not two independently assembled ones). Economic allocation is the ISO 14044 hierarchy's least-preferred partitioning method relative to system expansion (Ekvall et al., 2001; Weidema, 2010) — system expansion's core advantage being that it preserves the product system's physical mass, energy, and carbon balances intact, which any partitioning allocation, economic allocation included, generally does not (Weidema, 2010) — but this thesis selects it deliberately over system expansion for three stated reasons rather than by default. First, system expansion requires identifying a credible displaced (avoided) product for each of this thesis's eight-plus co-product streams, several of which (the premium fermented-beverage line, the spray-dried nutraceutical powder) have no single, uncontroversial market substitute to displace, so system expansion's own data requirements are not well met by this thesis's specific product portfolio. Second, Dominguez Aldama et al. (2023) find economic allocation is already the dominant attributional-LCA practice for agri-food co-product and food-waste-valorization systems specifically — this thesis's own product category — at 35% of the studies their systematic review examined, making it the more directly comparable choice against the closest existing literature. Third, and most specific to this thesis's own multi-output structure, this thesis's revenue-based allocation keys are structurally consistent with the value-density screening already performed in Section 3.4 and the a posteriori value-density interpretation of Section 8.5, in the specific sense

that all three use the same underlying market-price data — though this thesis states explicitly, per the standing methodological distinction established in Section 3.4 and reiterated here, that LCA allocation, a posteriori value-density interpretation, and formal multi-criteria decision-making selection (Section 8.5) remain three analytically distinct uses of that shared price data, not interchangeable applications of a single value-density concept.

Because the allocation-method literature is equally clear that no single method is uniformly preferred across all goal-and-scope contexts (Cherubini et al., 2011; Schrijvers et al., 2020), this thesis reports mass allocation and system expansion (the latter applied only to those co-product streams — bromelain, fiber/flour, the pulp-train lines — for which a credible displaced product can be identified from the market data of Chapter 2 and Chapter 3) as sensitivity checks rather than treating the economic-allocation result as the sole reportable figure, following Krexner et al. (2021) and Amponsah et al. (2023), whose comparisons show results shifting materially with the allocation method, and Sills et al. (2020), who recommend presenting multiple co-product handling methods, precisely because the choice is this consequential and no single method commands field-wide consensus. Figure 7.2 illustrates the three approaches conceptually as applied to this thesis's own hub-output structure.

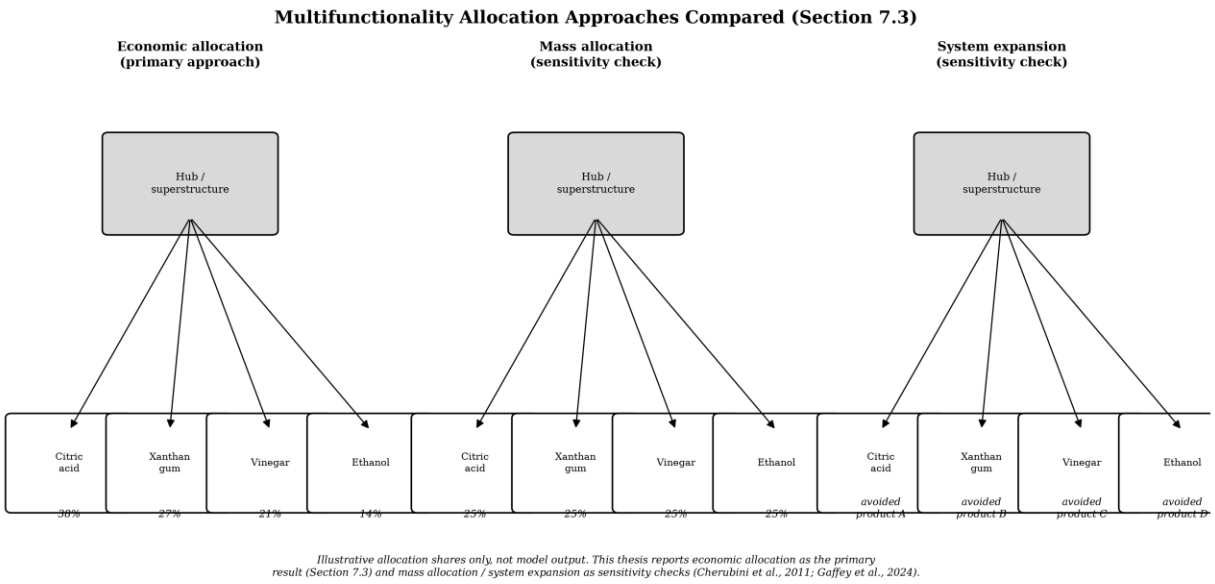


Figure 7.2. The three multifunctionality allocation approaches this thesis applies to the superstructure's shared upstream burden: economic allocation (primary result), and mass allocation and system expansion (sensitivity checks), illustrated conceptually against the four hub-output platforms.

This is an original diagram developed for this thesis, illustrating the allocation-method comparison established in the biorefinery LCA allocation literature (Cherubini et al., 2011; Ekvall et al., 2001) as applied to this thesis's own hub-output structure (Section 6.1, Figure 6.1); the percentage and avoided-product labels are illustrative only and do not represent computed model output.

7.4 Life Cycle Inventory (LCI) Development

The life-cycle inventory this thesis's LCA requires is drawn, by design rather than by convenience, from the same mass- and energy-balance data Section 4.8 already establishes for the techno-economic assessment: for each fraction train and candidate technology (Sections 4.2-4.7), the input flows of raw feedstock, process chemicals, enzymes, and membrane materials, and the utility flows of electricity and process heat drawn from the off-grid energy island (Chapter 5, Section 5.3); and, on the output side, the product and co-product mass flows Section 6.1's superstructure tracks, together with the air-emission, solid-residue, and wastewater streams each unit operation generates. This shared-data-source design is a deliberate methodological choice, not an incidental efficiency: it ensures the LCI is internally consistent with the mass and energy

balances the optimization itself enforces (Section 6.2's flow-balance constraints), rather than being separately estimated from generic literature defaults that might not respect the superstructure's own conservation laws.

Where this thesis's own process models (Chapter 4, Chapter 5) do not directly report an emission factor — the combustion emission profile of the downdraft gasifier's syngas-fired internal combustion engine (Section 5.1), or the diesel-backup generator's own combustion emissions (Section 5.2), most centrally — this thesis's LCI draws those factors from the closest available published characterization of comparable equipment at comparable scale, following the same evidentiary discipline established in Section 4.1 for pineapple-specific versus analogous bench data: syngas-fired internal-combustion-engine emission factors from the small-scale biomass-power life-cycle literature (Aberilla et al., 2019, who directly compare gasification, combustion, and anaerobic-digestion pathways against diesel generation for agricultural communities in developing-country contexts closely analogous to Bajo Atrato's own), and diesel-generator emission factors from the same comparative literature rather than from a generic transportation-diesel proxy, since stationary generator-set combustion characteristics differ materially from vehicular diesel combustion. Background (upstream) data — the embodied burden of constructing each candidate technology's equipment, and of the process chemicals, enzymes, and membrane materials consumed — will draw on established life-cycle-inventory databases at the implementation stage, with the specific database and version documented in the eventual computational report per this thesis's standing verification discipline (Section 6.9): reporting exactly which background data underlies a given result, not merely that background data was used, following the transparency the ISO 14044 standard itself requires (Finkbeiner et al., 2006) and the biorefinery-LCA literature finds too often absent in practice (Gaffey et al., 2024).

7.5 Life Cycle Impact Assessment (LCIA) Method and Impact Categories

This thesis adopts the ReCiPe 2016 method (Huijbregts et al., 2016) for life-cycle impact assessment, at the midpoint level and under the Hierarchist cultural perspective — ReCiPe's default and most broadly defensible perspective, balancing the Individualist perspective's short time horizon against the Egalitarian perspective's precautionary long horizon (van Zelm et al., 2025) — for three stated reasons. First, ReCiPe is widely applied in the comparative biorefinery and bioenergy literature this thesis draws on, including the closest multi-product biorefinery LCA (Amponsah et al., 2023), and offers the broadest set of midpoint categories among the major LCIA methods (Rashedi et al., 2020), making a ReCiPe-based result the most directly comparable to the closest existing published biorefinery assessments. Second, ReCiPe provides a full complement of midpoint impact categories beyond global warming potential alone — terrestrial and freshwater ecotoxicity, eutrophication, acidification, and fossil-resource scarcity chief among those most relevant to a gasification- and fermentation-based biorefinery — allowing this thesis to report a fuller environmental profile than a GWP-only assessment would, even though GWP alone is the specific metric Section 6.6's Pareto generation optimizes against. Third, comparative LCIA-method studies find that while absolute impact scores and, for some categories, category rankings can differ materially across methods (ReCiPe, CML, ILCD, IMPACT 2002+), the differences are most pronounced for toxicity-related and land-use categories and comparatively minor for climate change and ozone depletion (Owsianiak et al., 2014), although a comparison of European electricity mixes found ReCiPe 2016 to be the one method departing from the others on those two categories (Rybaczewska-Błażejowska et al., 2024) — the category this thesis's own Pareto-front objective depends on most directly — lending only qualified confidence that the GWP result reported here would not be qualitatively overturned by an alternative LCIA-method choice.

[NOTA — citas de 7.5 (Consensus, 4-nov). Rybaczewska-Błażejowska et al. (2024) reporta resultados coherentes en cambio climático y ozono «with the exception of the ReCiPe 2016 method», justo el método elegido: tu tercer argumento para ReCiPe necesita este matiz, o comparar el GWP con otro método como sensibilidad. Rashedi et al. (2020) estudia fotovoltaica, no biorrefinerías, y llama a ReCiPe «one of the most advanced», no el más usado «by a wide margin». Van Zelm et al. (2025, Nature Protocols) no apareció en Consensus: verifica la recomendación de punto medio y la perspectiva jerárquica en la fuente. Confirmados: Huijbregts et al. (2016), Owsianiak et al. (2014), Finkbeiner et al. (2006), Ekvall et al. (2001), Weidema (2010), Schrijvers et al. (2016, 2020), Sills et al. (2020), Kalbar et al. (2016/2017, indirectamente), Vance et al. (2022).]

This thesis reports midpoint results as the primary deliverable rather than aggregating to ReCiPe's endpoint (human health, ecosystem quality, resource scarcity) or single-score levels, following van Zelm et al.'s (2025) own methodological recommendation and the broader weighting-and-aggregation literature's documented concern that endpoint and single-score aggregation embeds value-laden weighting choices non-transparently into a single number (Kalbar et al., 2017), a concern directly analogous to, and consistent with, this thesis's own preference (Section 2.9, Section 8.5) for explicit, auditable multi-criteria decision-making over an implicit, opaque single-score collapse of multiple criteria. Where this thesis's Section 8.5 eventually requires a single environmental figure of merit for multi-criteria comparison against the economic objective, that figure is the midpoint GWP result specifically, not a ReCiPe endpoint or single score, keeping the MCDM step's own weighting choices explicit and separable from the LCIA method's internal aggregation.

7.6 Biogenic Carbon Accounting Treatment

A biomass-fed system's global warming potential result depends materially on how biogenic carbon — the CO2 released when biomass combusts or decomposes, having previously been fixed from atmospheric CO2 during the plant's growth — is accounted for, and this thesis

states its treatment explicitly rather than defaulting silently into the conventional carbon-neutrality assumption the LCA literature has increasingly, and for good reason, challenged. The neutrality assumption treats biogenic CO2 emissions as climate-neutral on the reasoning that an equivalent quantity of atmospheric carbon was fixed during the feedstock's growth, so that the two exactly offset within the assessment's system boundary; Wiloso et al. (2016) demonstrate this assumption is not a conservative simplification but an active source of bias, one whose direction and magnitude depend on system-boundary framing (cradle-to-gate versus cradle-to-grave), the specific form of carbon emitted (CO2 versus methane, which is not accounted for by a simple neutrality offset), and the relative economic valuation used to allocate biogenic carbon between co-products — three factors this thesis's own multi-output, cradle-to-gate system implicates directly, not hypothetically.

This thesis's specific case has one structural feature that substantially, though not completely, mitigates the biogenic-carbon-accounting concern documented for forestry and dedicated-energy-crop bioenergy systems specifically (Zanchi et al., 2012; Booth, 2018; Liu et al., 2018): the leaves/crown fraction gasified in the energy island (Section 5.1) is an annual-crop agricultural residue, not standing forest biomass or a long-rotation energy crop, so the carbon-debt payback-time concern central to the forest-bioenergy carbon-neutrality critique — the multi-decade lag between a harvested tree's combustion emission and its replacement growth's re-absorption — does not apply in the same form here: pineapple's annual-to-biennial cultivation cycle (Chapter 3) re-fixes the carbon released by one harvest cycle's gasification within, at most, a few years, not the decades-to-centuries timescale forest bioenergy's critics document (Alessandro et al., 2014). This thesis nonetheless does not invoke this structural difference as license to adopt an unexamined neutrality assumption: following Wiloso et al.'s (2016) recommended complete-inventory approach, this thesis's LCI (Section 7.4) tracks biogenic carbon flows explicitly —

carbon fixed during feedstock growth, carbon released at gasification and at fermentation off-gassing, and carbon retained in unconverted residues (ash, spent biomass) — rather than netting them to zero by assumption, so that the eventual GWP result is auditable against the underlying carbon balance rather than resting on an assumption this thesis's own system characteristics happen to make less consequential but do not eliminate.

[NOTA — citas de 7.6 (Consensus, 4-nov). El informe del JRC (doi:10.2788/2960) no aparece en Consensus, y «Alessandro» es probablemente el nombre de pila del primer autor: la cita habitual es Agostini, Giuntoli y Boulamanti (2013/2014). Corrige la entrada si se confirma. Booth (2018) analiza justamente residuos quemados para bioenergía y pone como condición que su destino alternativo se descomponga pronto: tu argumento del residuo anual encaja con esa condición, pero cítala así, no solo como crítica a la bioenergía forestal. Zanchi et al. (2012) apoya que los residuos de descomposición rápida dan beneficios desde el inicio. Wiloso et al. (2016), Liu et al. (2018) y Aberilla et al. (2019) confirmados.]

The diesel-backup fuel's combustion emissions (Section 5.2), by contrast, carry no biogenic-neutrality question at all: diesel is a fossil fuel, and its combustion CO2 is counted in full against the GWP result without any biogenic offset, following the standard fossil-carbon treatment uncontested across the LCA literature. This asymmetry — biogenic carbon requiring an explicit accounting decision, fossil carbon requiring none — is itself part of why this thesis's Section 5.4 Flexibility Index and Section 6.7 economic-robustness treatment both matter to this chapter's own result: a design that draws more heavily on diesel backup under an adverse feedstock-availability scenario (Section 5.4) will show a correspondingly worse GWP result under this chapter's accounting than the nominal-design figure would suggest, a coupling this thesis's Section 7.9 makes explicit rather than leaving as an unstated implication.

7.7 Comparative Baseline: Diesel-Only Off-Grid Reference Scenario

This thesis's central environmental claim is comparative, not absolute: that an off-grid, biomass-energy-autonomous biorefinery (Chapter 5, Chapter 6) offers a global-warming-potential

advantage over the counterfactual against which any Bajo Atrato investment decision would actually be evaluated — continued or expanded diesel-generator-based off-grid electricity supply (Section 1.2's ZNI framing; Bustos et al., 2014) serving an equivalent processing throughput without the energy island's gasification-ICE contribution. This thesis therefore defines a diesel-only reference scenario, holding the process-technology side of the superstructure (Chapter 4, Chapter 6) fixed at whatever configuration a given Pareto point selects and substituting a diesel-generator-supplied electricity and heat balance for Chapter 5's gasification-ICE-plus-diesel-backup energy island, so that the comparison isolates the energy-system choice's environmental contribution from the process-technology choice's own contribution, rather than conflating the two into a single undifferentiated comparison.

The comparative bioenergy-versus-diesel LCA literature for exactly this off-grid, developing-region context is well established and consistently, though not uniformly, favorable to the biomass alternative: Aberilla et al. (2019) find gasification-based small-scale power has up to twelve-fold lower life-cycle impacts per kWh than direct combustion, and 85% to two-fold lower impacts than diesel generation across most (though, notably, not all — eutrophication, ecotoxicity, and human toxicity are the documented exceptions) of the eighteen impact categories their study examines; the closest single direct precedent to this thesis's own energy-island design, a biomass-gasification combined-heat-and-power retrofit for an off-grid olive-oil mill (Sánchez-Lozano et al., 2023), reports specifically the kind of agro-industrial off-grid context this thesis's own Bajo Atrato case instantiates. This thesis's Section 7.9 protocol commits to reporting this comparison across the same full ReCiPe midpoint category set established in Section 7.5, not GWP alone, precisely because Aberilla et al.'s (2019) finding that the biomass alternative is not uniformly superior across every category is a documented empirical pattern this thesis's own result must be

checked against rather than assumed away by only reporting the category most favorable to the thesis's own design.

7.8 Uncertainty, Sensitivity, and Interpretation Protocol

Consistent with this thesis's standing discipline (Section 6.9) of stating verification and reporting protocols in advance of the computational results they will govern, rather than reporting results before the corresponding checks exist, this section commits to the uncertainty and sensitivity analysis this thesis's LCA will apply once implemented, rather than reporting numerical uncertainty bounds that do not yet exist. Gaffey et al.'s (2024) finding that 39% of the biorefinery LCA studies their review examined conducted no uncertainty or sensitivity analysis whatsoever is treated here as a specific standard this thesis's own assessment is constructed not to fall short of, not merely as a general methodological aspiration.

Three uncertainty sources receive explicit treatment. First, parameter uncertainty in the process-model yields and emission factors themselves (Section 7.4), propagated through the same documented composition and yield ranges Chapter 3 and Chapter 4 already carry forward (Section 3.5) into a Monte Carlo or scenario-based sensitivity band on the eventual GWP and other midpoint results, rather than a single deterministic point estimate reported without a range. Second, methodological uncertainty in the allocation approach (Section 7.3), addressed by this thesis's own commitment to report economic allocation, mass allocation, and system expansion as parallel results rather than a single allocation-method figure, following the multi-method reporting practice Krexner et al. (2021) and Amponsah et al. (2023) recommend specifically because allocation-method choice can shift results by a larger margin than parameter uncertainty typically does. Third, biogenic-carbon-accounting uncertainty (Section 7.6), addressed by reporting the complete-inventory result alongside a conventional-neutrality-assumption result as an explicit sensitivity

bound, following Wiloso et al.'s (2016) own recommended reporting practice, so that a reader can see directly how much of this thesis's reported climate benefit depends on which biogenic-carbon convention is adopted.

Interpretation, the fourth phase ISO 14044 requires of any LCA study (Finkbeiner et al., 2006), will proceed by checking the resulting GWP and midpoint impact-category results for internal consistency against the mass and energy balances of Section 4.8 (a completeness check analogous in spirit to Section 6.9's degrees-of-freedom verification), by identifying which unit operations or streams dominate each reported impact category (a standard LCA hot-spot analysis this thesis commits to performing rather than reporting only aggregate totals), and by explicitly stating the limitations Section 7.1 and this section have already flagged — prospective rather than measured data, background-database vintage and regional representativeness, and the allocation-method sensitivity documented above — as bounded qualifications on the result's interpretation rather than omissions discovered only by a critical reader.

7.9 Coupling to the MINLP Superstructure: GWP as the Section 6.6 Objective

Section 6.6 already established that this thesis's ε-constraint Pareto-front generation bounds the superstructure's environmental performance, measured as GWP, as the constrained objective against which expected NPV (or its CVaR variant, Section 6.7) is maximized. This section closes the present chapter by stating precisely how this chapter's LCA methodology supplies that GWP term without collapsing this chapter's own independent methodological standing into a mere optimization input, the specific conflation Section 2.8's critique of the prior superstructure-biorefinery literature warns against.

The coupling is structural, not incidental: because this chapter's functional unit (1 tonne of feedstock processed, Section 7.2) and economic allocation approach (Section 7.3) are both defined

to remain stable across every candidate configuration the superstructure can generate, a GWP-per-functional-unit figure computed under this chapter's methodology is well defined and comparable across the full range of ε values Section 6.6's Pareto sweep evaluates — a property this thesis verified was necessary, not merely convenient, when Section 7.2 rejected a product-based functional unit specifically because it would not remain stable across configurations with different output mixes. Practically, this means each MINLP instance Section 6.6 solves at a given ε bound carries an implicit GWP-per-functional-unit value computed from that specific configuration's technology-selection and flow-allocation decisions (Section 6.2-6.4) via this chapter's LCI structure (Section 7.4) and LCIA method (Section 7.5), not a single pre-computed GWP figure applied uniformly regardless of configuration — the specific mechanism by which this chapter's LCA methodology and Chapter 6's optimization remain two distinct disciplines in genuine dialogue, per Section 2.8's own framing, rather than one discipline's output silently substituting for the other's proper development.

This chapter's biogenic-carbon treatment (Section 7.6) and diesel-baseline comparison (Section 7.7) are not, by contrast, direct inputs to Section 6.6's optimization: the optimization itself operates on the complete-inventory GWP figure Section 7.6 establishes as this thesis's primary accounting convention, while the neutrality-assumption sensitivity bound and the diesel-only comparative baseline are both post-optimization interpretive devices, applied to the resulting Pareto front's endpoints and to the eventual recommended configuration of Section 8.5, respectively, rather than embedded within the MINLP's own constraint structure. The chapter thus answers the conflation it opened with (Section 2.8): environmental performance enters the optimization as one well-defined term, computed configuration by configuration on a functional unit that stays stable across the Pareto front, while the fuller midpoint profile, the biogenic-carbon

bound, and the diesel-only comparison remain an assessment in their own right rather than a by-product of the economic model. For a Bajo Atrato decision-maker the separation matters beyond method: it keeps the environmental case the plant is meant to make — a GWP advantage over the diesel-only alternative a ZNI site would otherwise rely on (Section 7.7) — reportable across the whole Pareto front (Section 9.2), not only at the configuration the economics favor. What the front still lacks is its other axis in usable form: Chapter 8 supplies the cost and revenue content of the NPV term and carries the front through to a single recommended configuration.

Chapter 8. Techno-Economic Assessment

An optimization model that maximizes NPV is only as credible as the costs and prices inside it, and it does not, by itself, tell anyone what to build. Chapter 6 left both questions open: its NPV objective is a function awaiting capital-cost, operating-cost, and revenue content — much of it for first-of-a-kind equipment whose cost estimates carry documented error — and its ε-constraint sweep returns a set of non-dominated NPV-GWP configurations, with GWP supplied by Chapter 7, rather than a recommendation. This chapter closes both gaps: it develops the techno-economic assessment (TEA) that gives the NPV objective its content, and it carries the resulting Pareto front through to a single recommended configuration whose economic risk it then quantifies. Sections 8.1-8.4 build the cost model: the discounted-cash-flow framework, capital and operating expenditure, revenue, and profitability metrics. Section 8.5 narrows the Pareto front to a recommended configuration through value-density reinterpretation and multi-criteria selection, kept as two separate exercises; Section 8.6 quantifies that configuration's risk by Monte Carlo analysis, distinct from the CVaR treatment of Section 6.7; and Section 8.7 reconciles the chapter's outputs with Chapters 6 and 7.

8.1 TEA Methodology and the Discounted Cash Flow Framework

Techno-economic assessment for biorefinery process design is, across the literature this thesis draws on, built around a discounted cash flow (DCF) framework: capital expenditure is incurred up front (or over a short construction period), operating cash flows accrue annually over the project horizon as the difference between revenue and operating expenditure, and both are discounted to a present value at a specified discount rate to yield net present value, internal rate of return, and payback period as the standard reported profitability metrics (Ellamla et al., 2020;

Scown et al., 2021). Figure 8.1 depicts this timeline schematically as this thesis will apply it. This thesis's specific DCF implementation follows the discounted-cash-flow-rate-of-return (DCFROR) convention most consistently documented in the biofuel and biorefinery TEA literature (Ellamla et al., 2020), rather than a simplified payback-only or non-discounted accounting treatment, precisely because Section 6.6 and Section 6.7's optimization-embedded NPV objective already requires a discounted formulation — this chapter's DCF methodology is the specific procedure by which that objective's numerical content (capital cost, operating cost, revenue, at whatever discount rate this thesis specifies) is populated, not a separate or looser economic treatment developed independently of Chapter 6's formal objective.

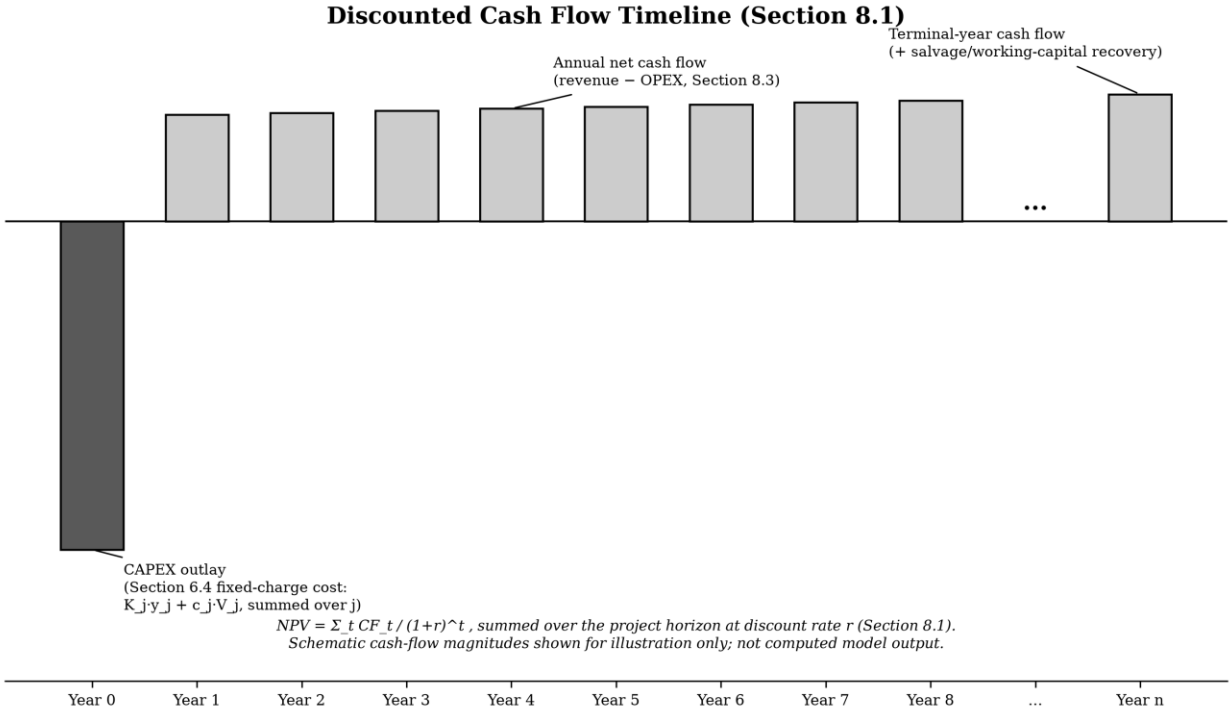


Figure 8.1. The discounted cash flow timeline this thesis's TEA methodology applies: a capital-expenditure outlay at project year zero, annual net operating cash flows over the project horizon, and a terminal-year cash flow including salvage value and working-capital recovery.

This is an original diagram developed for this thesis, illustrating the standard discounted-cash-flow-rate-of-return (DCFROR) structure documented in the biorefinery TEA literature (Ellamla et al., 2020; Scown et al., 2021); the specific cash-flow magnitudes shown are illustrative only and do not represent computed model output.

This thesis departs from the single-product minimum-selling-price (MSP) convention that dominates the comparative TEA literature this section otherwise follows (Aui et al., 2021; Patel et al., 2025, among the broader lignocellulosic-biorefinery TEA literature reviewed there) for a reason specific to this thesis's own superstructure structure, not as an unexplained methodological deviation: MSP is well defined for a single-product or single-dominant-product facility, where a breakeven selling price for that one product can be solved for holding all other parameters fixed, but this thesis's superstructure (Section 6.1) generates up to eight simultaneously marketable product and co-product streams whose relative output mix is itself an optimization output, not a fixed recipe — precisely the same structural feature that led Section 7.2 to reject a product-based functional unit for the LCA. A single MSP figure would therefore require an arbitrary choice of which product's price to solve for while holding the others fixed, silently privileging one output over the rest of the portfolio. This thesis instead reports portfolio-level NPV, IRR, and payback period at fixed (market-derived) prices for every output, consistent with Section 6.6's own portfolio NPV objective, and reserves single-product breakeven pricing only as a diagnostic sensitivity check on any one output within Section 8.6's protocol, not as this chapter's primary profitability metric.

8.2 Capital Expenditure Estimation

This chapter's capital expenditure (CAPEX) figure is not independently derived; it is the direct evaluation, at whatever technology-selection and capacity-installation solution (y, V) a given Chapter 6 optimization instance returns, of the fixed-charge cost function already formalized in Section 6.4: $CAPEX = \Sigma_j (K_j \cdot y_j + c_j \cdot V_j)$, summed across every candidate technology j in the superstructure, whether or not it is ultimately activated. This chapter's role is to populate the K_j (fixed) and c_j (variable, capacity-proportional) cost coefficients themselves from defensible

cost-estimation sources, not to reformulate the cost structure Section 6.4 already established — restating that structure here would risk the same chapter-boundary confusion Section 6.9 and Section 7.9 both explicitly guard against elsewhere in this thesis.

Populating K_j and c_j for a novel or first-of-a-kind biorefinery technology carries a documented, and non-trivial, estimation risk: Tsagkari et al.'s (2016) comparative benchmarking of six early-stage capital-cost-estimation heuristics for biorefinery equipment specifically found that extrapolating conventional-petroleum-refinery cost correlations to biorefinery-specific unit operations can violate the correlations' own underlying scale and technology-maturity assumptions, in some cases producing capital-cost estimates that diverge by more than a factor of two depending on which heuristic is applied to the same underlying equipment. This thesis's own candidate technologies span a wide maturity range — ceramic microfiltration and spray drying are mature, broadly deployed unit operations with correspondingly reliable vendor-quotation-based cost data, while the shared sugar hub's own bilinear-pooling infrastructure and the specific enzymatic-hydrolysis train configuration this thesis proposes have comparatively little first-of-a-kind cost precedent — so this thesis's CAPEX estimation protocol commits to reporting, technology by technology, which cost-estimation class (vendor quotation, published correlation, or heuristic scale-up from a smaller precedent) underlies each K_j and c_j value, following Tsagkari et al.'s (2016) own recommendation that a biorefinery TEA disclose this provenance rather than present a single blended CAPEX figure of unstated reliability.

[NOTA — LIBROS (4/5), vacíos del CAPEX. El capítulo no menciona capital de trabajo, contingencia ni la clase de precisión de la estimación. Burk (2026): (a) §6.1, p. 77, clases de estimación: orden de magnitud (peor que ±30 %), estudio (±30 %), preliminar (±20 %), definitiva (±10 %), detallada (±5 %); declara la tuya (clase estudio) y úsala como rango del Monte Carlo de 8.6. (b) §11.2, p. 193: contingencia del 10-50 % de ISBL + OSBL, mayor para tecnologías menos establecidas; diferénciala (alta para el hub y la hidrólisis, baja para membranas y secado), lo que cumple el compromiso de este párrafo. (c) §11.1, p. 189: capital de trabajo (inventarios); con estacionalidad y añejamiento del vino puede ser

importante. También: factor de Lang (p. 194) y exponentes de escala (§10.2, pp. 166-176) como verificación de orden de magnitud.]

Where a candidate technology's cost cannot be sourced from a direct vendor quotation or a well-validated published correlation at a comparable scale — the specific condition Tsagkari et al. (2016) find most exposed to estimation error — this thesis's protocol additionally commits to reporting that technology's CAPEX as a range rather than a point estimate, propagated into Section 8.6's Monte Carlo uncertainty treatment as a wider parameter distribution than a mature, well-quoted technology would carry, consistent with this thesis's broader standing discipline (Section 4.1, Section 7.4) of carrying documented uncertainty ranges into downstream analysis rather than collapsing them to a single number by unstated convenience.

8.3 Operating Expenditure and Revenue Modeling

Operating expenditure (OPEX) aggregates, across every activated candidate technology, the raw-material and feedstock-handling costs (Chapter 3), process chemical, enzyme, and membrane consumable costs (Chapter 4), utility costs met internally by the off-grid energy island rather than purchased externally (Chapter 5) — a structural feature of this thesis's design that this chapter's OPEX line explicitly credits, since an off-grid facility does not pay a utility a grid-connected facility would — and labor, maintenance, and general overhead costs scaled to the installed capacity following standard biorefinery TEA cost-scaling conventions (Ellamla et al., 2020; Scown et al., 2021). The diesel-backup fuel cost (Section 5.2) is carried as an operating cost specifically dependent on the Chapter 5 Flexibility Index analysis's implied backup-utilization rate, not a fixed annual allowance, so that a configuration relying more heavily on diesel backup under adverse feedstock conditions correctly shows a higher OPEX, not merely a worse Chapter 7 GWP result — the same asymmetry Section 7.6 already flagged between fossil and biogenic carbon

accounting, here manifesting as an economic rather than environmental consequence of the identical underlying design choice.

Revenue is computed from each output platform's market-derived unit price (Section 2.4, Section 3.4) applied to its optimized production volume, using the identical price dataset Section 7.3 already uses to construct its economic-allocation keys — a shared-data-source design this thesis states explicitly, per the same discipline applied to the LCI in Section 7.4, so that a reader can trust the TEA revenue figures and the LCA allocation keys are mutually consistent rather than independently and potentially divergently sourced. Where a candidate output platform's market price itself carries a documented range rather than a single point estimate (several of the specialty-fermentation product lines reviewed in Chapter 2 report price ranges spanning a factor of two or more across geographic markets and purity grades), this thesis's revenue model carries that range forward explicitly into Section 8.6's uncertainty treatment, following the same range-preservation discipline as Section 8.2's CAPEX treatment, rather than collapsing to an unstated midpoint assumption.

8.4 Profitability Metrics and Discount Rate

This thesis reports three profitability metrics for the primary (expected-NPV-optimal, Section 6.6) and CVaR-optimal (Section 6.7) configurations alike, rather than NPV alone: net present value itself, at the discount rate specified below; internal rate of return, the discount rate at which NPV equals zero, reported as a scale-independent profitability check against NPV's own scale-dependence; and payback period, the simple (undiscounted) and discounted time to cumulative cash-flow breakeven, reported specifically because a Bajo Atrato investment decision-maker (Section 7.1) is likely to weight near-term capital recovery more heavily than a pure NPV ranking would suggest, a practical consideration this thesis's Section 8.5 formal MCDM treatment

can and does accommodate as an explicit decision criterion rather than leaving it as an unstated background concern.

The discount rate this thesis applies is not treated as a single uncontested constant: following the sensitivity patterns documented across the comparative TEA literature (Iglesias et al., 2021, in the bio-oil TEA context; the discount-rate sensitivity finding recurring across the DCFROR literature reviewed in Section 8.1), this thesis's NPV and IRR results are reported at a base-case discount rate reflecting a defensible weighted-average cost of capital for a Bajo Atrato-scale agro-industrial investment, together with a sensitivity band spanning a plausible range of financing conditions, rather than a single undisclosed rate whose influence on the headline NPV figure a reader could not otherwise assess — discount-rate sensitivity is, empirically, one of the more consequential single-parameter drivers of TEA outcomes across the literature this thesis reviews (Pandit et al., 2025, finding internal rate of return among the most influential sensitivity parameters for a comparable biomass-to-liquid-fuel case), so treating it as a fixed, unexamined input would understate a real source of this thesis's own result's variability.

8.5 Value-Density Reinterpretation and Multi-Criteria Decision-Making Selection

This section performs two analytically distinct exercises on the Pareto front Section 6.6 generates, and this thesis's standing methodological discipline — stated first in Section 3.4, reiterated in Section 7.3, and enforced here for the third and final time — requires that the two not be conflated into a single undifferentiated 'value' judgment, since they answer different questions using the value-density concept in structurally different ways.

The first exercise is a posteriori value-density reinterpretation. Section 3.4 used value density — market price per unit of the limiting upstream resource each candidate technology consumes — as an a priori screening heuristic, narrowing the candidate-technology set before the

superstructure of Chapter 6 was even formalized, on the explicit understanding stated there that a screening heuristic is a computationally cheap filter, not a substitute for full optimization. This section now asks the reverse-direction question: does the Chapter 6 superstructure's actual optimized technology selection and flow allocation, arrived at by solving the full nonconvex MINLP rather than by the Section 3.4 heuristic, agree with what the value-density screening would have predicted? Where the two agree, this thesis treats that agreement as a mild corroborating check on the optimization's plausibility (a solution that contradicts a simple, defensible heuristic without an identifiable reason to do so would warrant closer scrutiny of the model formulation, per Section 6.9's own verification discipline); where they disagree — most plausibly at the shared sugar hub, whose bilinear coupling (Section 4.6, Section 6.3) is exactly the structural feature a simple per-technology value-density heuristic cannot capture, since value density is computed technology-by-technology and has no mechanism for representing a shared resource's cross-technology opportunity cost — this thesis treats the disagreement as substantive evidence that the full MINLP formulation is doing genuine work the heuristic screening could not, which is, after all, the entire methodological justification (Section 1.4, Section 6.1) for building the superstructure in the first place rather than resting on Section 3.4's screening result alone.

The second, entirely separate exercise is formal multi-criteria decision-making selection: choosing a single recommended configuration from the full Pareto-optimal set Section 6.6 generates, trading off NPV against GWP (and, where Section 6.7's CVaR-optimal front diverges from the expected-NPV front, potentially trading off expected performance against downside risk as a third criterion). Following Section 2.9's comparative review of the MCDM literature, and Sorkhab et al.'s (2026) broader finding that biorefinery-specific MCDM applications most commonly favor the Analytic Hierarchy Process for criteria weighting and the weighted-sum

method for aggregation despite outranking methods' documented advantage when criteria are not fully compensatory, this thesis selects an outranking method — PROMETHEE, following the same reasoning Section 2.9 already develops — specifically because this thesis's own two-to-three-criterion decision problem (NPV, GWP, and optionally downside risk) is exactly the kind of partially-compensatory decision context outranking methods are documented to handle better than pure ranking methods (Opricovic & Tzeng, 2007): a catastrophic GWP result should not be treated as fully redeemable by an outstanding NPV result, a value judgment a weighted-sum or TOPSIS-style compensatory aggregation would not enforce without an additional, separately justified constraint.

[NOTA: «Sorkhab et al. (2026)» no aparece en Consensus ni en la lista de referencias. Lo que se le atribuye (AHP como método de ponderación más común y suma ponderada como agregación más común en el MCDM de biorrefinerías) coincide exactamente con Bose et al. (2026, Biomass and Bioenergy, doi:10.1016/j.biombioe.2026.109036), una revisión de 61 estudios. Comprueba si es la misma obra (¿Sorkhab es coautor?) y cita al primer autor. La «ventaja documentada» de los métodos de superación no está en ese resumen: es tu argumento de 2.9.]

This thesis's MCDM criteria weights — the relative importance assigned to NPV versus GWP versus downside risk — are, per this thesis's own standing preference for auditable rather than opaque weighting (Section 7.5), derived via a stated, disclosed weighting procedure (following the entropy-weighting or Analytic-Hierarchy-Process conventions reviewed in Section 2.9, with the specific choice and its justification to be stated once this chapter's implementation stage is reached) rather than presented as an unweighted or implicitly-weighted single recommendation. This chapter's final recommended configuration — the specific point on the Pareto front, and the specific technology-selection and flow-allocation decisions it embodies — is the deliverable this formal MCDM procedure produces, and it is that configuration, not the Pareto

front's NPV-optimal or GWP-optimal extreme points alone, that Section 8.6's uncertainty analysis and Chapter 9's synthesis both take as their subject.

8.6 Uncertainty and Risk Analysis

This section's uncertainty treatment is methodologically distinct from, and applied at a different stage than, Section 6.7's CVaR formulation, and this thesis states that distinction explicitly rather than leaving two superficially similar risk analyses to be conflated by a reader, following the same discipline already applied to the Section 5.4/Section 6.7 flexibility-versus-robustness distinction. Section 6.7's CVaR treatment is embedded within the MINLP itself: it is a decision-stage formulation, evaluated over a finite, discrete scenario set S with associated probabilities, that shapes which configuration the optimization selects in the first place by penalizing the lower tail of the scenario-dependent NPV distribution. This section's Monte Carlo treatment, by contrast, is a post-selection validation exercise, applied only to the single configuration Section 8.5's MCDM procedure has already recommended, propagating continuous (not discretized) parameter uncertainty — in CAPEX (Section 8.2), OPEX and revenue (Section 8.3), and the discount rate (Section 8.4) — through that fixed configuration's cash-flow model via repeated random sampling, following the Monte Carlo TEA convention now standard across the biorefinery literature (Barahmand et al., 2022, whose scoping review of TEA and life-cycle-cost uncertainty studies finds Monte Carlo the dominant treatment specifically because it accommodates continuous, correlated parameter distributions that a discrete scenario set does not represent as naturally).

This thesis's Monte Carlo protocol follows the reporting convention established across the closest comparable biorefinery and bioenergy TEA precedents: Iglesias et al. (2021) report an 80% probability of NPV exceeding zero and an 80% probability of IRR exceeding 10% for a

eucalyptus-residue bio-oil biorefinery; Colantoni et al. (2021), in the closest single precedent to this thesis's own energy-island economics, report the probability of a positive NPV for biomass-gasification combined-heat-and-power systems specifically, finding that probability rises with plant scale, from 66% to 90% — a finding this thesis's own protocol will check its energy-island-coupled configurations against, since Chapter 5 and Chapter 6 jointly determine this thesis's own effective scale. Following this precedent, this thesis's Monte Carlo protocol commits to reporting the full NPV probability distribution (not a point estimate alone), the probability of a positive NPV, and the probability of the IRR exceeding the discount-rate threshold established in Section 8.4, once the underlying process and cost models are fully implemented.

[NOTA — citas del Cap. 8 (Consensus, 4-nov). Iglesias et al. (2021) confirmado (80 % VPN>0, 80 % TIR>10 %), pero en su análisis de sensibilidad lo que más pesa es la capacidad de planta, no la tasa de descuento: en 8.4 se cita como ejemplo de sensibilidad a la tasa; Pandit et al. (2025) sí pone la TIR/tasa en segundo lugar, después del costo de materia prima. Colantoni et al. (2021) confirmado (66-90 %, mayor en plantas grandes; gasificador + motor). Confirmados también: Pérez-López et al. (2018), Aui et al. (2021), Patel et al. (2025), Barahmand et al. (2022), Zang et al. (2020). Ellamla et al. (2020) es un capítulo de libro (Wiley), sin resumen en Consensus.]

Figure 8.2 illustrates, schematically, the tornado-diagram sensitivity analysis this thesis's protocol additionally commits to reporting: a one-at-a-time ranking of which individual parameters most influence the recommended configuration's NPV, holding all others at their base-case value, following the standard single-parameter sensitivity convention documented throughout the TEA literature this section reviews (Pandit et al., 2025; Zang et al., 2020, among others already cited in Section 8.1 and Section 8.4). This thesis expects — though does not yet confirm, since no computation has been executed — that feedstock availability (via the leaves/crown allocation split α, Section 6.1, Section 6.3) and hub-output-platform prices (Section 7.3) will rank among the most influential parameters, on the reasoning that both directly govern the shared sugar hub's bilinear-coupled economics (Section 4.6, Section 6.3) that this thesis's entire superstructure argument treats

as structurally central; this expectation is stated here as a hypothesis Section 8.6's eventual computational implementation will test, not as a result already established, following the same evidentiary discipline Section 6.9 and Section 7.8 both apply to their own as-yet-unexecuted protocols.

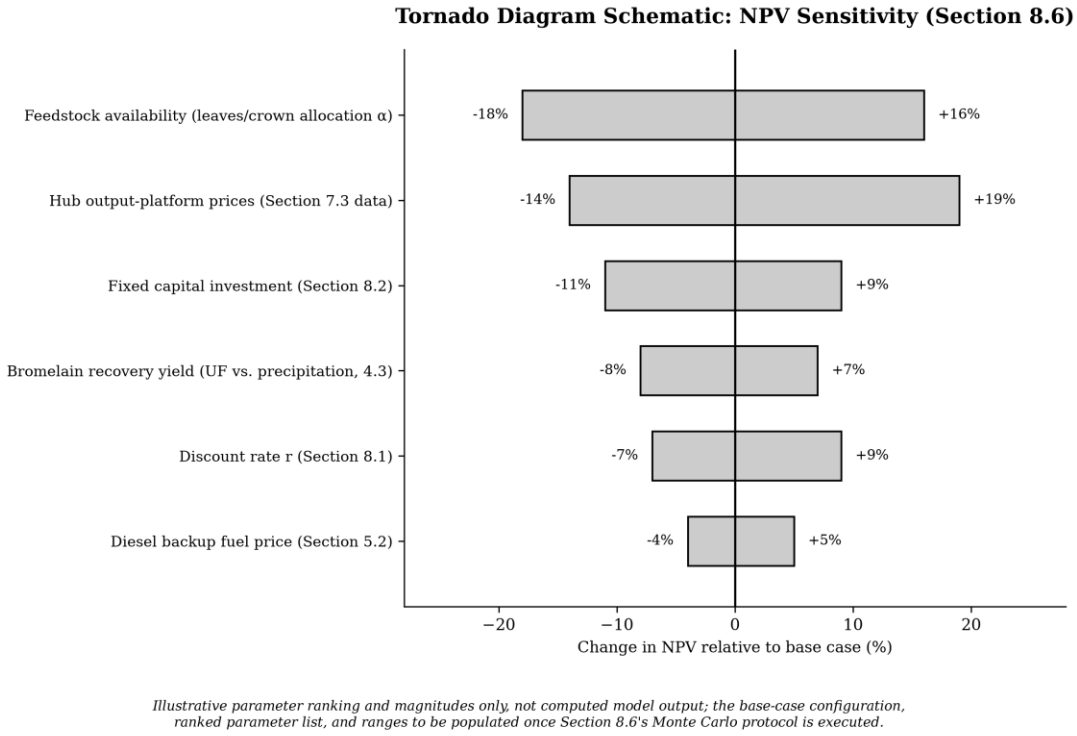


Figure 8.2. A schematic tornado diagram illustrating the one-at-a-time NPV sensitivity analysis this thesis's uncertainty protocol will apply to the Section 8.5 recommended configuration, ranking candidate parameters by their individual influence on NPV.

This is an original diagram developed for this thesis, illustrating the standard tornado-diagram sensitivity-analysis convention documented throughout the biorefinery TEA literature (Pandit et al., 2025; Zang et al., 2020); the parameter ranking and percentage ranges shown are illustrative only and do not represent computed model output.

Following Pérez-López et al.'s (2018) demonstration that a joint Monte Carlo treatment of environmental and economic indicators from a shared set of underlying uncertain parameters can reveal correlations invisible to either analysis performed separately — they report a strong positive

correlation between GWP and minimum selling price for an algal biorefinery case, both driven by the same underlying biomass-productivity uncertainty — this thesis's Section 8.6 protocol commits to sampling its Monte Carlo parameter uncertainty from a set that overlaps explicitly with Section 7.8's own LCA uncertainty treatment wherever a parameter (feedstock yield, process conversion efficiency) genuinely drives both the economic and environmental result, so that the correlation structure between this chapter's NPV distribution and Chapter 7's GWP distribution can itself be reported, rather than the two chapters' uncertainty analyses being conducted as though the underlying parameter uncertainty were independent between them when it is not.

8.7 Reconciling This Chapter with Chapters 6 and 7

This closing section states explicitly what this chapter has, and has not, done relative to the optimization of Chapter 6 and the environmental assessment of Chapter 7, following the same reconciliation discipline Section 7.9 already applied to the Chapter 6-Chapter 7 relationship. This chapter has not re-optimized anything: the CAPEX, OPEX, and revenue models developed in Sections 8.2-8.3 are the specific numerical content that populates Section 6.6's NPV objective and Section 6.7's CVaR objective at the point those objectives are evaluated within the MINLP itself, so Chapter 6's optimization and this chapter's TEA methodology are one integrated economic model viewed from two documentation angles, not two independently-arrived-at economic results that happen to agree.

What this chapter has done, and what is genuinely additional to Chapter 6's own scope, is threefold: first, Section 8.4's supplementary profitability metrics (IRR, payback period) beyond the NPV figure the optimization itself reports; second, Section 8.5's formal, two-part post-optimization treatment — value-density reinterpretation and MCDM selection — that narrows the full Pareto front to the single recommended configuration Chapter 6's optimization alone does not

and structurally cannot produce, since an ε-constraint sweep by design returns a set of non-dominated points, not a single recommendation; and third, Section 8.6's Monte Carlo uncertainty and risk validation of that specific recommended configuration, a post-hoc robustness check genuinely distinct from Section 6.7's decision-embedded CVaR treatment, as this section has already stated. Chapter 9's synthesis takes this chapter's Section 8.5 recommended configuration, together with its Section 8.6 uncertainty bounds and Chapter 7's parallel environmental assessment of that same configuration, as the specific deliverable this dissertation's design and optimization work is ultimately answerable to — the point at which the methodological apparatus developed across Chapters 4 through 8 is built to converge on a single, defensible answer to the research question Chapter 1 posed.

Chapter 9. Techno-Economic-Environmental Synthesis

Chapters 6 through 8 developed three methodologically independent but structurally coupled analyses: the MINLP superstructure and its NPV-GWP Pareto front (Chapter 6), the life-cycle assessment methodology that supplies the GWP term and a fuller environmental profile beyond it (Chapter 7), and the techno-economic assessment that supplies the NPV term, narrows the Pareto front to a single recommended configuration via formal multi-criteria decision-making, and validates that configuration's robustness under continuous parameter uncertainty (Chapter 8). Built to be independent, the three analyses can disagree — a Flexibility Index implying infeasibility at a feedstock deviation the CVaR scenario set treats as routine, for instance (Section 9.3) — and a recommendation drawn from any one of them would hide that disagreement. This chapter is where those three analyses are read together rather than separately. It is not a fourth independent methodology — it introduces no new model, no new optimization, and no new impact-assessment method — but a synthesis in the literal sense: placing the outputs Chapters 6-8 already define next to one another, checking that they are mutually consistent, and using them jointly to evaluate the research hypotheses Section 1.5 posed before any of Chapters 4-8 were written. Figure 9.1 depicts this chapter's own structure: three methodological inputs converging on a recommended configuration (Section 9.3), which this chapter then interrogates from three angles (Sections 9.4-9.5) before a fourth, methodological rather than numerical, synthesis step (Section 9.6) situates Hypothesis H1's own test against Section 6.10's benchmarking of this thesis's superstructure paradigm, and this configuration is handed forward to Chapter 10's policy framing and Chapter 11's conclusions.

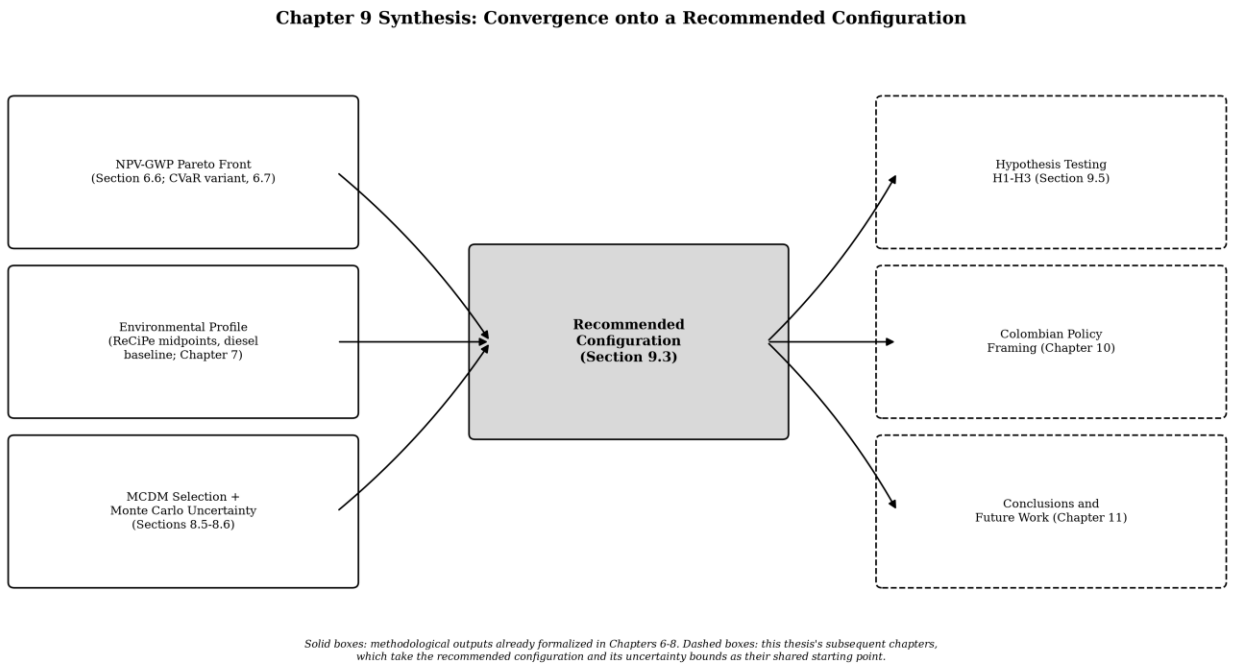


Figure 9.1. This chapter's synthesis structure: the Pareto front (Section 6.6), the environmental profile (Chapter 7), and the MCDM-selected recommended configuration with its uncertainty bounds (Sections 8.5-8.6) converge here before being carried forward into hypothesis testing (Section 9.5), Chapter 10's policy framing, and Chapter 11's conclusions.

This is an original diagram developed for this thesis, summarizing the chapter-to-chapter data flow already established across Chapters 6-8 and this chapter's own place within it.

Consistent with the evidentiary discipline this thesis has applied at every point where a computational result has not yet been produced — Section 6.9's verification protocol, Section 7.8's uncertainty and interpretation protocol, and Section 8.6's Monte Carlo protocol chief among them — this chapter states how each synthesis step will be performed and what it will mean once the underlying MINLP has actually been solved, rather than asserting specific Pareto-front coordinates, NPV figures, or GWP values that do not yet exist. A synthesis chapter is, by its nature, more exposed than a methodology chapter to the temptation to write as though results were already in hand; this chapter resists that temptation for the same reason every prior chapter has: an

unearned numerical claim would not merely be imprecise, it would misrepresent this thesis's actual present state as a design and methodology dissertation to any reader relying on it.

9.1 The NPV-GWP Pareto Front: Structure and Interpretation Protocol

Section 6.6 established that this thesis's ε-constraint sweep generates a set of non-dominated configurations trading off expected NPV (or its CVaR variant, Section 6.7) against GWP (Chapter 7), each corresponding to a distinct combination of technology-selection binaries, hub-flow allocations, and output-platform activations. This section states the protocol by which that front, once generated, is read: first, confirming non-dominance and non-degeneracy directly against Hypothesis H3 (Section 9.5) — that is, checking computationally that no returned point is in fact dominated by another (a check against numerical or convergence artifacts in the ε-constraint sweep itself, following Section 6.8's solution-strategy verification discipline) and that the front spans a genuine range of trade-offs rather than clustering at a single point, which would indicate either a genuinely dominant configuration or an under-resolved sweep; second, characterizing the front's shape (convex, nonconvex, or mixed) and relating that shape explicitly to the hub's bilinear nonconvexity (Section 4.6, Section 6.3), since Section 6.6 already predicted, rather than merely accommodated, the possibility of a nonconvex front on exactly this structural basis; and third, identifying the front's two anchor points — the NPV-optimal and GWP-optimal extremes — as the two configurations against which every intermediate point, including the eventual MCDM-recommended one (Section 9.3), is naturally compared.

This section also states, in advance of computation, the specific comparison this thesis's Hypothesis H1 requires: a sequential-design baseline, constructed by optimizing each of the four fraction trains' technology selection and capacity independently — each train solved as its own smaller MINLP, drawing on the same unit-operation models of Chapter 4 but with no shared sugar

hub linking them, so that each train's candidate output either sells directly at the market prices of Section 7.3/Section 8.3 or, for the three hub-bound streams (peel juice, hydrolysate, UF permeate), is priced at an assumed pass-through value reflecting its stand-alone disposal or sale value absent the hub's pooling benefit — and then summing the four trains' independently optimized NPVs. This sequential-baseline NPV is the quantity Hypothesis H1 predicts the simultaneous superstructure's own NPV-optimal Pareto anchor will exceed; Section 9.5 states exactly how that comparison will be reported once both figures exist.

9.2 Environmental Profile of the Pareto Front

Chapter 7 established that this thesis's LCIA (Section 7.5) reports a fuller ReCiPe midpoint profile than GWP alone, and that the diesel-only reference scenario (Section 7.7) supplies the comparative baseline against which this thesis's central environmental claim is evaluated. This section's protocol extends that reporting across the full Pareto front, not only its GWP-optimal anchor: for a representative subset of Pareto-front configurations spanning the NPV-GWP trade-off space (at minimum, the two anchor points and the eventual MCDM-recommended configuration of Section 9.3), this thesis will report the complete midpoint impact-category profile, not GWP in isolation, following the same reasoning Section 7.5 already gives for reporting midpoints generally: a configuration that improves GWP by drawing more heavily on one process route could, in principle, worsen a different midpoint category (eutrophication or ecotoxicity, most plausibly, given the fermentation-heavy product portfolio Section 6.1's superstructure spans), and Aberilla et al.'s (2019) finding that biomass-based off-grid power is not uniformly superior to diesel across every impact category is the specific, literature-grounded reason this thesis does not assume a GWP improvement implies an across-the-board environmental improvement without checking.

This section also states how the biogenic-carbon sensitivity bound of Section 7.6 and the diesel-baseline comparison of Section 7.7 will be reported across the front: as a banded range around each reported GWP figure (complete-inventory accounting as the primary value, conventional-neutrality-assumption accounting as the sensitivity bound) and as a percentage improvement over the diesel-only counterfactual configuration at matched process-technology selection, respectively — both applied consistently across every Pareto-front point this chapter reports on, not only the GWP-optimal anchor, so that a reader comparing two intermediate configurations sees the same accounting convention applied to both rather than a convention that silently shifts across the comparison.

9.3 The Recommended Configuration: Techno-Economic and Environmental Profile

Section 8.5 established the two-part post-optimization procedure — a posteriori value-density reinterpretation, then formal PROMETHEE-based multi-criteria decision-making — by which this thesis narrows the full Pareto front to a single recommended configuration. This section is where that configuration's complete profile is assembled and reported as a single coherent object, rather than left distributed across Chapters 6 through 8's separate treatments: its technology-selection and flow-allocation decisions (Section 6.2-6.4), its expected NPV and CVaR-adjusted NPV (Section 6.6-6.7), its IRR and payback period (Section 8.4), its Monte Carlo NPV distribution and probability of a positive NPV (Section 8.6), its complete ReCiPe midpoint environmental profile and diesel-baseline comparison (Section 9.2), and its Flexibility Index result against the energy-autonomy constraint (Section 5.4) — the last of these included here specifically because Section 5.4 and Section 6.7 both established that a configuration's economic performance and its categorical operability under feedstock uncertainty are separate questions, so a complete profile of

the recommended configuration must report both, not treat a good NPV result as implying operational robustness it has not been separately verified to have.

Assembling this profile is itself a check this thesis's methodology imposes on itself: because the profile draws on six distinct prior sections' outputs (Sections 5.4, 6.6, 6.7, 7.5, 8.4, 8.6), any internal inconsistency among them — a Flexibility Index result implying infeasibility at a feedstock deviation the CVaR scenario set treats as routine, for instance — would surface here as a direct contradiction rather than remaining hidden across chapters a reader might not cross-reference. This section's reporting protocol accordingly commits to flagging, not silently resolving, any such inconsistency discovered once the underlying computations are executed, consistent with Section 6.9's standing verification discipline that a discovered problem is reported as a problem, not quietly patched over in the final write-up.

9.4 Robustness Synthesis: Flexibility, Stochastic/CVaR, and Monte Carlo Read Together

This thesis has, at three separate points, developed a distinct treatment of a distinct kind of uncertainty, and has at each point insisted the three not be conflated: the Flexibility Index (Section 5.4) asks whether the recommended configuration's fixed design remains operable across documented feedstock-parameter deviations, independent of cost; the CVaR-adjusted objective (Section 6.7) asks which configuration performs best under a risk-averse criterion applied to price and yield uncertainty, evaluated over a discrete scenario set at the point the MINLP itself is solved; and the Monte Carlo protocol (Section 8.6) asks, for the single configuration already selected, how much its reported NPV varies under continuous parameter uncertainty in cost and price inputs, evaluated after selection as a validation rather than a decision-shaping exercise. This section is not a fourth uncertainty treatment; it is the explicit reading of these three together, because a complete

answer to how robust the recommended configuration actually is requires all three, and no single one of them alone.

Read together, the three treatments answer three genuinely different practical questions a Bajo Atrato investment decision-maker would ask in sequence: Can this design physically operate across the feedstock variability Chapter 3 documents (Flexibility Index)? Was the economic case for this specific design already constructed to hedge against adverse price and yield draws, or does it depend on optimistic point estimates (CVaR)? And, given the specific cost and price data this thesis's TEA populated the model with, how much confidence should be placed in the resulting NPV figure (Monte Carlo)? This section's synthesis protocol commits to answering all three explicitly for the recommended configuration of Section 9.3, in that order, rather than reporting a single aggregate 'robustness' statement that would obscure which of the three distinct questions it was actually answering.

9.5 Testing the Research Hypotheses

Section 1.5 stated three hypotheses this dissertation set out to test. This section states, for each, the specific comparison this thesis's methodology (Chapters 4-8) has been constructed to make, and what result would confirm or disconfirm it — the test protocol, not a result, since the underlying MINLP has not yet been solved.

H1 (simultaneous superstructure exceeds sequential single-fraction design). This hypothesis is tested by the direct comparison Section 9.1 already set up: the simultaneous superstructure's NPV-optimal Pareto anchor (Section 6.6) against the independently-optimized sequential-baseline sum (Section 9.1's protocol). Confirmation requires the simultaneous figure to exceed the sequential sum; the specific mechanism Hypothesis H1 attributes that gap to — hub-allocation trade-offs invisible to sequential, train-by-train design, following the simultaneous

superstructure/utility co-design precedent of Kong et al. (2016) already cited in Section 1.5 — is separately checked by examining whether the gap concentrates specifically at the three hub-bound streams (peel juice, hydrolysate, UF permeate) rather than being diffusely spread across all four trains, since a gap concentrated exactly where the bilinear hub coupling operates would corroborate the specific causal story H1 tells, not merely the aggregate NPV comparison. Section 9.6 below states explicitly what confirming H1 would, and would not, establish about this thesis's specific hub-and-outputs architecture relative to other integration topologies a designer could in principle have chosen.

H2 (the energy-autonomy constraint is binding). This hypothesis is tested directly by Section 5.4's Flexibility Index result: a Flexibility Index below the value that would certify feasibility across the full documented feedstock-uncertainty range (Section 3.5) confirms the constraint is binding in the specific, quantified sense Section 5.4 defines; a critical vertex θ_c (Section 5.4, Figure 5.3) identified at an economically plausible, not merely mathematically extreme, feedstock deviation further corroborates that the binding constraint is one a real operating facility could plausibly encounter, not an artifact of an unrealistically wide uncertainty box. This thesis additionally tests H2's magnitude claim by comparing the recommended configuration's feasible product portfolio (Section 9.3) against a hypothetical unconstrained variant of the same superstructure solved without Chapter 5's energy-autonomy coupling — the same comparison logic Section 1.2 and Section 1.4 use qualitatively to motivate the energy-autonomy constraint in the first place, made quantitative here.

H3 (the Pareto set is non-dominated and non-degenerate). This hypothesis is tested by Section 9.1's own front-verification protocol: non-dominance is a direct, checkable property of the returned ε-constraint solution set (Section 6.6, Section 6.8), and non-degeneracy — meaningful

trade-offs existing rather than a single point dominating on both criteria — is assessed by the front's spread on both axes relative to each anchor point's own value, following the same non-degenerate-front precedent Section 1.5 cites for hydrocarbon and algal biorefinery superstructures solved by the identical ε-constraint method (Wang et al., 2013; Gong & You, 2014). A degenerate front — one collapsing to a single dominant point — would not falsify this thesis's superstructure formulation itself, but it would falsify H3 specifically and would substantially simplify Section 9.3's MCDM step, since a dominant configuration requires no trade-off adjudication; this thesis states that possibility here rather than presupposing H3's confirmation.

This section's role is to state these three tests precisely enough that their eventual outcome — confirmation, partial confirmation, or disconfirmation — can be reported unambiguously in Chapter 11's conclusions once the underlying computation exists, following exactly the same discipline that governs every other as-yet-unexecuted protocol in this thesis: the test is fully specified now; its result is reported when, and only when, it has actually been obtained.

9.6 The Corsano et al. Benchmark and Hypothesis H1: What Confirmation Would, and Would Not, Establish

Section 6.10 benchmarked this thesis's superstructure paradigm against Corsano et al.'s (2009) NLP-superstructure and Multiplant Complex frameworks and, in doing so, surfaced a comparison this thesis has not run: its own shared, hub-centered architecture (Section 6.1) against a hypothetical set of separate, dedicated processing trains, one per output platform, with no shared bilinear hub at all. Hypothesis H1's own test (Section 9.5) is a comparison on the SAME underlying question — integration architecture — but at a different, and more basic, point on it, and this section states explicitly, before Chapter 11 reports whichever of the two comparisons this thesis has actually run, why the two must not be conflated. Three architectures, not two, are in play once

Section 6.10's benchmark is read alongside Section 9.5's test: fully independent, sequentially designed trains sharing no resources at all (H1's sequential baseline, Section 9.1); this thesis's own hub-connected architecture, which shares the pooled sugar RESOURCE across trains without sharing processing EQUIPMENT between them (Section 6.1, Section 6.3) — the same structural role Corsano et al.'s (2009) mother-plant-to-derivative-plants resource balance plays in their Multiplant Complex (MC) framework (Section 6.10); and a hypothetical fully integrated multiproduct plant sharing EQUIPMENT as well as resources across products — the role Corsano et al.'s (2009) Multiproduct Plant (MP) alternative plays in their own benchmark, and the architecture this thesis has not modeled at all.

Confirming H1 would establish the first-to-second step on this spectrum: that this thesis's shared-sugar-hub architecture outperforms treating the four fraction trains as fully independent, sequentially designed plants, corroborating this dissertation's central Section 2.6 critique of a sequential or pre-resolved-allocation treatment of shared upstream resources. It would NOT, on its own, establish that the second point — this thesis's specific hub-and-four-output-platforms topology — is itself the best available point on the integration spectrum, since the third point (a Corsano-style fully integrated multiproduct plant) is never constructed or evaluated by H1's test protocol at all. Corsano et al.'s (2009) own result for the analogous second-to-third comparison in their Sugar Cane case study — MC and MP economically close but not identical (a ~19% investment-cost difference) alongside a materially different labor-demand implication (15 versus 9 workers) — is the concrete precedent motivating this caution: in a structurally similar problem, moving from resource-sharing to full equipment-sharing integration was a non-negligible, not a foregone-conclusion, difference, and there is no basis in this thesis's own analysis for assuming its

own hub architecture would show a negligible difference against the analogous, currently unmodeled comparison.

Practically, this means the recommended configuration Section 9.3 assembles, and whatever hypothesis-test outcome Chapter 11 reports for H1, should both be read as the best result THIS THESIS'S CHOSEN TOPOLOGY can produce, not as evidence that no better topology exists — a scope qualification stated here explicitly rather than left for a reader to infer, consistent with the same discipline Section 9.3's own closing paragraph applies to internal inconsistencies among its six input sections. This synthesis point does not weaken H1's test or this thesis's hub formulation (Section 6.3's case for representing the hub's bilinear pooling exactly, rather than approximating it away, is independent of which broader plant topology surrounds that hub); it sharpens what a confirmed H1 result is actually evidence FOR, and reinforces, rather than duplicates, the future-work item Section 11.3 ("Seventh") already commits to: running the analogous MC-versus-MP-style architectural comparison for this thesis's own case, once Section 6.9's verification protocol and H1's own test have both been executed.

Chapter 10. Colombian Policy Framing and Implementation

Chapters 6 through 9 developed and synthesized a techno-economic-environmental case for the Bajo Atrato biorefinery on its own engineering and economic terms — as though the recommended configuration of Section 9.3 existed in a policy-neutral vacuum. It does not. Section 1.2 already established that this thesis's entire off-grid framing is itself a policy fact, not merely a technical constraint: Bajo Atrato is a Zona No Interconectada (ZNI) precisely because of a set of financial, geographic, and institutional conditions Colombian electricity policy has not resolved (Bustos et al., 2014; Garces et al., 2021), and any biorefinery sited there inherits the regulatory, subsidy, and institutional environment that framing implies, whether or not this thesis's own modeling chapters treat it explicitly. This chapter makes that environment explicit. Section 10.1 develops a scenario and what-if analysis of how policy support and market volatility jointly bear on the recommended configuration's performance. Section 10.2 develops the public-policy implications of this thesis's design framework, and of the configuration it is built to recommend, for rural development, bioeconomy strategy, and energy access in ZNI territories specifically. Section 10.3 closes with an honest accounting of the barriers this thesis's recommended configuration would actually face in Bajo Atrato, the enablers already available to offset them, and a phased implementation roadmap.

This chapter is deliberately not a repetition of Chapter 8's uncertainty and sensitivity analysis under a different name. Section 8.6's Monte Carlo protocol propagates continuous parameter uncertainty — cost, price, yield — through a fixed configuration's cash-flow model to characterize technical and market risk. This chapter's scenario analysis (Section 10.1) instead varies a small number of discrete policy and macro-conditions (incentive availability, subsidy design, market volatility regime) to characterize policy risk and policy leverage specifically, a

distinct question Section 8.6's continuous-parameter framework was not built to answer and does not attempt to.

10.1 Scenario and What-If Analysis

This section examines how the recommended configuration's techno-economic performance (Section 9.3) would plausibly shift under different combinations of two policy-relevant axes: the degree of ZNI energy and bioeconomy policy support available to a Bajo Atrato investment, and the degree of feedstock-price and market volatility the project faces independent of that policy environment. Figure 10.1 organizes this as a two-by-two scenario matrix, not because policy support and market volatility are the only two relevant dimensions of uncertainty this thesis has identified — Section 8.6 and Section 6.7 already treat several others quantitatively — but because these two specifically determine which of this thesis's already-developed configurations (expected-NPV-optimal versus CVaR-optimal, Section 6.6-6.7) is the more defensible recommendation under a given policy regime, a question the purely technical chapters do not themselves resolve.

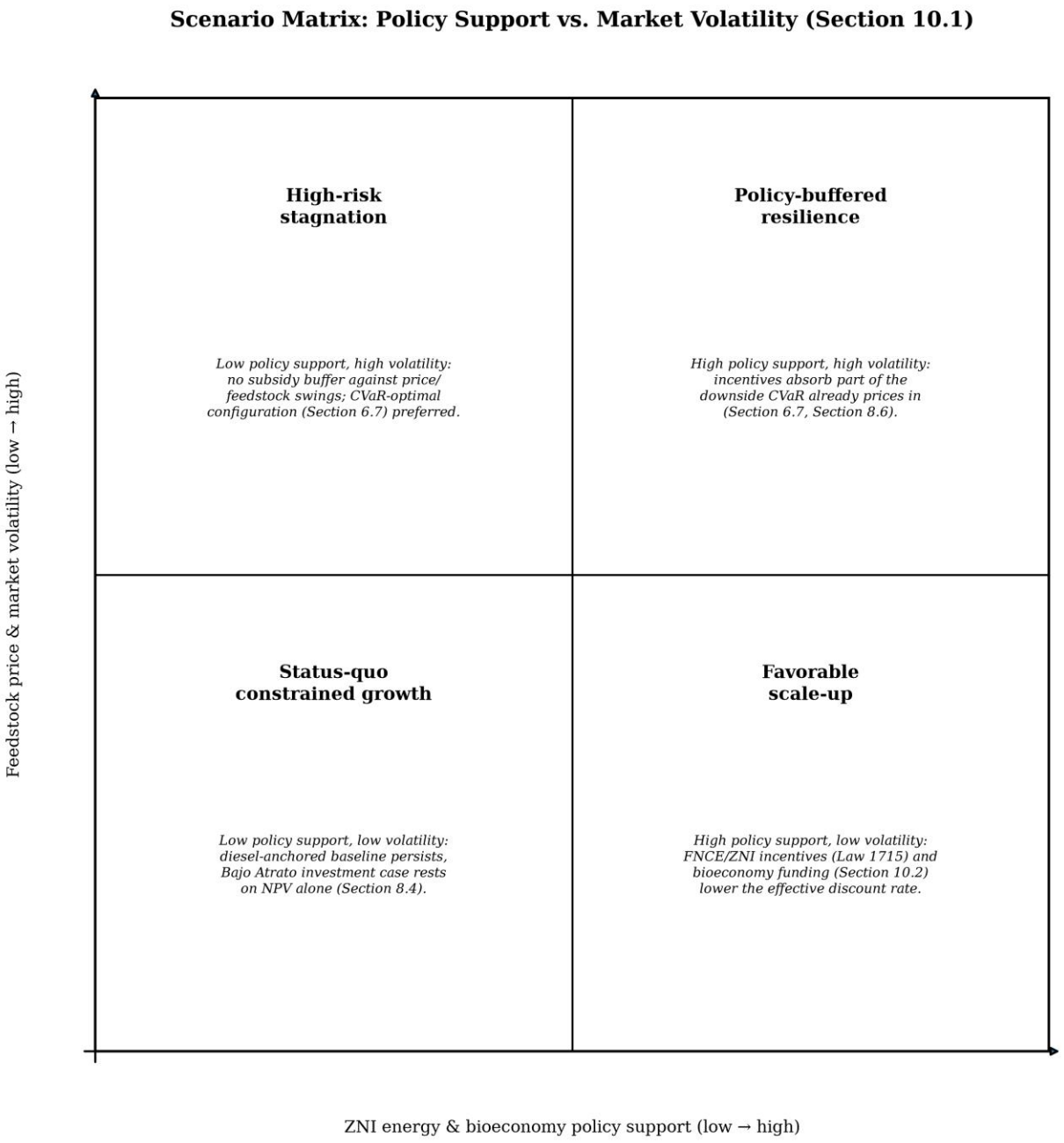


Figure 10.1. A scenario matrix crossing ZNI energy and bioeconomy policy support against feedstock-price and market volatility, indicating which of this thesis's already-developed configurations (expected-NPV-optimal or CVaR-optimal) is the more defensible recommendation in each quadrant.

This is an original diagram developed for this thesis, applying a standard two-axis scenario-planning structure to the specific policy and market variables Section 10.1 identifies as most consequential for the Bajo Atrato investment case.

The low-policy-support, low-volatility quadrant ("status-quo constrained growth") is the scenario under which this thesis's investment case rests most heavily on Section 8.4's NPV figure standing on its own, without a subsidy or incentive buffer and without a market-volatility risk this thesis's CVaR treatment (Section 6.7) needs to price in heavily; here, the deciding question is simply whether the expected-NPV-optimal configuration's return is attractive enough on unsubsidized terms; the high-policy-support, low-volatility quadrant ("favorable scale-up") is the scenario in which Colombia's non-conventional-renewable-energy incentive framework — Law 1715 of 2014 and its associated FNCE (Fuentes No Convencionales de Energía) provisions (Rodríguez-Urrego et al., 2018) — plausibly lowers the effective cost of capital or offsets a share of capital expenditure (Section 8.2), improving the NPV case without this thesis's model needing to be re-solved, since a lower discount rate or reduced CAPEX is already within the sensitivity range Section 8.4 and Figure 8.2 characterize. The two high-volatility quadrants are where this thesis's CVaR-optimal configuration (Section 6.7), rather than the expected-NPV-optimal one, becomes the more defensible recommendation: in the low-policy-support/high-volatility quadrant ("high-risk stagnation") specifically, because no subsidy exists to backstop a bad price or yield draw, so the risk-averse configuration's protection against the downside tail carries more practical weight than its lower expected value costs; in the high-policy-support/high-volatility quadrant ("policy-buffered resilience"), because policy support and CVaR's own downside protection partially substitute for one another, and this thesis's recommendation there depends on the specific magnitude of each, a quantitative question Section 10.1's protocol defers to the eventual joint execution of Section 6.7's CVaR computation and a specific, named incentive-program costing rather than resolving qualitatively here.

This section's what-if analysis is explicitly a qualitative-to-semi-quantitative overlay on results Chapters 6 and 8 already generate, not a new optimization: no new MINLP is solved for this chapter, and no policy parameter is added to Section 6.2's variable set. Where a specific policy instrument (a capital grant, a feed-in tariff for surplus electricity, a tax holiday) can be represented as a modification to an existing Section 8.2-8.4 parameter — a reduced effective discount rate, a reduced K_j capital-cost coefficient — this thesis's existing Monte Carlo and tornado-diagram machinery (Section 8.6, Figure 8.2) is the correct tool to quantify its effect, and Section 10.1's role is to identify which existing parameter a given real Colombian policy instrument maps onto, not to build a parallel policy-optimization model duplicating Chapter 6's own.

10.2 Public-Policy Implications: Rural Development, Bioeconomy, and Energy Access in ZNI Territories

Colombia's national bioeconomy strategy explicitly frames biological-resource valorization as a development lever for regions whose economies remain concentrated in extractive industry and primary production (Rojas-Jiménez, 2021), and empirical assessments of the strategy's regional reach have found its translation from national policy document to actual regional funding and niche development uneven and inconsistent across Colombia's departments (Balanzó Guzmán et al., 2021) — a finding this thesis's own Bajo Atrato/Chocó siting makes directly relevant rather than abstract, since Chocó is not among the departments where bioeconomy policy attention and investment have concentrated to date. This thesis's recommended configuration (Section 9.3), once computed, would be, in the specific sense Grilli et al.'s (2024) Colombian bioeconomy-transition decision-support framework formalizes, exactly the kind of regionally grounded, stakeholder-relevant bioeconomy investment case that framework argues is needed to correct that uneven translation: a fully specified, quantitatively evaluated project

proposal for a specific territory, rather than a general policy aspiration awaiting a concrete instantiation.

For rural development specifically, the OECD's 2022 review of Colombian rural policy identifies the bioeconomy and renewable energy as among the under-tapped development opportunities available to Colombia's rural regions, alongside persistent structural barriers — land-tenure informality, low connectivity, uneven access to education and health services — that a single biorefinery investment cannot itself resolve but that a biorefinery investment's own labor demand, produce market, and energy infrastructure can incrementally address (OECD, 2022). Chocó's specific post-conflict context sharpens this framing further: Colombia's National Bioeconomy Strategy, launched in the wake of the 2016 Peace Agreement, frames the sustainable use of biodiversity as a means to achieve both sustainability and peace (Huddart et al., 2022), and Bajo Atrato is among the historically underserved, conflict-affected regions that framing addresses, so this thesis's recommended configuration is not merely an industrial siting decision but one that intersects directly with Colombia's own stated post-conflict rural development objectives — a framing this thesis states as context for Section 10.3's roadmap, not as a claim this thesis itself has evaluated the project's post-conflict development impact with the same rigor Chapters 6-9 apply to its techno-economic and environmental performance.

[NOTA — citas de los Cap. 9-11 (Consensus, 4-nov). Huddart et al. (2022) vincula la paz con la Estrategia Nacional de Bioeconomía, no con el Acuerdo de Paz de 2016, que según el resumen «increased access» a los ecosistemas; que Bajo Atrato pertenezca a esas regiones es afirmación tuya (correcta, pero sin esa fuente). Grilli et al. (2024) pide ampliar los nodos de conocimiento y los mecanismos financieros: no sirve para decir que ya están orientados a proyectos como este. Balanzó Guzmán et al. (2021): Consensus registra como autor a J. P. Centeno; revisa la autoría. Confirmados: Alviar et al. (2021), OECD (2022, 2023), Rodríguez-Urrego et al. (2018, Ley 1715/2014), Ortiz Jara (2019), Salvador et al. (2022), Romero-Piñeiro et al. (2025; eucalipto, no residuo anual), Garces et al. (2021) y Aberilla et al. (2019) en 10.2. Rojas-Jiménez (2021) es un capítulo de libro (Palgrave), sin resumen en Consensus.]

For energy access specifically, this thesis's off-grid energy-island design (Chapter 5) is directly responsive to the documented failure mode Garces et al. (2021) identify in Colombian last-mile electrification policy: an institutionally complex subsidy environment that has continued prioritizing diesel-generator supply through supply-side subsidies even where renewable alternatives demonstrate better outcomes, without those better-performing pilot experiences being systematically disseminated or scaled. This thesis's own energy-autonomy constraint (Section 1.2, Section 5.4) is, in effect, a design decision to opt out of that diesel-subsidy-dependent status quo at the level of a single facility, and the specific evidence base Section 5.1-5.2 and Section 7.7 already assemble — that biomass gasification-ICE generation outperforms diesel generation across most environmental impact categories in directly comparable off-grid, developing-region contexts (Aberilla et al., 2019) — is precisely the kind of pilot-level evidence Garces et al. (2021) find Colombian ZNI policy has so far failed to disseminate systematically. This thesis's contribution to that dissemination gap is necessarily modest — one fully specified case study, not a policy program — but it is stated here as a genuine, not merely rhetorical, policy contribution this dissertation's technical chapters make available to a policy audience beyond its own academic one.

10.3 Barriers, Enablers, and Implementation Roadmap for Bajo Atrato

This section states, without minimizing them, the specific barriers a real Bajo Atrato implementation of this thesis's recommended configuration would face, following the same disclosure discipline this thesis has applied to every prior chapter's own limitations (Section 6.9, Section 7.8, Section 8.2's cost-estimation-provenance protocol). Three barriers are structural rather than incidental. First, capital-cost estimation uncertainty (Section 8.2): several of this thesis's candidate technologies, the shared sugar hub's own infrastructure chief among them, lack first-of-

a-kind cost precedent, and Tsagkari et al.'s (2016) documented factor-of-two divergence across costing heuristics for exactly this kind of novel biorefinery equipment means a real financing process would need to close that estimation gap — via vendor engagement, pilot-scale costing, or both — before Section 8.4's NPV figure could be treated as bankable rather than indicative. Second, institutional complexity in the ZNI subsidy and regulatory framework itself: Garces et al. (2021) and Ortiz Jara (2019) both document a fragmented, multi-actor ZNI institutional environment that has historically hindered, rather than facilitated, exactly the kind of renewable-energy-substituting investment this thesis's energy island represents, so navigating that framework is itself a project risk distinct from and additional to the technical and market risks Section 8.6 already quantifies. Third, feedstock-logistics scale-up: Section 3.3 already noted, and this thesis's Chapter 9 synthesis did not revisit, that this thesis's design basis assumes co-located, single-site feedstock sourcing, and that a geospatial multi-site logistics model was judged out of scope at the current design stage; a real implementation growing beyond that single-site assumption would need to develop the logistics-optimization capability Section 3.3 deferred, a genuine barrier this thesis's own scope decision has not yet addressed rather than one external to this dissertation's own choices.

Three enablers partially offset these barriers, and this thesis states them with the same evidentiary specificity rather than as generic optimism. First, Colombia's Law 1715/2014 FNCE incentive framework (Rodríguez-Urrego et al., 2018) and the broader distributed-renewable-energy investment-support instruments the OECD has catalogued for Colombian ZNI contexts specifically (OECD, 2023) provide a concrete, named policy channel through which Section 10.1's "favorable scale-up" scenario could actually be realized, not merely hypothesized. Second, Colombia's national bioeconomy funding and knowledge-hub infrastructure, while unevenly

distributed regionally (Balanzó Guzmán et al., 2021), does exist and is oriented toward regionally grounded, biodiversity-based value addition (Alviar et al., 2021), although Grilli et al. (2024) find that its knowledge hubs and financial mechanisms still need to be expanded, giving this project a stronger claim on that funding than a similarly performing project without an explicit bioeconomy-strategy alignment would have. Third, the broader Latin American circular-bioeconomy literature's own documented barriers — Salvador et al. (2022) identify policy fragmentation, scaling difficulty, and weak inter-institutional collaboration as the region's most consistent obstacles — are barriers this thesis's specific project can partially route around precisely because it is a single, already fully specified facility proposal rather than a diffuse regional strategy awaiting implementation detail, the same advantage Grilli et al.'s (2024) decision-support framework argues a concrete case study holds over an abstract policy aspiration.

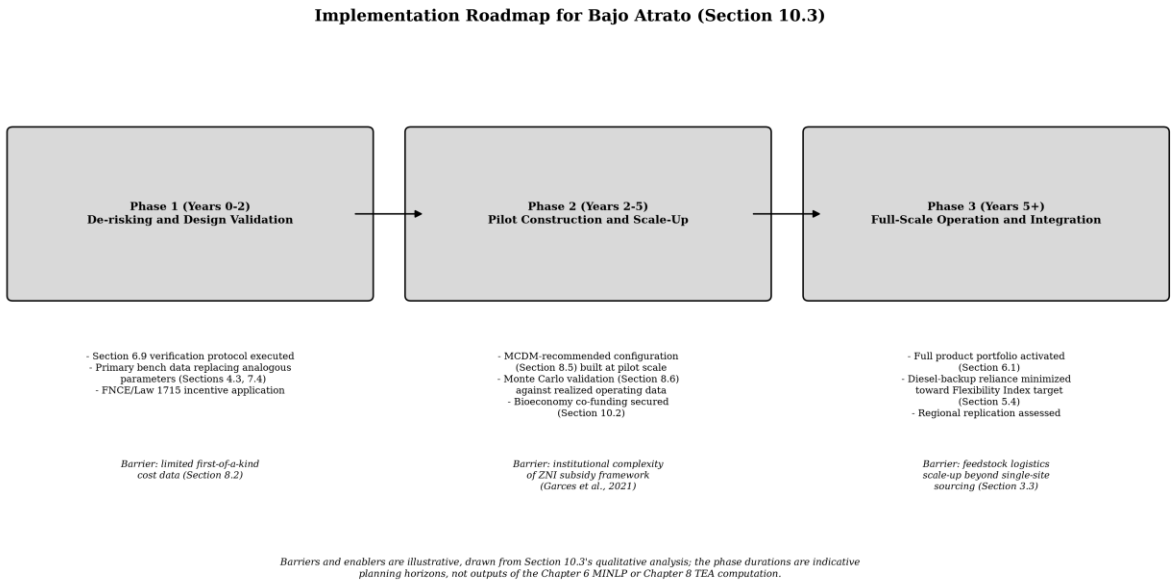


Figure 10.2. A phased implementation roadmap for the Bajo Atrato biorefinery, mapping each phase's technical milestones (drawn from this thesis's own verification and validation protocols) against the specific barrier most salient to that phase.

This is an original diagram developed for this thesis; the phase durations are indicative planning horizons based on the scale of comparable off-grid biomass-energy and biorefinery implementation precedents reviewed in Chapter 2 and this chapter, not an output of the Chapter 6 MINLP or Chapter 8 TEA computation.

Figure 10.2's three-phase roadmap is deliberately built around milestones this thesis has already specified elsewhere, rather than introducing new implementation criteria invented for this chapter alone: Phase 1 closes with Section 6.9's verification protocol actually executed and with the specific analogy-sourced parameters Section 4.3 and Section 7.4 flagged as carrying wider uncertainty replaced by primary bench or pilot data; Phase 2 closes with Section 8.6's Monte Carlo validation checked against real, not merely modeled, operating data from a pilot-scale build of the Section 8.5 recommended configuration; and Phase 3 targets the full activation of Section 6.1's product portfolio with diesel-backup reliance reduced toward the operability margin Section 5.4's Flexibility Index defines, at which point this thesis's own energy-autonomy design intent (Section 1.2) is realized in an operating facility rather than only in its design basis. This roadmap's role is to make explicit that this dissertation's own chapters already define the technical content of what "successful implementation" would concretely mean, so that Chapter 10's policy framing and Chapter 11's conclusions are answerable to the same standard of specificity this thesis has applied throughout, rather than closing on a generic call for future investment.

Chapter 11. Conclusions and Future Work

This dissertation set out, in Section 1.5, to design and optimize a whole-plant pineapple biorefinery for the off-grid Bajo Atrato context via an MINLP superstructure, and to assess its techno-economic and environmental performance under uncertainty. Chapters 3 through 10 developed the seven specific objectives that general aim decomposes into, framed by three explicit hypotheses whose tests Section 9.5 specifies, and situated the resulting design within Colombia's own policy landscape. This closing chapter states what has, and has not, been established against each of those seven objectives (Section 11.1), states this dissertation's limitations plainly rather than leaving them scattered for a reader to reconstruct (Section 11.2), and closes with the specific research directions this thesis's own unfinished business defines (Section 11.3).

11.1 Conclusions Against Each Specific Objective

Figure 11.1 summarizes, objective by objective, what this dissertation has established at its present design-and-methodology stage. The pattern across all seven is consistent and worth stating once, explicitly, rather than repeating seven times: every objective has been methodologically completed — characterized, formulated, or specified to the point of computational readiness — while the numerical results that formulation would produce once solved (a specific Pareto front, a specific NPV, a specific GWP figure, a specific Flexibility Index value) remain pending the implementation and verification protocol Section 6.9 lays out, with one bounded exception: a reduced instance of the model, solved and verified as Section 6.9 describes, whose results are stated after the objectives below. This is not a hedge added retroactively to this closing chapter; it is the same evidentiary discipline this dissertation applied at every point from Section 6.9 onward, restated here as a summary rather than introduced for the first time.

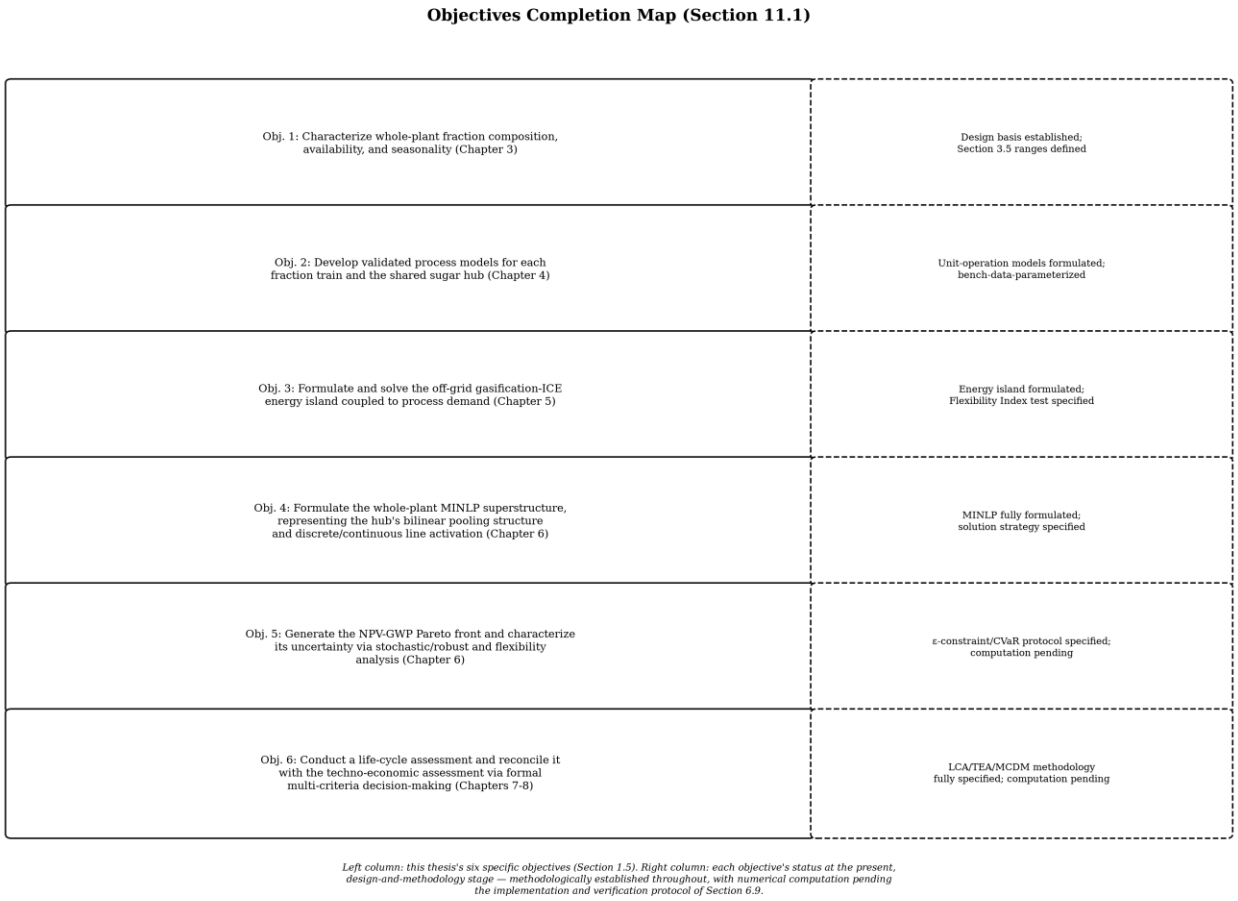


Figure 11.1. This dissertation's six specific objectives (Section 1.5) mapped against their status at the present design-and-methodology stage: methodologically established throughout, with numerical computation pending.

[NOTA: la Figura 11.1 y su pie muestran seis objetivos. Si conservas el objetivo (7), agrégalo a la figura y cambia «six» por «seven» en el pie; no edité la imagen.]

This is an original diagram developed for this thesis, summarizing the objective-by-objective conclusions this section states in prose.

Objective 1 (feedstock characterization, Chapter 3) is complete in the sense this thesis's design basis requires: composition, seasonality, and availability ranges for all four whole-plant fractions are established from the pineapple-specific and closely analogous bench-scale literature (Chapter 2, Chapter 3), each parameter's evidentiary provenance stated rather than left implicit, and the documented ranges — not single point estimates — are what Chapters 4 through 8 carry

forward. Objective 2 (validated process models, Chapter 4) is complete as a modeling exercise: every candidate technology in every fraction train, and the shared sugar hub's bilinear pooling formulation specifically, is derived from either direct pineapple-specific bench data or explicitly flagged analogous data, with SuperPro Designer established as the simulation environment on stated methodological grounds (Section 4.1). Objective 3 (the off-grid energy island, Chapter 5) is complete as a coupled formulation: the gasification-ICE system is sized against process heat/power demand, the Flexibility Index test is formalized against documented feedstock-parameter uncertainty (Section 5.4), and — critically, per this thesis's own standing methodological distinction — that flexibility test is kept structurally separate from the economic-uncertainty treatment of Chapter 6, not conflated with it.

Objective 4 (the MINLP superstructure, Chapter 6) is complete as a formulation: sets, parameters, and variables are fixed (Section 6.2); the hub's bilinear nonconvexity is represented exactly, not linearized away, with a stated McCormick-relaxation-plus-spatial-branch-and-bound solution strategy (Section 6.8) rather than an unexamined solver call; and the semicontinuous technology-activation and mutual-exclusivity structure (Sections 6.4-6.5) is fully specified. Objective 5 (the Pareto front and its uncertainty characterization, also Chapter 6) is complete as a protocol: the ε-constraint/AUGMECON generation method is justified against the weighted-sum alternative specifically because of this problem's own documented nonconvexity (Section 6.6), and the stochastic/CVaR economic-uncertainty treatment (Section 6.7) is kept explicitly distinct from Objective 3's flexibility analysis, per the same standing distinction just restated above. Objective 6 (life-cycle assessment reconciled with techno-economic assessment via formal MCDM, Chapters 7-8) is complete as a dual methodology: Chapter 7's LCA stands as an independently developed assessment, not a Pareto-term afterthought, with its own goal-and-scope, allocation, and

biogenic-carbon-accounting treatments each separately justified; Chapter 8's TEA supplies the NPV objective's actual cost and revenue content and narrows the resulting Pareto front to a single recommended configuration via a two-part, non-conflated procedure — a posteriori value-density reinterpretation and formal PROMETHEE-based MCDM selection (Section 8.5) — that Chapter 9 then synthesizes into a single coherent configuration profile. Objective 7 (synthesis and policy framing, Chapters 9-10) is complete as a specification: Chapter 9 states, before any computation, the comparison that will test each hypothesis and what would confirm or disconfirm it (Section 9.5), and Chapter 10 places the configuration that procedure is built to recommend within Colombia's ZNI policy environment and a phased implementation roadmap (Section 10.3).

The reduced instance of Section 6.9 adds three conclusions that the formulation alone could not. First, whether the plant is worth building turns on the wine excise tax before it turns on technology: with the tax, no configuration of the six lines is profitable; without it, the best one earns USD 2.3 million in NPV. Second, the wine line is robust to the measured evidence: it stays profitable in every sample drawn from the ranges of Campos et al. (2020) and Seguí Gil and Fito Maupoey (2018), and the measured pulp share (0.48-0.50 instead of the assumed 0.35) raises rather than lowers its value. Third, the bromelain line, present in every profitable certified design, is the conclusion least supported by evidence: at USD 30 per kg it requires an ultrafiltration concentration factor of at least eight and a powder of at least 1,290 GDU/g, both above what has been measured for pineapple liquor. Its inclusion in any recommended configuration should therefore be read as conditional on that demonstration, or on a price near USD 60 per kg.

Reading these seven conclusions together yields this dissertation's own answer, at the methodological level available at this stage, to the research question Section 1.5 posed: a simultaneous MINLP superstructure formulation of this kind can be built, is internally coherent

across its energy, economic, and environmental sub-models, and is constructed specifically to expose — rather than obscure — the trade-offs and nonconvexities a sequential or linearized treatment would hide. Whether the resulting configurations are, in fact, more profitable than a sequential baseline (Hypothesis H1), whether the energy-autonomy constraint is in fact binding at a quantified magnitude (Hypothesis H2), and whether the resulting Pareto set is in fact non-degenerate (Hypothesis H3) are empirical questions this methodology has been built to answer precisely, per Section 9.5's test protocol, but which remain open until that protocol is actually executed. One scope qualification on that first hypothesis is worth stating now rather than only once it is tested: per Section 9.6's synthesis of Section 6.10's benchmarking exercise, an eventual H1 confirmation would establish that this thesis's shared-hub architecture outperforms fully independent, sequentially designed trains, but would not by itself establish that this thesis's specific hub-and-four-output-platforms topology is the best available integration architecture — a further, more deeply integrated point on the same architectural spectrum (a single multiproduct plant sharing processing equipment, not merely the pooled sugar resource, across output platforms, in the sense Corsano et al., 2009, benchmark against their own comparable Multiplant Complex architecture) is never constructed or tested by H1's protocol at all.

[NOTA: H1 solo compara NPV. «and no less environmentally favorable» venía de la pregunta antigua, que el 24-sep se cambió por eso mismo; lo quité aquí para que pregunta, H1 y conclusión digan lo mismo. Si quieres la comparación ambiental contra la línea base secuencial, añádela a H1 o H3 y a 9.5, no solo aquí.]

11.2 Limitations

This dissertation's limitations have been flagged individually, at the point each arose, throughout Chapters 3-10; this section collects them in one place rather than leaving that collection as an implicit exercise for the reader. Five limitations are structural, in the sense that they follow

from this dissertation's own scope decisions rather than from an oversight this thesis failed to notice.

First, and most consequentially, this is a design and methodology dissertation, not a completed computational study: the full MINLP has not been solved end-to-end (only a reduced instance without the hub, the energy island, or the GWP objective has; Section 6.9), so every quantitative claim this thesis could in principle produce about the full design — the Pareto front's actual shape, the recommended configuration's actual NPV and GWP, the Flexibility Index's actual value — remains a protocol rather than a result. This is stated here not as an apology but as the single limitation every other conclusion in this dissertation is conditioned on. Second, this thesis's parameter base mixes direct pineapple-specific bench data with analogous data drawn from related fruit-processing or biomass-conversion literature where pineapple-specific evidence does not exist (flagged explicitly in Sections 4.3, 4.5, and 7.4 at each point it occurs); the resulting model is defensible but not equivalent in evidentiary strength to one built entirely from primary, pineapple-specific measurement. A direct comparison of the reduced instance's parameters with two measurement studies (Campos et al., 2020; Seguí Gil & Fito Maupoey, 2018) quantifies the point: five of the fourteen parameters checked lie outside the measured ranges, three of them on the optimistic side (a by-product share that includes the crown, the fermentation efficiency of the liquor, and the activity of the bromelain powder). Third, this thesis's design basis assumes co-located, single-site feedstock sourcing (Section 3.3), deliberately excluding a geospatial multi-site logistics-optimization model on the judgment that its marginal contribution would not justify its modeling cost at this thesis's assumed scale — a judgment this thesis states as a scope decision, not a limitation invisible to its own author, and one that would need revisiting if the biorefinery's scale grew materially beyond this thesis's reference scenario.

Fourth, this thesis's cradle-to-gate (not full cradle-to-grave) LCA boundary and its economic (not system-expansion) primary allocation choice (Chapter 7) are each defensible, stated, and justified methodological choices, but each is also a choice the LCA literature itself documents as consequential to reported results (Cherubini et al., 2011; Gaffey et al., 2024) — this thesis's sensitivity checks (mass allocation, system expansion, biogenic-carbon-neutrality bound) partially, but do not completely, offset the residual uncertainty that choice introduces. Fifth, this thesis's temporal scope treats seasonality as parametric uncertainty within a representative annual reference scenario rather than as an explicit multi-period scheduling problem (Section 1.6); a full campaign-scheduling extension, of the kind the production-scheduling literature this thesis's own methodological foundation draws on would support, was judged outside this dissertation's scope.

A sixth limitation is worth stating separately because it is not a scope decision but an open methodological question this thesis has not fully resolved: Section 8.5's MCDM criteria weights (the relative importance of NPV, GWP, and downside risk) are to be derived via a disclosed weighting procedure whose specific numerical values this thesis has not yet fixed. This thesis's commitment to a transparent, auditable weighting procedure (Section 7.5, Section 8.5) is itself a methodological position, but it does not eliminate the underlying difficulty that any such weighting embeds a value judgment about how to trade economic performance against environmental performance — a judgment this thesis makes explicit rather than resolves definitively, since no purely technical procedure can substitute for that judgment being made and owned by a decision-maker.

A seventh limitation, surfaced only once this thesis's superstructure paradigm was benchmarked against an independent methodology (Section 6.10) and read together with Hypothesis H1's own test protocol (Section 9.6), concerns the scope of what this dissertation's

central architectural comparison can establish. This thesis's superstructure commits, from Chapter 6 onward, to a single shared-hub topology and tests, via Hypothesis H1, only whether that topology outperforms fully independent, unconnected trains — not whether it outperforms a more deeply integrated alternative, a single multiproduct plant sharing processing equipment, not merely the pooled sugar resource, across output platforms. Corsano et al.'s (2009) own benchmark for a structurally similar sugar-cane-derivatives case found this further comparison economically non-negligible (a ~19% investment-cost difference between their Multiplant Complex and Multiproduct Plant configurations) rather than a foregone conclusion, so this dissertation's own architectural commitment should be read as the best-performing topology among those it actually tests, not as demonstrated optimal across every topology a designer could in principle construct. Section 11.3 ("Seventh") states the concrete follow-on study this limitation motivates.

11.3 Recommendations and Future Research Directions

The most immediate direction for future work is the computational execution this dissertation's methodology has been built toward but has not yet performed: implementing the full MINLP in a solver environment (BARON, following Section 6.8's stated choice), executing Section 6.9's three-tier verification protocol, generating the actual NPV-GWP Pareto front, and testing Hypotheses H1-H3 against the specific comparisons Section 9.5 already defines. Every other future-work direction stated below is properly sequenced after, not instead of, this one, since several of them depend on knowing which region of the Pareto front and which specific recommended configuration this thesis's methodology actually selects.

Second, replacing this thesis's analogous-data parameters with primary, pineapple-specific bench or pilot data — the reduced instance of Section 6.9 ranks first the ultrafiltration concentration factor and powder activity of the stem+core liquor and second the pulp share of local

fruit, ahead of the spray-drying (Section 4.3) and leaves/crown pretreatment (Section 4.5) parameters this thesis had flagged — would narrow the parameter ranges Section 8.6's Monte Carlo protocol currently must carry as wide, analogy-sourced bands, and is a natural, bounded experimental research program a follow-on study could undertake independent of the full computational implementation. Third, extending this thesis's Section 7.6 biogenic-carbon treatment from a static complete-inventory accounting to a dynamic LCA framework — following the dynamic biogenic-carbon modeling precedent Romero-Piñeiro et al. (2025) apply to energy-crop-derived hydrogen production, adapted to this thesis's own annual-cycle pineapple-residue context — would sharpen this thesis's own Section 7.6 argument that pineapple's short cultivation cycle substantially, though not completely, mitigates the forest-bioenergy carbon-debt concern, by replacing that qualitative argument with an explicit time-resolved carbon-flux calculation.

[NOTA: protocolo de medición sugerido (paso 4 de la ruta de rigor), replicable con los de Campos (2020) y Seguí (2018): seis lotes de nueve frutas del Bajo Atrato; fracciones de pulpa, cáscara, corazón y corona, y si la corona llega a la planta; rendimiento de líquido del prensado con y sin 0,1 % de Ca(OH)2; °Brix y azúcares del líquido; proteína (Lowry y Bradford); fracción de retenido y factor de concentración alcanzables en la UF; GDU por g del polvo y su contenido de proteína (el dato que Seguí no informó).]

Fourth, if this thesis's recommended configuration (once computed) or a subsequent investment decision warrants scale beyond single-site feedstock sourcing, developing the geospatial multi-site logistics-optimization model this thesis's Section 3.3 explicitly deferred becomes a well-defined next research task, with this thesis's own superstructure providing the fixed process-technology layer such a logistics model would need to sit alongside rather than redesign. Fifth, extending Section 1.6's single-period reference-scenario treatment to an explicit multi-period or campaign-scheduling formulation — drawing on the state-task-network and resource-task-network scheduling frameworks this thesis's own methodological foundation already establishes

for related discrete-manufacturing contexts — would allow this thesis's superstructure to represent seasonal feedstock availability as a scheduling decision rather than only as parametric uncertainty, a genuine model extension rather than a reformulation of what already exists.

Sixth, once a physical pilot facility exists (Section 10.3's Phase 2), validating this thesis's process models, energy-island sizing, and TEA/LCA parameters against real operating data — closing the loop this thesis's own prospective, design-stage assessment has necessarily left open (Section 7.1, Section 11.2) — would convert this dissertation's methodology from a validated design tool into a validated operating-facility model, the single most consequential form of future validation available and one no amount of further desk-based modeling can substitute for. Seventh, and following directly from Section 6.10's benchmarking exercise against Corsano et al.'s (2009) Multiplant Complex framework, running the analogous architectural comparison for this thesis's own hub-and-outputs superstructure — a hypothetical set of separate, dedicated processing trains, one per output platform, benchmarked against the shared, hub-centered architecture Chapter 6 adopts as its sole configuration — would test, rather than assume, that this thesis's central architectural commitment is the economically preferable one; Corsano et al.'s (2009) own Sugar Cane case study found the comparable choice economically close but not identical (a ~19% investment-cost difference alongside a materially different labor-demand implication neither their cost model nor this thesis's own prices), so the analogous result for this thesis's own case is a genuine open question, not a foregone conclusion, and one well-scoped to run once Section 6.9's verification protocol and this future-work list's first item are complete. Eighth, and finally, this thesis's specific methodological contributions — the explicit bilinear hub formulation, the joint energy-autonomy-constrained superstructure, and the integrated MCDM-reconciled TEA/LCA procedure (Section 1.7) — are not specific to pineapple or to Bajo Atrato by construction, and a

natural extension is testing their transferability to other whole-plant, multi-fraction agro-industrial residues sited in other Colombian or Latin American Zonas No Interconectadas, following the same broader circular-bioeconomy opportunity Salvador et al. (2022) and Grilli et al. (2024) both identify as under-realized across the region: this dissertation's own case study as the first fully specified instance of a methodology this thesis argues, but has not yet demonstrated across multiple sites, is generalizable in its structure even where its specific parameter values are not.

This dissertation closes where Section 1.3 began: with the claim that an off-grid, whole-plant pineapple biorefinery for Bajo Atrato is not merely a technically feasible idea but a problem whose genuine difficulty — a nonconvex shared-resource hub, a hard energy-autonomy constraint, and a joint economic-environmental objective — rewards being confronted directly, with the full mathematical and methodological apparatus that difficulty requires, rather than simplified away for tractability. Chapters 3 through 10 have built that apparatus, chapter by chapter, each one's methodological choices stated and justified rather than assumed. What remains — the computation itself, and everything Section 11.3 lists as following from it — is substantial, but it is a well-defined continuation of a design this dissertation has, at every step, taken care to make ready for exactly that continuation.

The consequence reaches beyond pineapple and beyond Bajo Atrato. Section 1.2 argued that biorefinery techno-economics has carried a hidden boundary condition — grid access — that silently removes most of rural Colombia from the set of sites any feasibility study would consider. The methodology developed here removes that condition: it makes energy autonomy a constraint to be designed for rather than a reason to exclude a site. Any agro-residue in a Zona No Interconectada could be assessed on the same terms — a transferability Section 11.3 still lists as

to be demonstrated — and the question for Colombian bioeconomy policy would shift from where the grid reaches to what each territory's own biomass can sustain.

[NOTA: el cierre afirmaba que cualquier residuo en cualquier ZNI «can be assessed»; 11.3 («Eighth») dice que la transferibilidad se argumenta pero no se ha demostrado. Calibré el verbo; la frase sigue terminando en la idea que sube de nivel. Recuerda revisar y reformular con tu voz todo lo verde antes de entregar.]

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[NOTA: Consensus registra este artículo con un solo autor (F. Hossain); confirma los coautores. Su resumen no da las cifras que 1.1 le atribuye («exceeding 25 million tonnes», «more than 80 producing countries»): verificalas en el texto completo o busca la fuente (p. ej., FAOSTAT).]

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